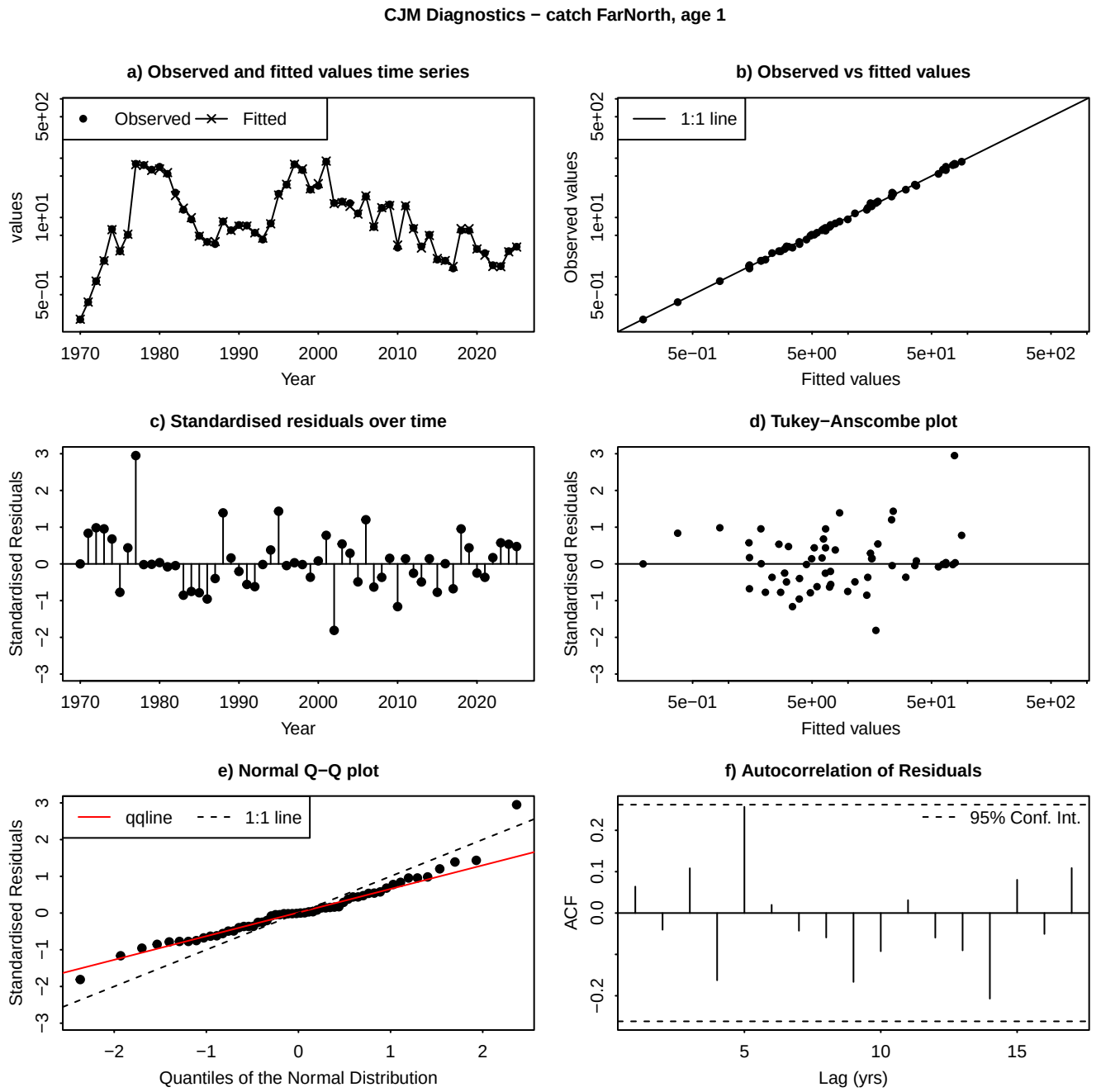
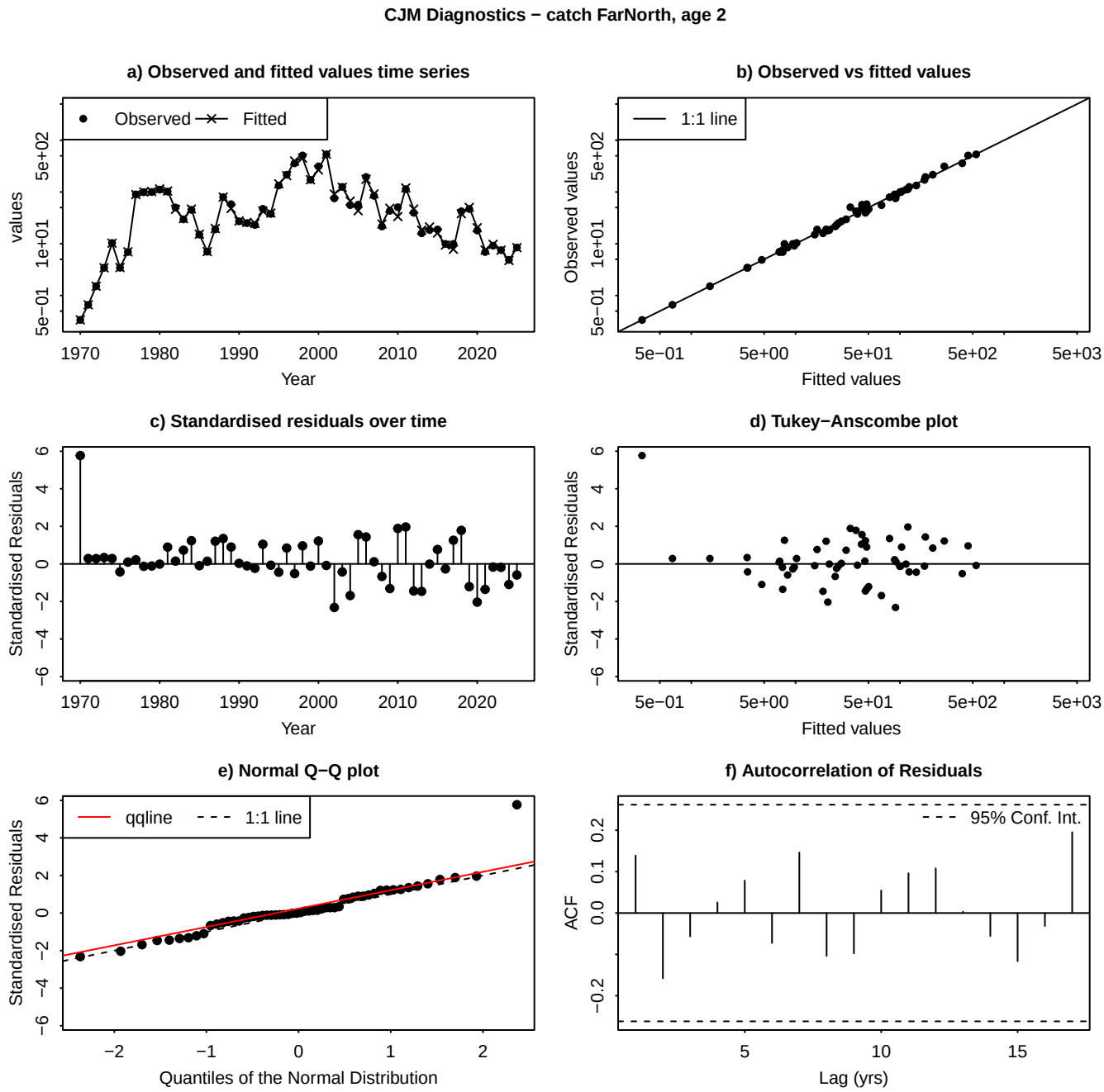
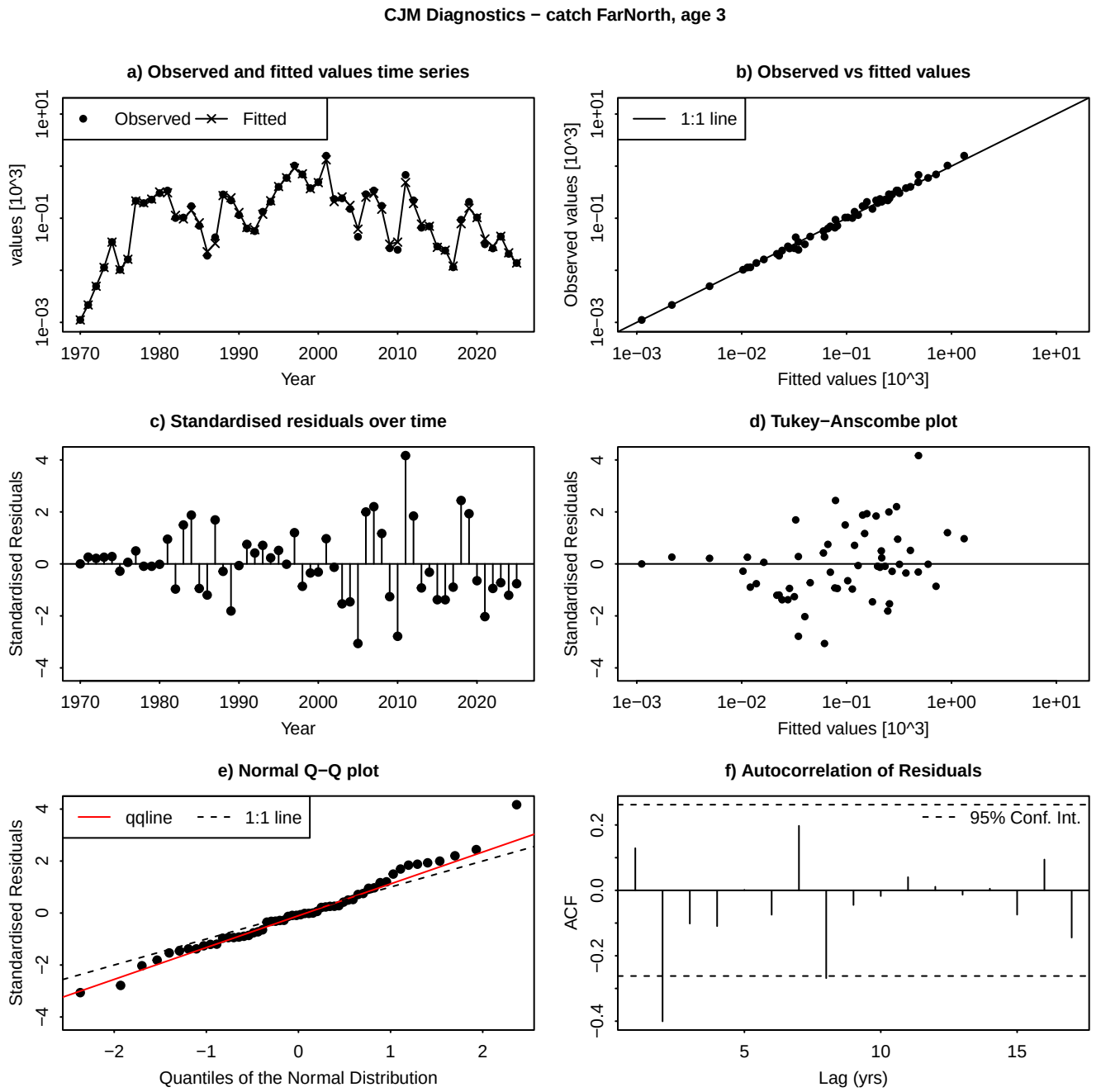
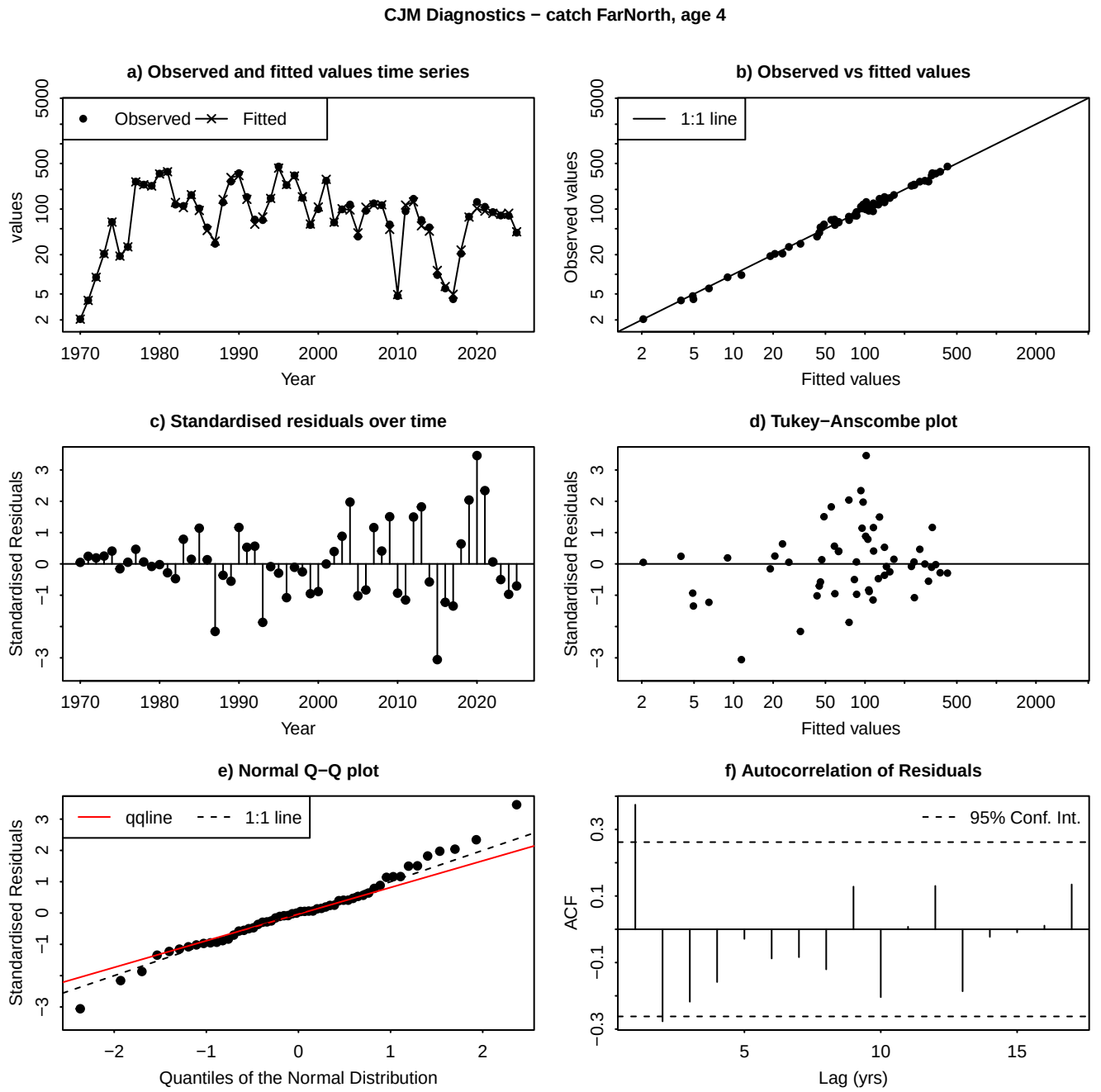
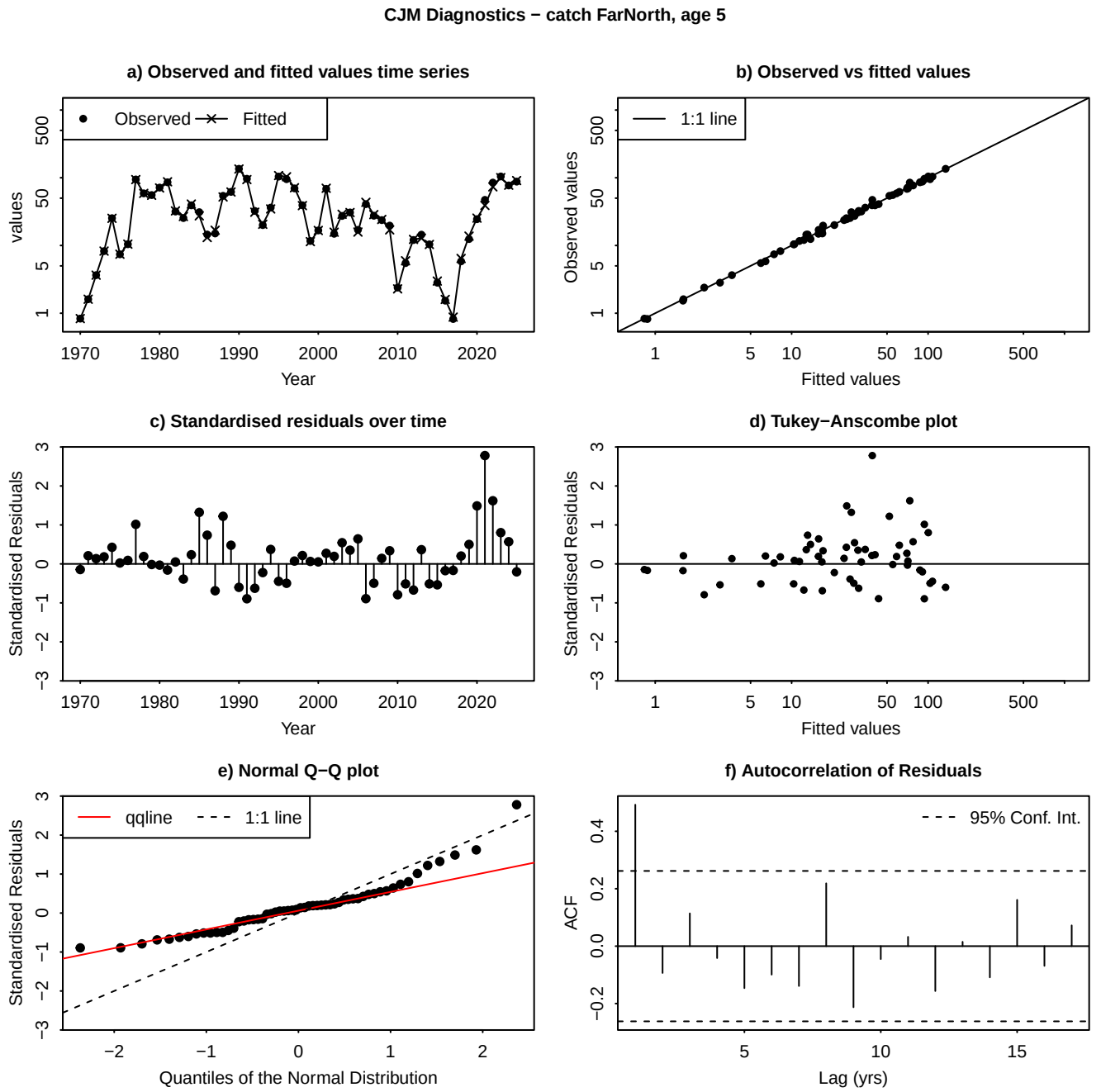
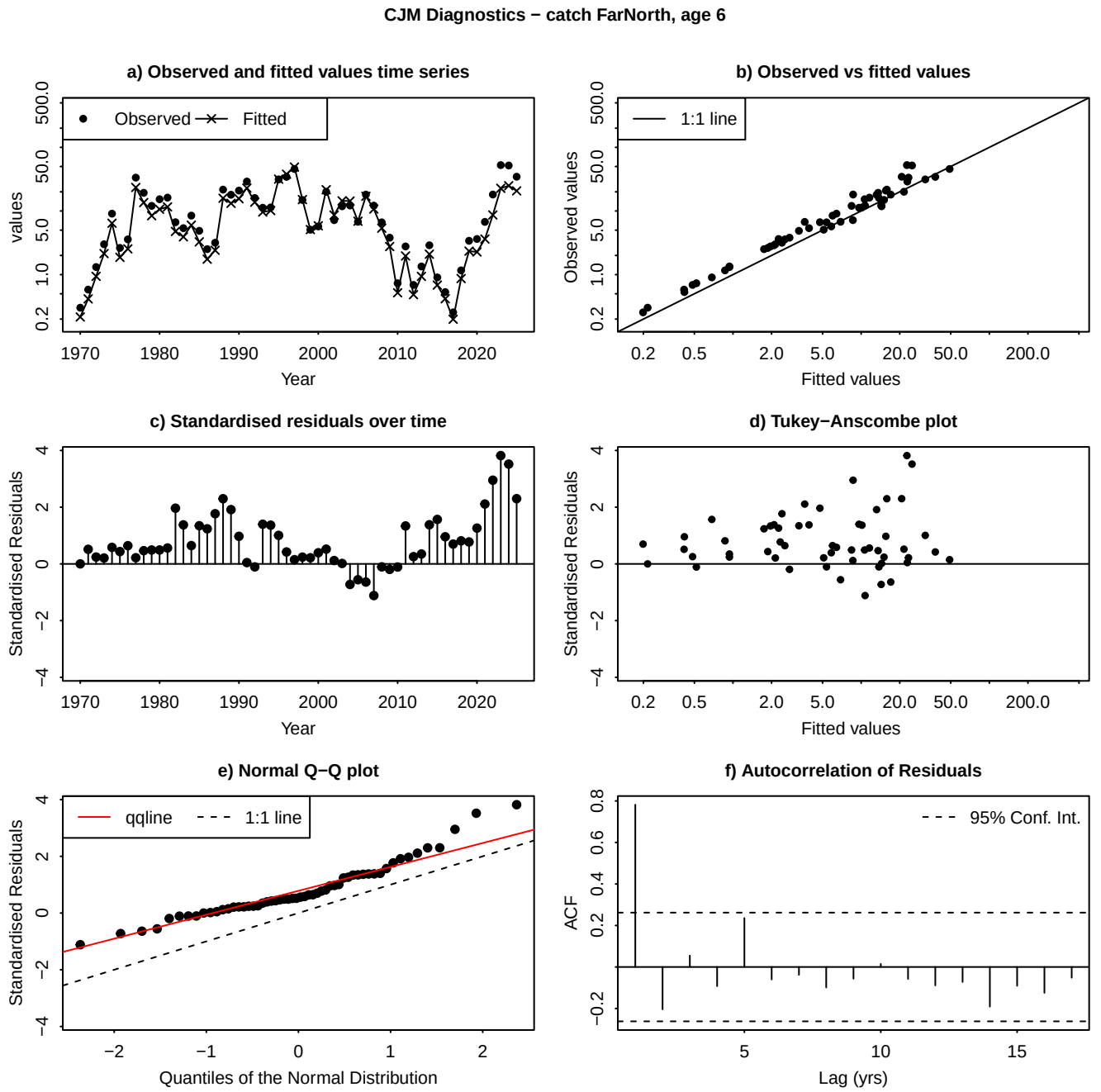

CJM Diagnostics − catch FarNorth, age 1
a) Observed and fitted values time series
Observed
Fitted
values
5e−01
1e+01
5e+02
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
5e−01
1e+01
5e+02
5e−01
5e+00
5e+01
5e+02
Fitted values
c) Standardised residuals over time
Standardised Residuals
−3
−2
−1
0
1
2
3
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
−3
−2
−1
0
1
2
3
5e−01
5e+00
5e+01
5e+02
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
−3
−2
−1
0
1
2
3
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.0
0.1
0.2
5
10
15
Lag (yrs)

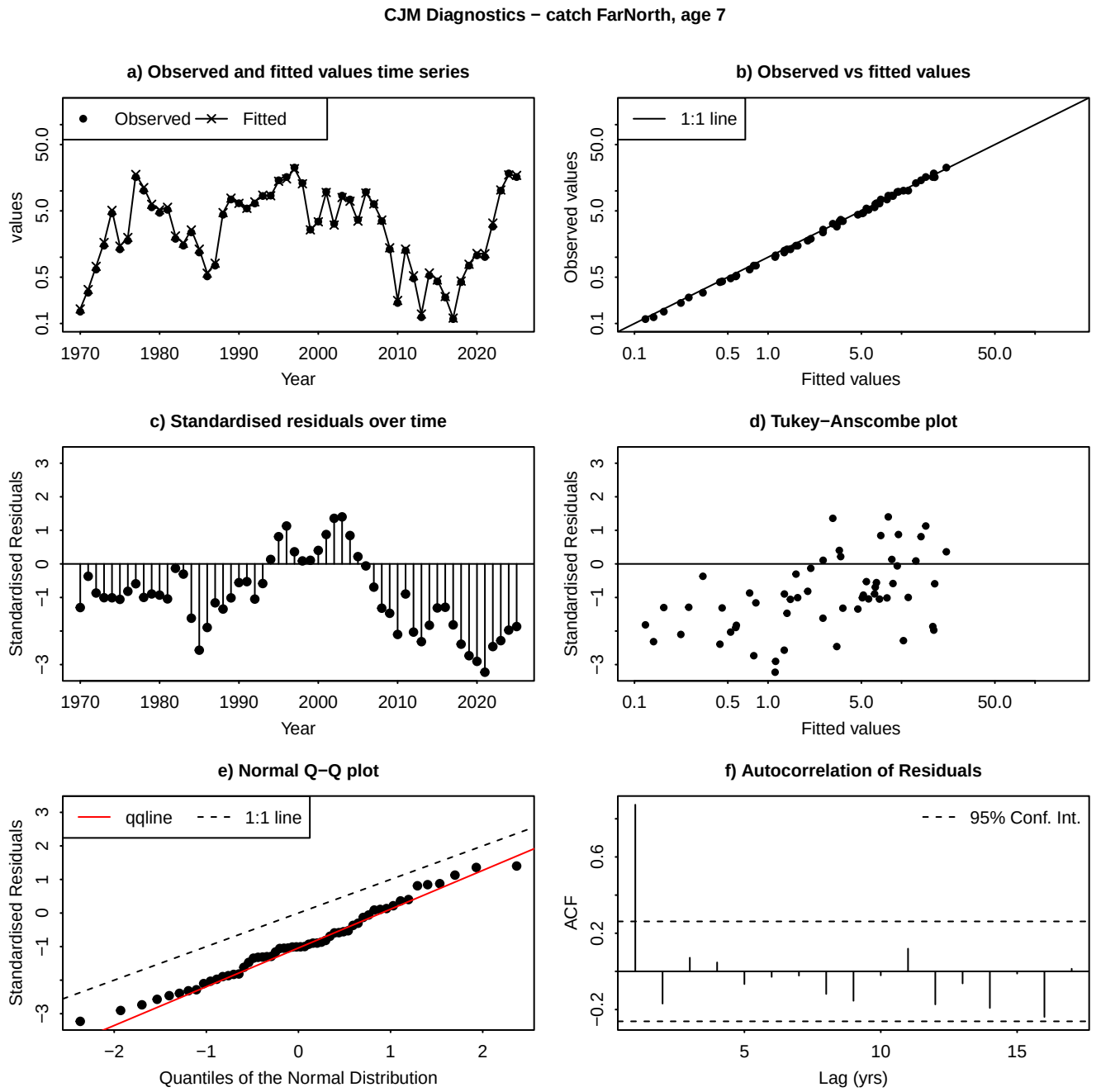

CJM Diagnostics − catch FarNorth, age 2
a) Observed and fitted values time series
Observed
Fitted
values
5e−01
1e+01
5e+02
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
5e−01
1e+01
5e+02
5e−01
5e+00
5e+01
5e+02
5e+03
Fitted values
c) Standardised residuals over time
Standardised Residuals
6
4
2
0
−2
−4
−6
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
6
4
2
0
−2
−4
−6
5e−01
5e+00
5e+01
5e+02
5e+03
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
6
4
2
0
−2
−4
−6
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
0.2
0.1
0.0
−0.2
5
10
15
Lag (yrs)

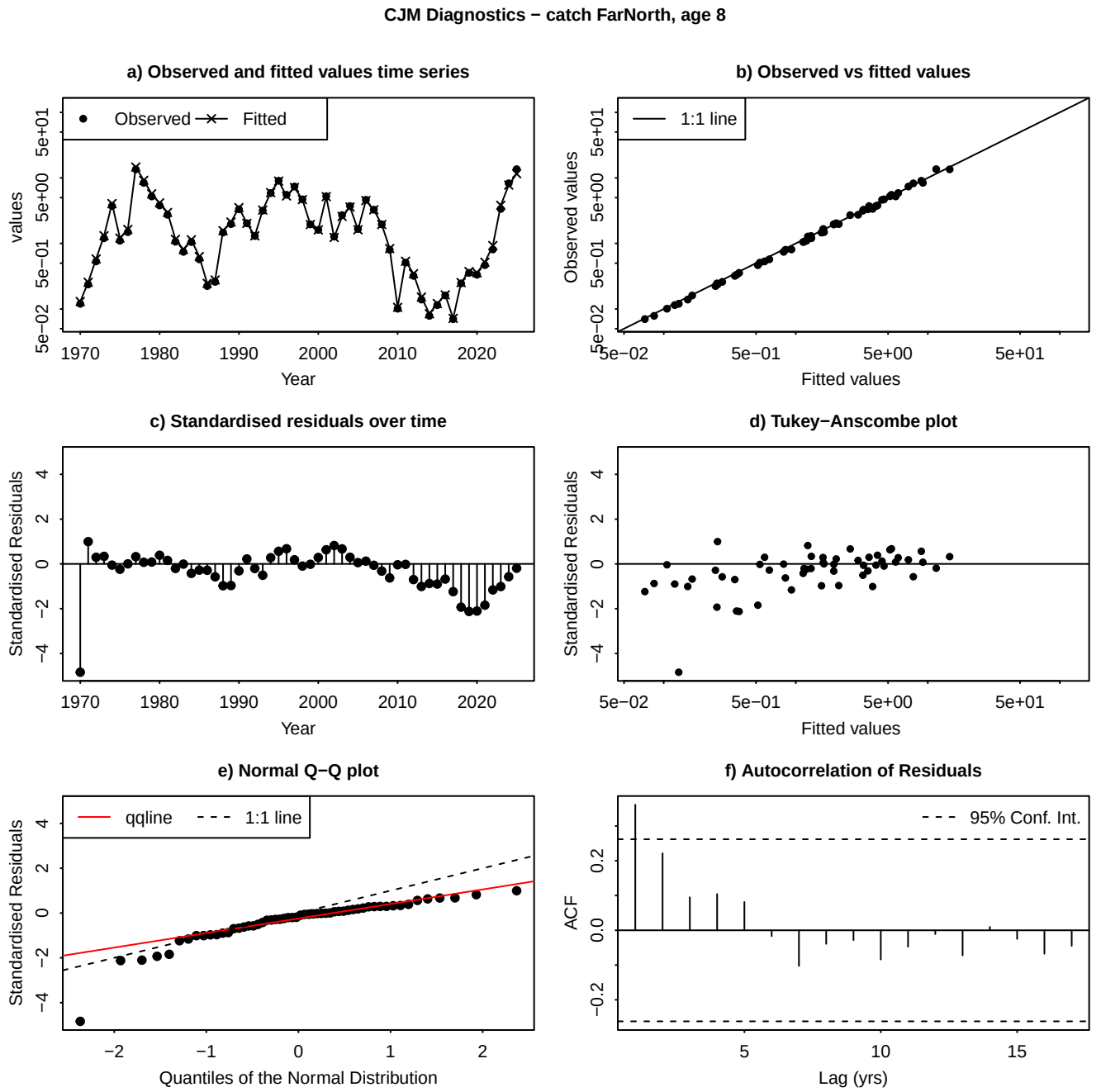

CJM Diagnostics − catch FarNorth, age 3
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
1e+01
1e−01
1e−03
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
1e+01
1e−01
1e−03
1e−03
1e−02
1e−01
1e+00
1e+01
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
4
2
0
−2
−4
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
4
2
0
−2
−4
1e−03
1e−02
1e−01
1e+00
1e+01
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
4
2
0
−2
−4
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
0.2
0.0
−0.2
−0.4
5
10
15
Lag (yrs)

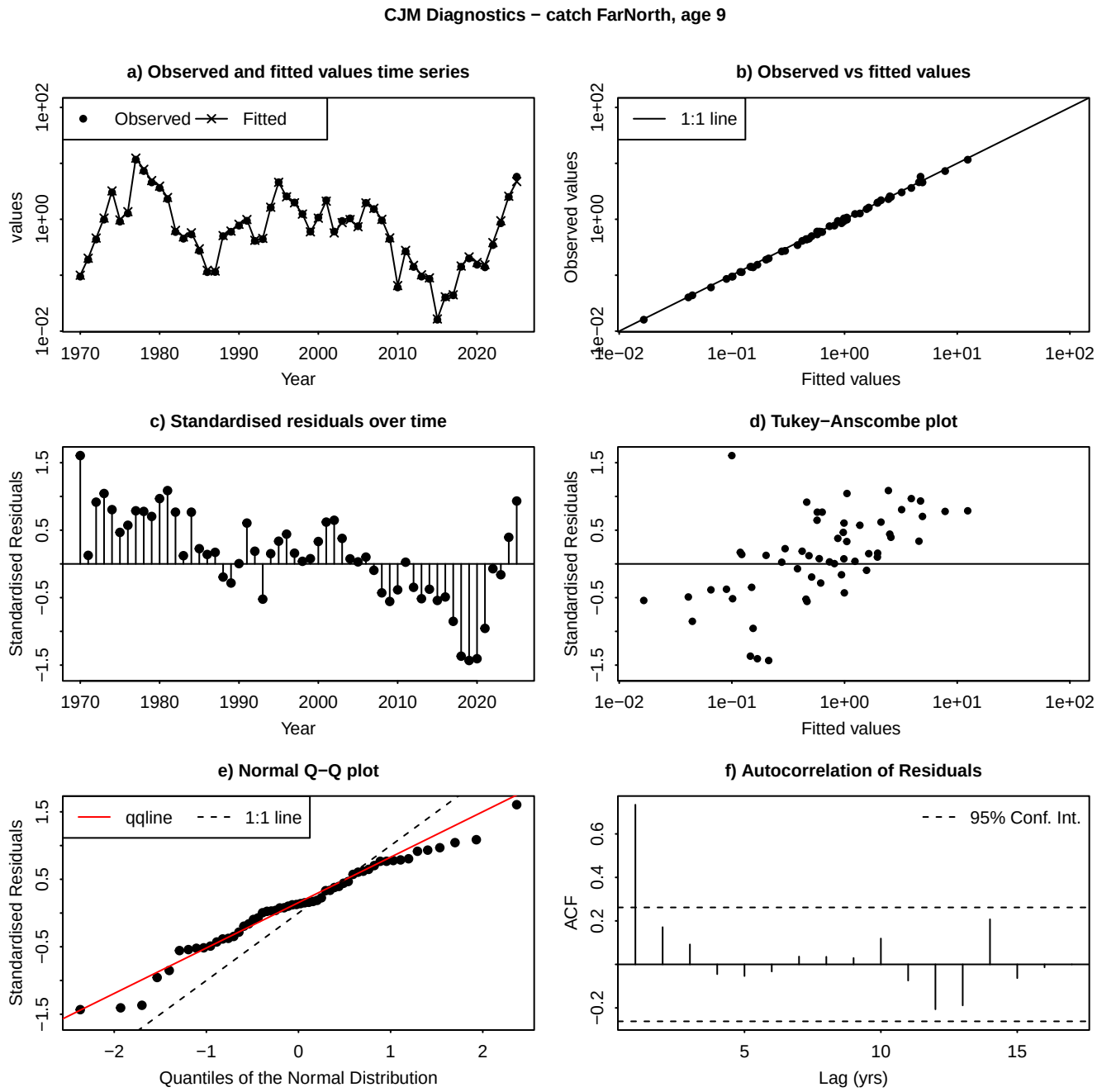

CJM Diagnostics − catch FarNorth, age 4
a) Observed and fitted values time series
Observed
Fitted
values
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

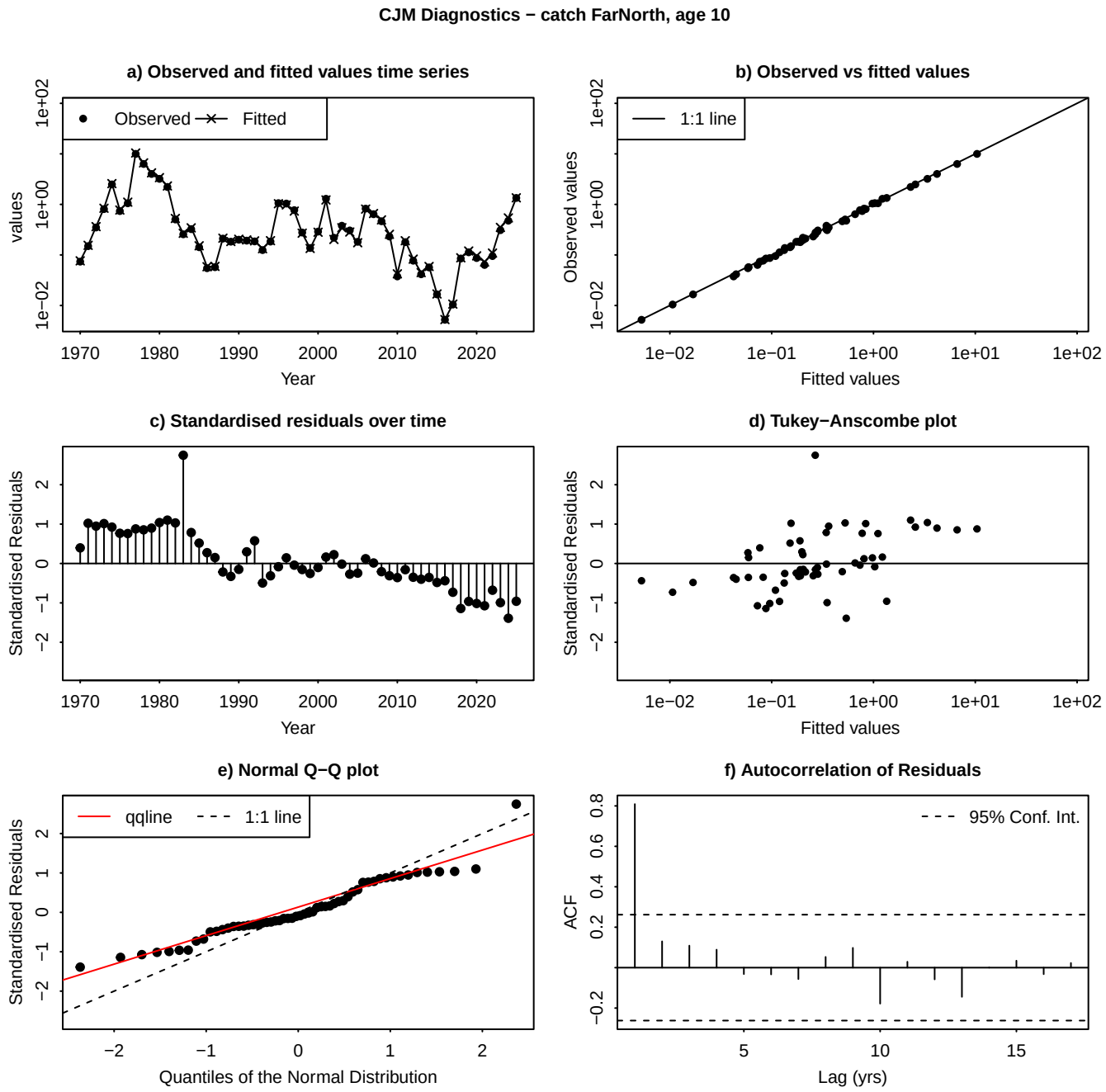

CJM Diagnostics − catch FarNorth, age 5
a) Observed and fitted values time series
Observed
Fitted
values
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

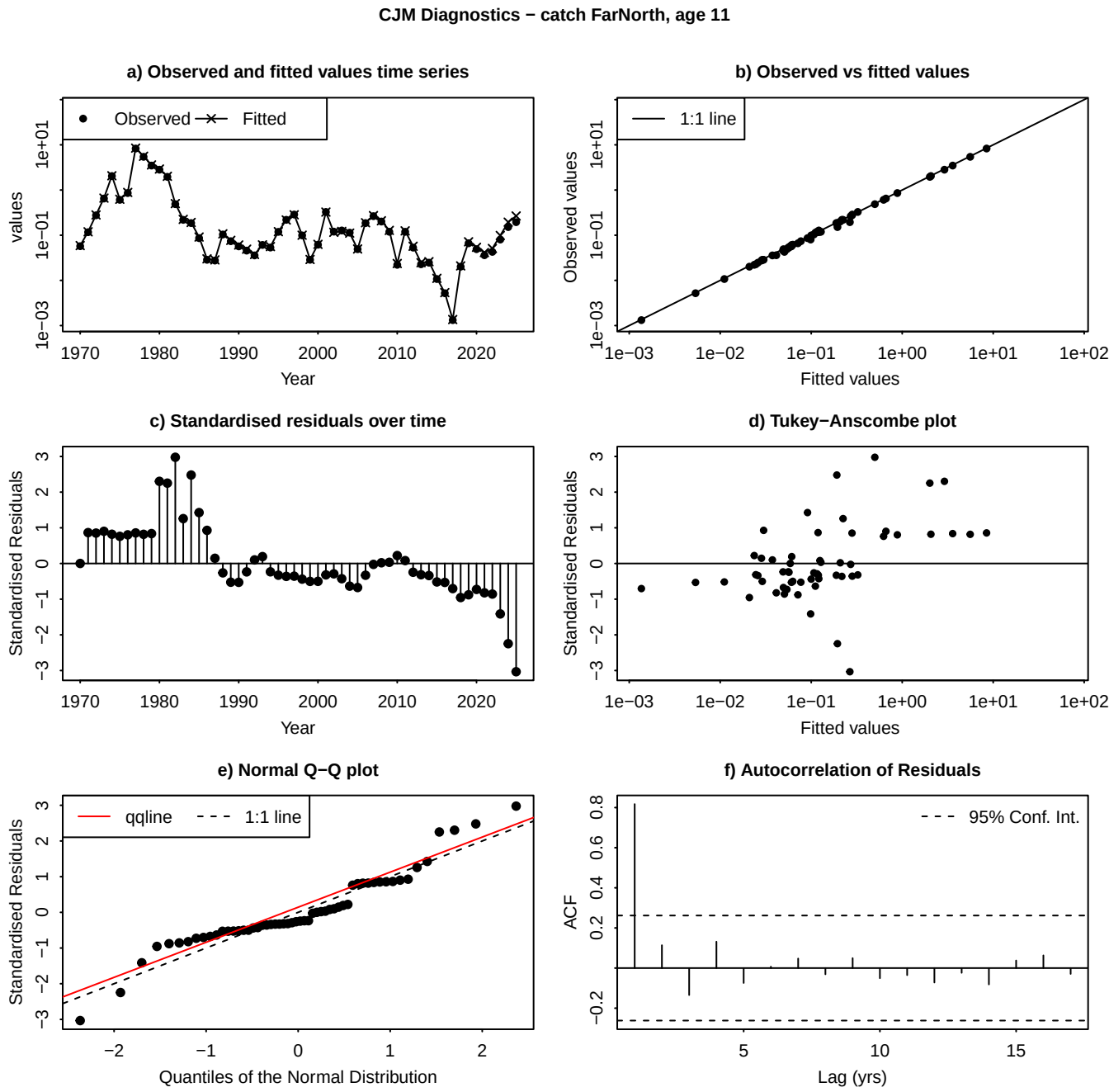

CJM Diagnostics − catch FarNorth, age 6
a) Observed and fitted values time series
Observed
Fitted
values
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

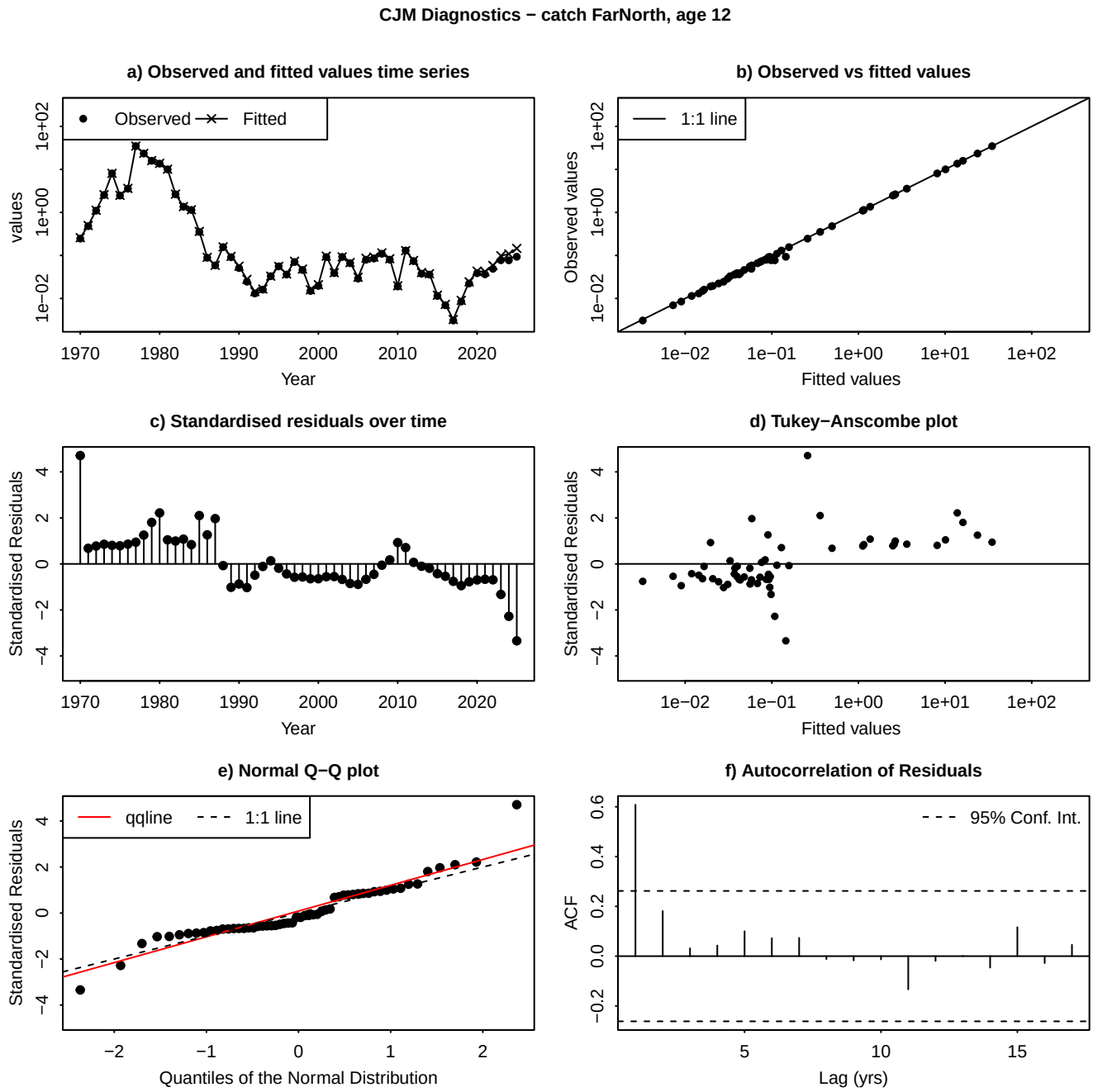

CJM Diagnostics − catch FarNorth, age 7
a) Observed and fitted values time series
Observed
Fitted
values
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

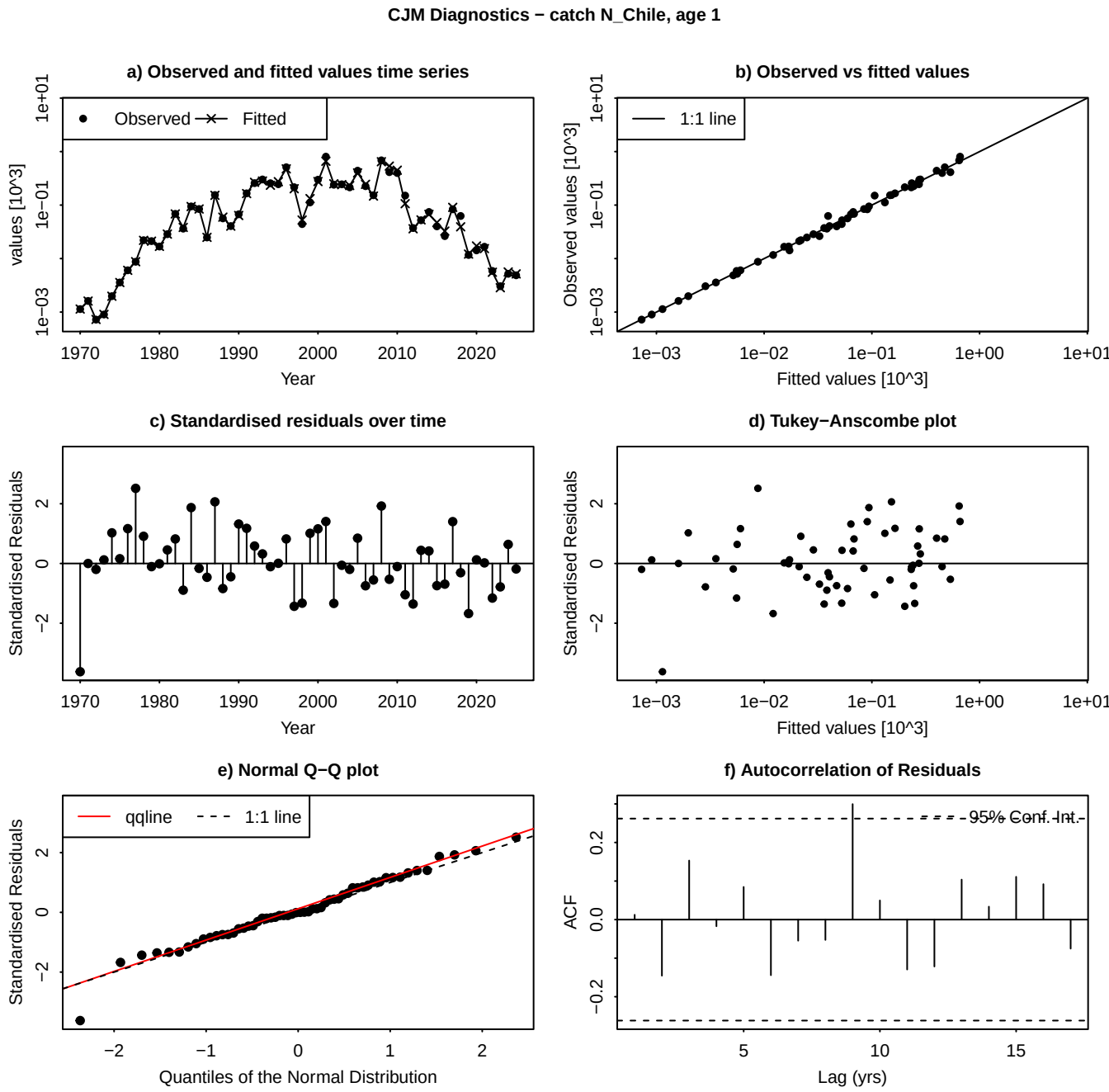

CJM Diagnostics − catch FarNorth, age 8
a) Observed and fitted values time series
Observed
Fitted
values
5e−02
5e−01
5e+00
5e+01
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
c) Standardised residuals over time
Standardised Residuals
−4
−2
0
2
4
Year
d) Tukey−Anscombe plot
e) Normal Q−Q plot
qqline
1:1 line
Quantiles of the Normal Distribution
−2
−1
0
1
2
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.0
0.2
5
10
15
Lag (yrs)

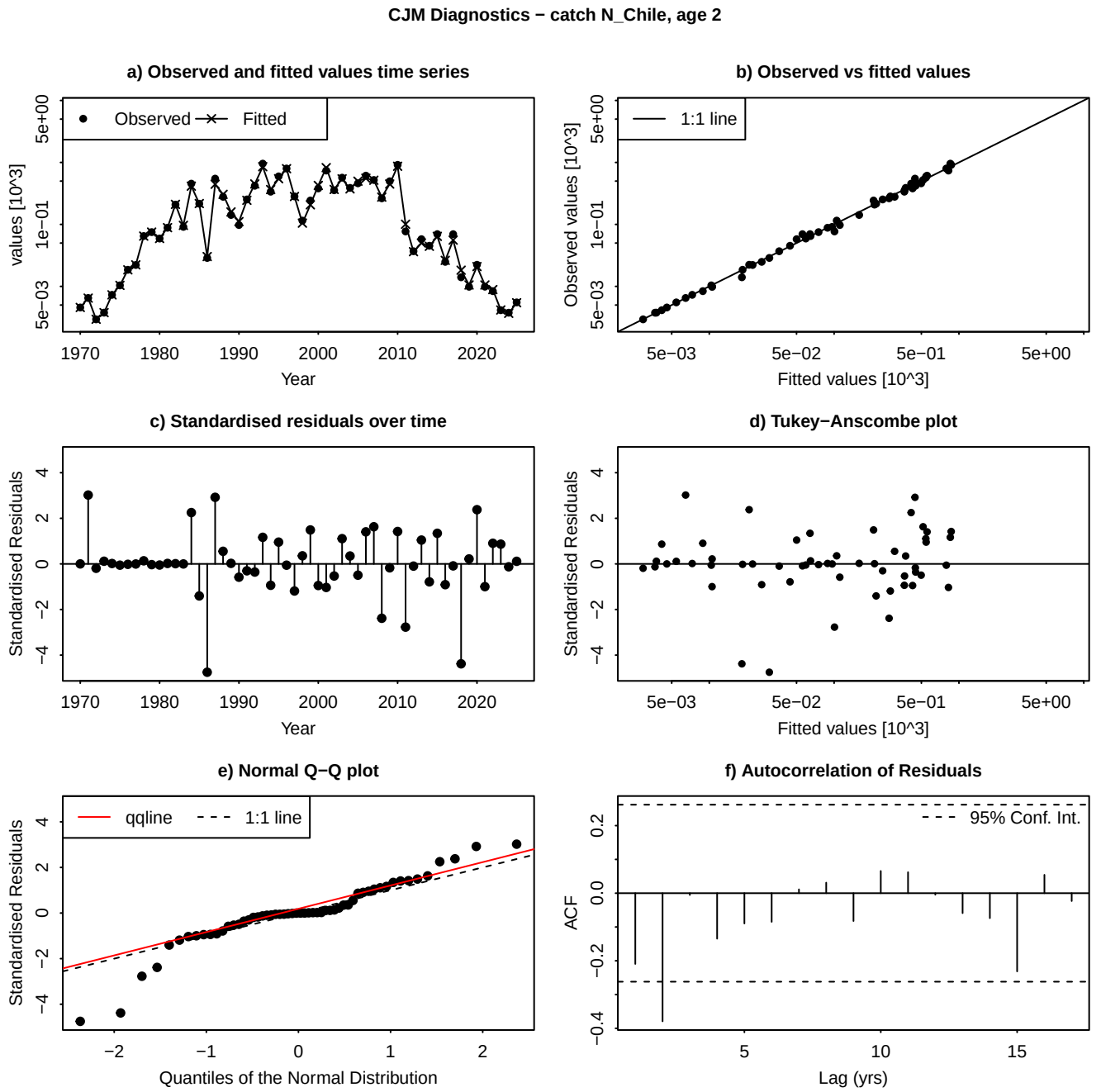

CJM Diagnostics − catch FarNorth, age 9
a) Observed and fitted values time series
Observed
Fitted
values
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

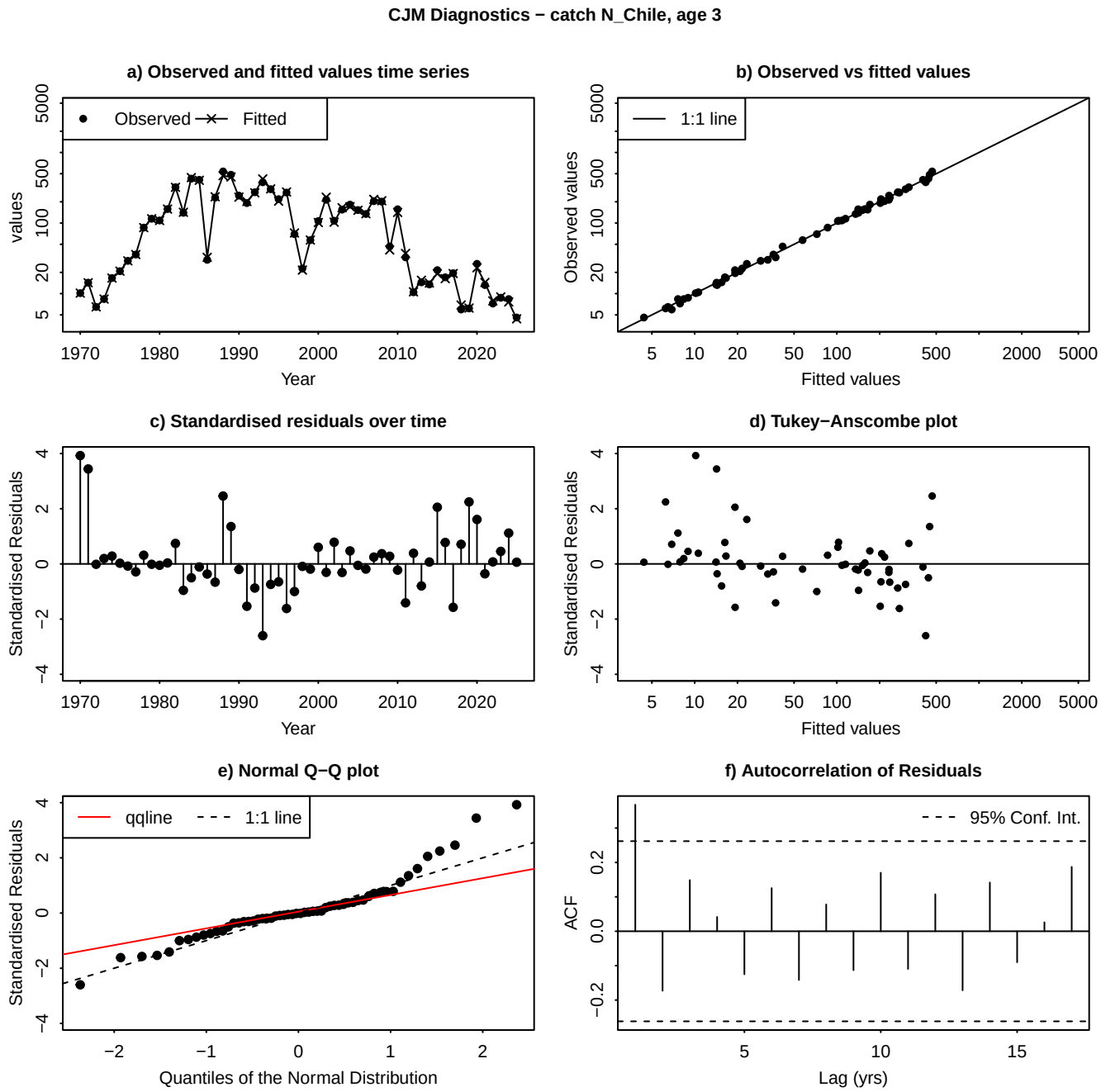

CJM Diagnostics − catch FarNorth, age 10
a) Observed and fitted values time series
Observed
Fitted
values
1e+02
1e+00
1e−02
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
1e+02
1e+00
1e−02
1e−02
1e−01
1e+00
1e+01
1e+02
Fitted values
c) Standardised residuals over time
Standardised Residuals
2
1
0
−1
−2
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
2
1
0
−1
−2
1e−02
1e−01
1e+00
1e+01
1e+02
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
2
1
0
−1
−2
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
0.8
0.6
0.4
0.2
−0.2
5
10
15
Lag (yrs)

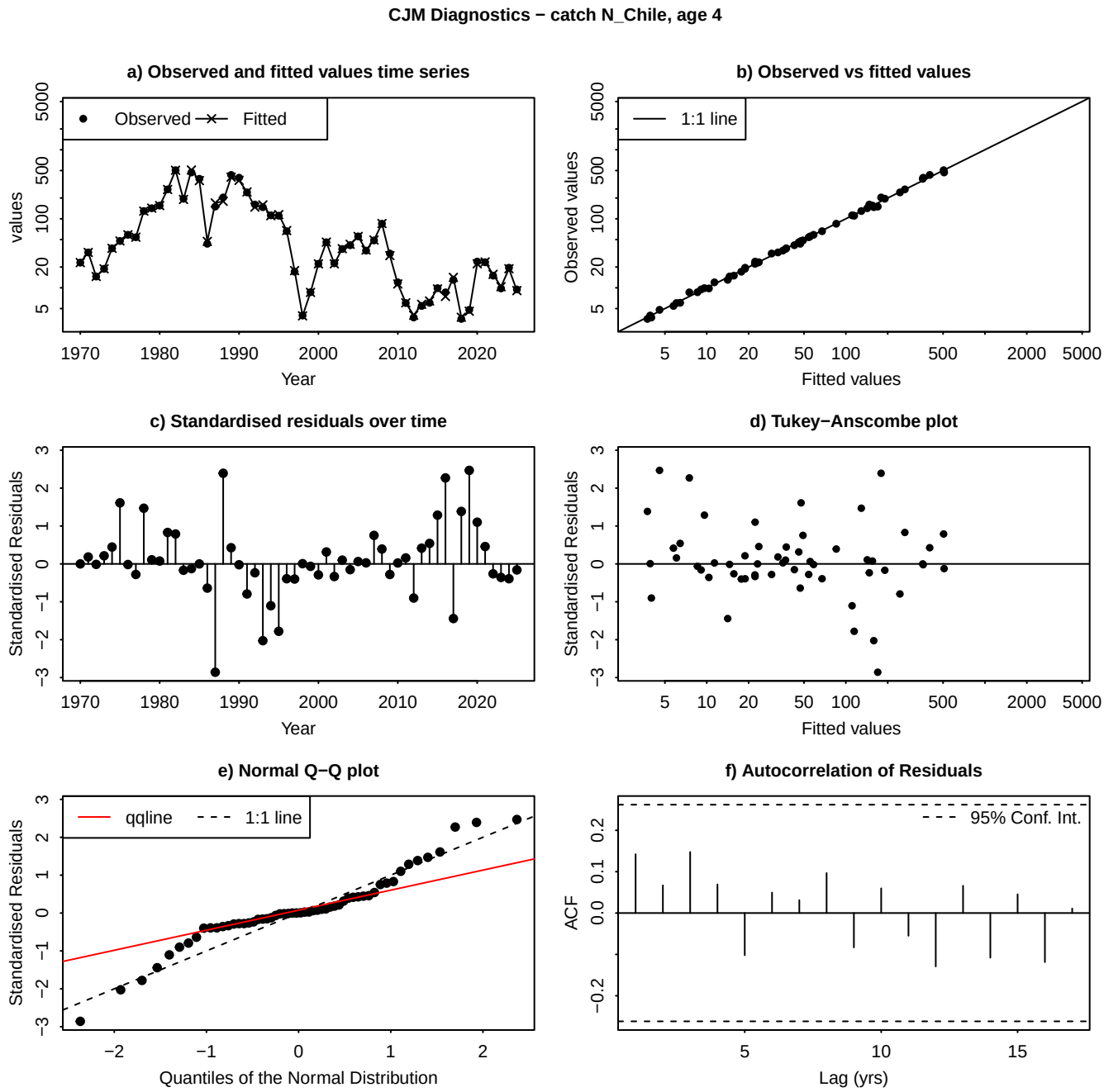

CJM Diagnostics − catch FarNorth, age 11
a) Observed and fitted values time series
Observed
Fitted
values
1e−03
1e−01
1e+01
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
1e−03
1e−02
1e−01
1e+00
1e+01
1e+02
Fitted values
c) Standardised residuals over time
Standardised Residuals
−3
−2
−1
0
1
2
3
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.2
0.4
0.6
0.8
5
10
15
Lag (yrs)

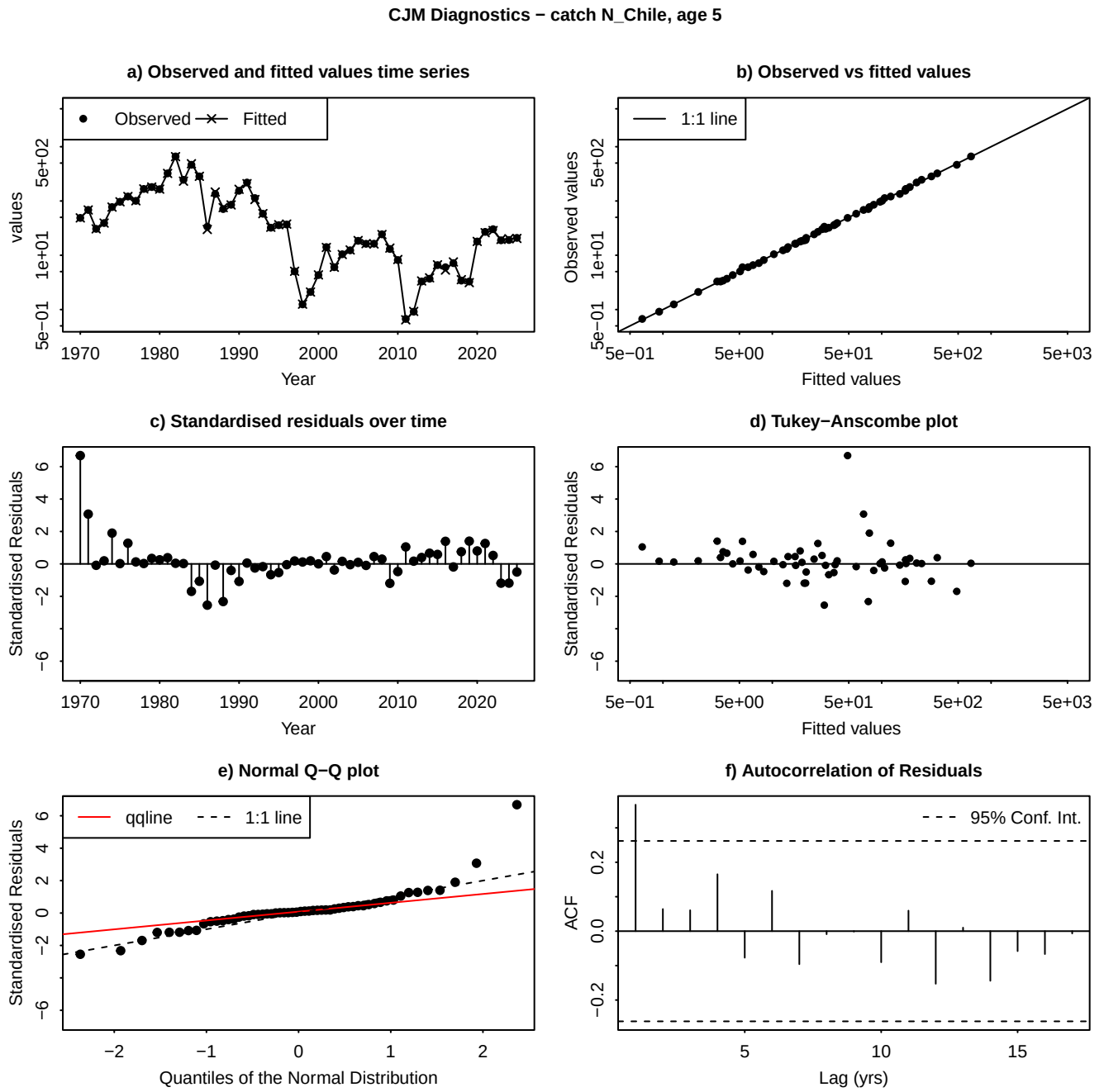

CJM Diagnostics − catch FarNorth, age 12
a) Observed and fitted values time series
Observed
Fitted
values
1e−02
1e+00
1e+02
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
1e−02
1e−01
1e+00
1e+01
1e+02
Fitted values
c) Standardised residuals over time
Standardised Residuals
−4
−2
0
2
4
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.0
0.2
0.4
0.6
5
10
15
Lag (yrs)

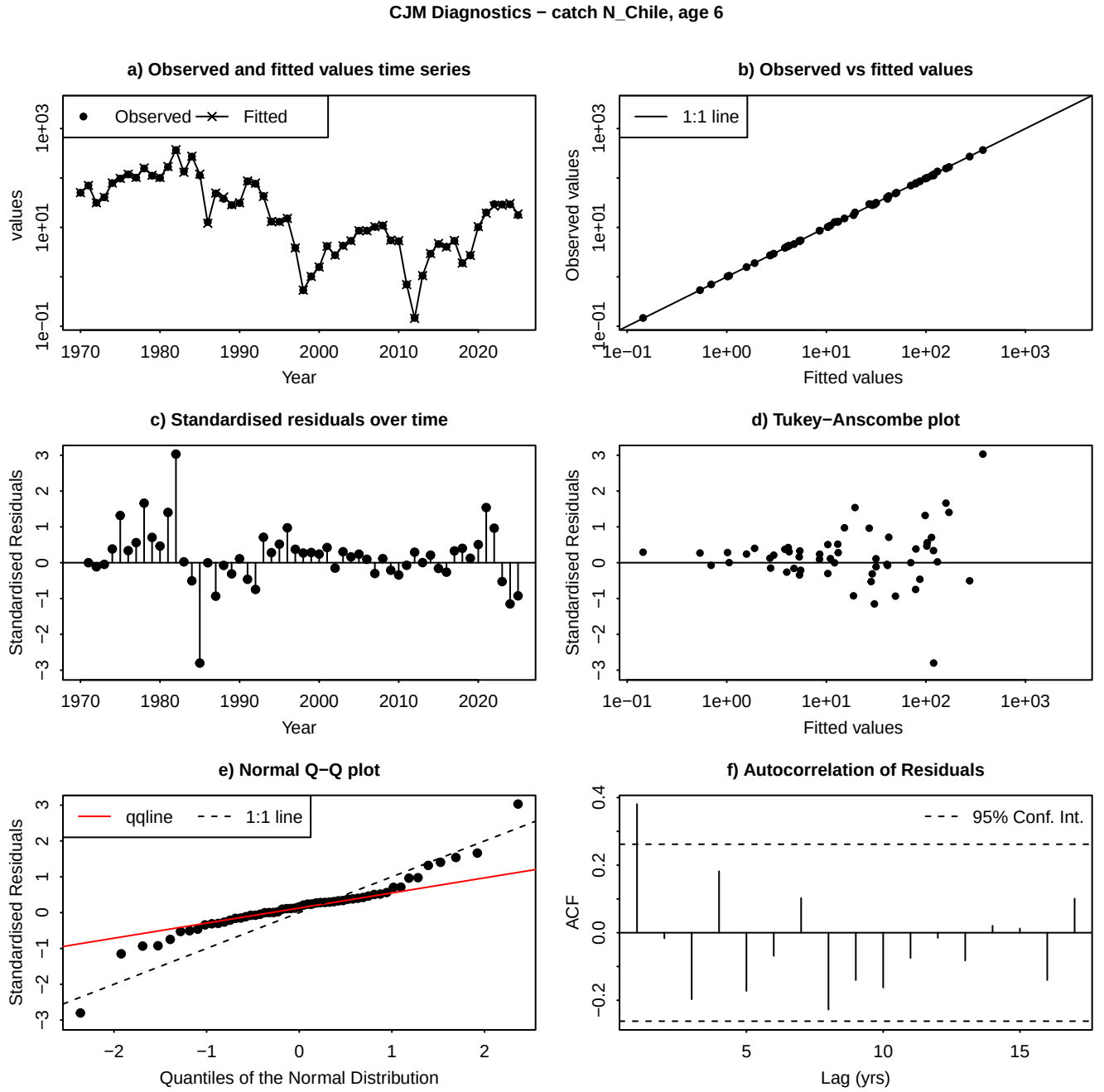

CJM Diagnostics − catch N_Chile, age 1
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
1e+01
1e−01
1e−03
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
1e+01
1e−01
1e−03
1e−03
1e−02
1e−01
1e+00
1e+01
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
2
0
−2
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
2
0
−2
1e−03
1e−02
1e−01
1e+00
1e+01
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
2
0
−2
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
0.2
0.0
−0.2
5
10
15
Lag (yrs)

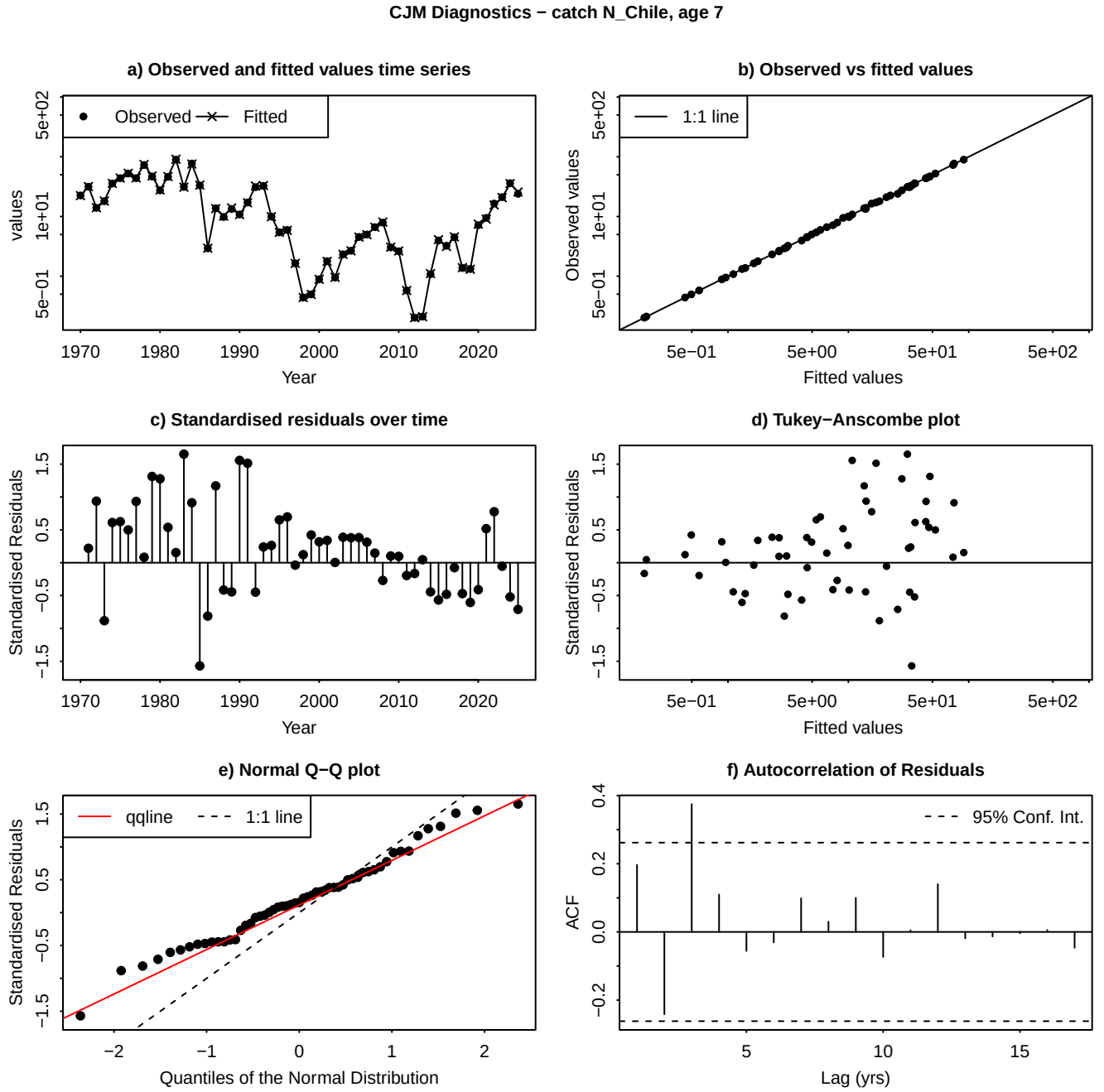

CJM Diagnostics − catch N_Chile, age 2
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
5e+00
1e−01
5e−03
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
5e+00
1e−01
5e−03
5e−03
5e−02
5e−01
5e+00
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
4
2
0
−2
−4
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
4
2
0
−2
−4
5e−03
5e−02
5e−01
5e+00
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
4
2
0
−2
−4
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
0.2
0.0
−0.2
−0.4
5
10
15
Lag (yrs)

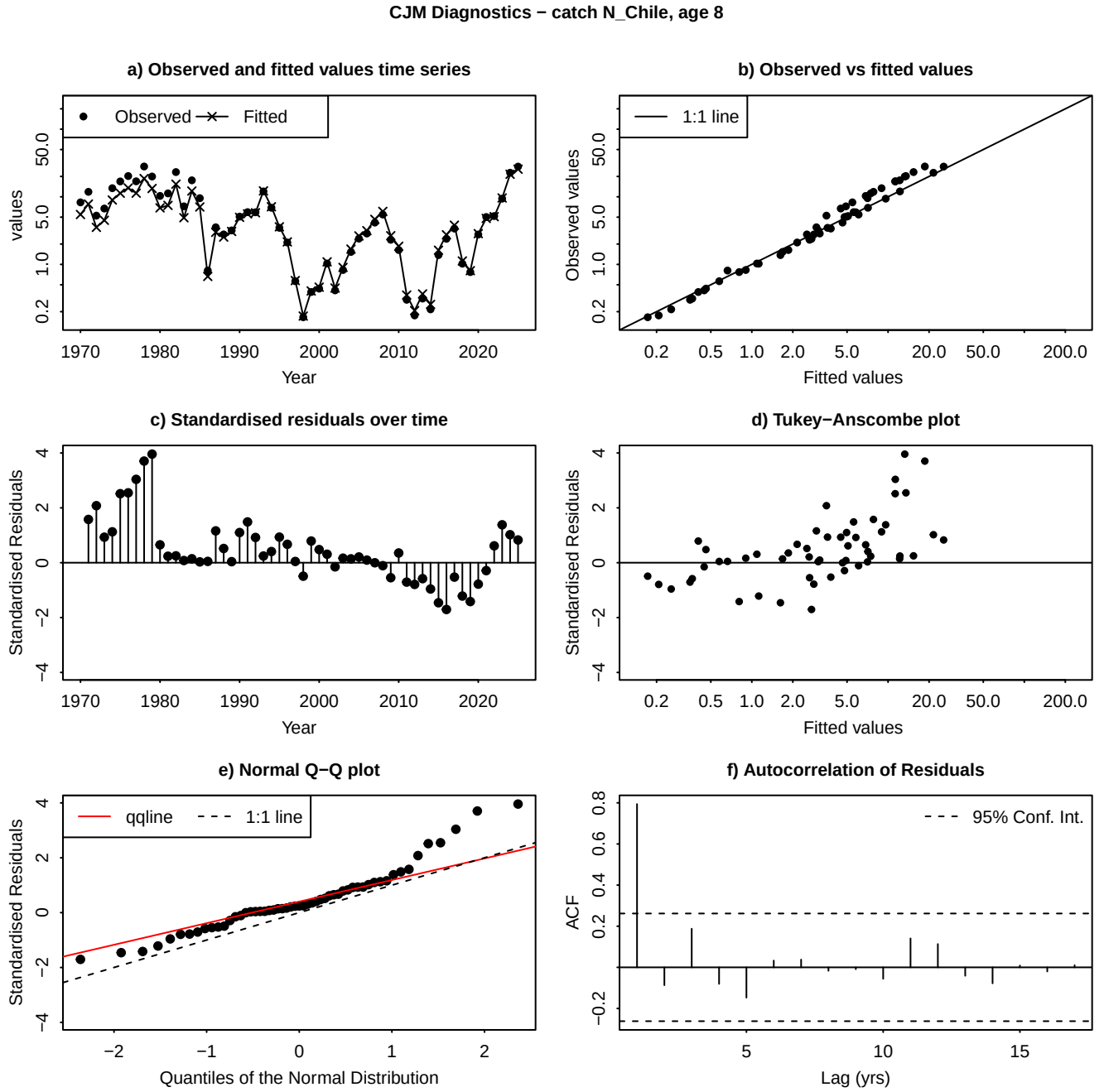

CJM Diagnostics − catch N_Chile, age 3
a) Observed and fitted values time series
Observed
Fitted
values
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

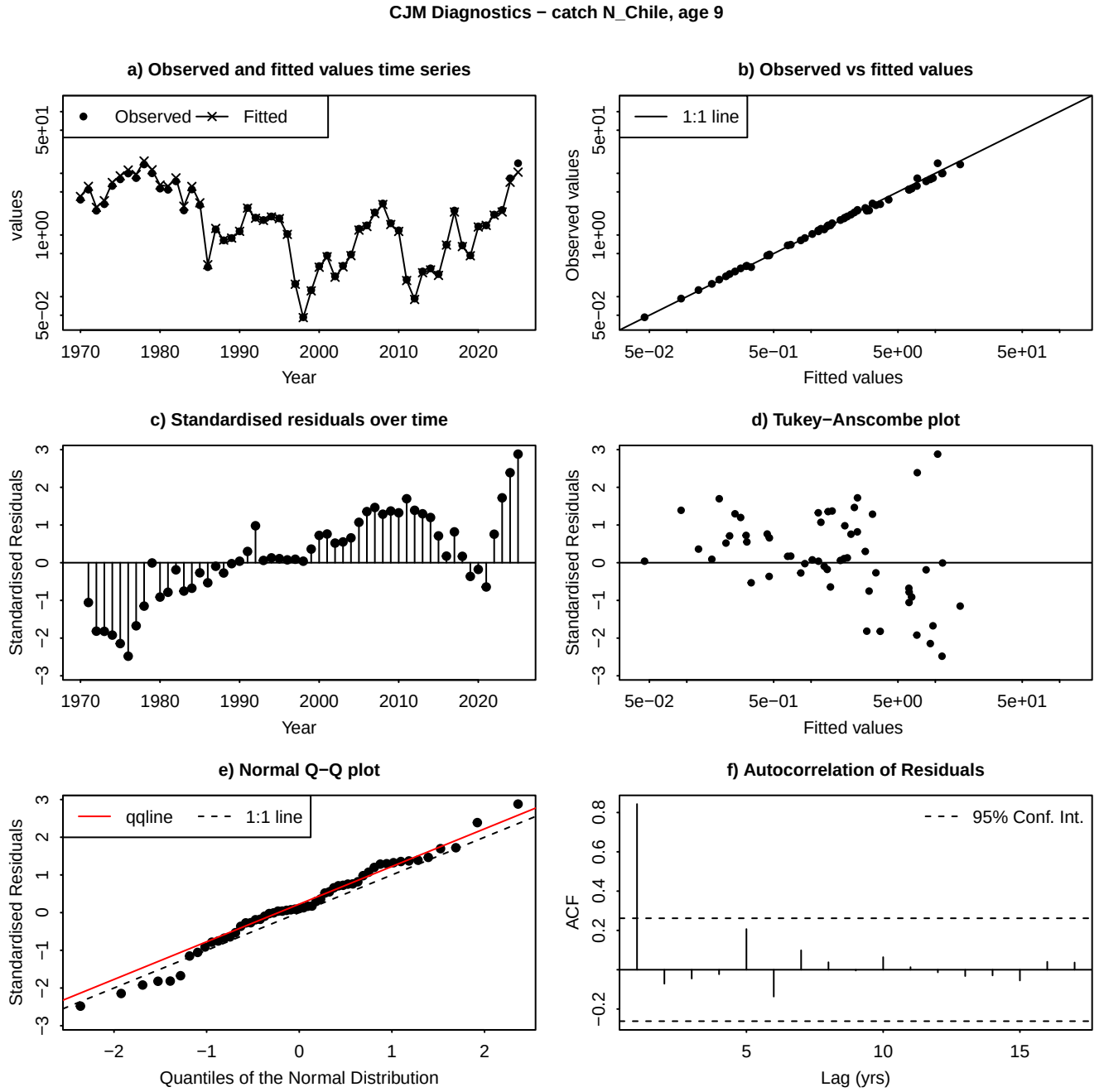

CJM Diagnostics − catch N_Chile, age 4
a) Observed and fitted values time series
Observed
Fitted
values
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

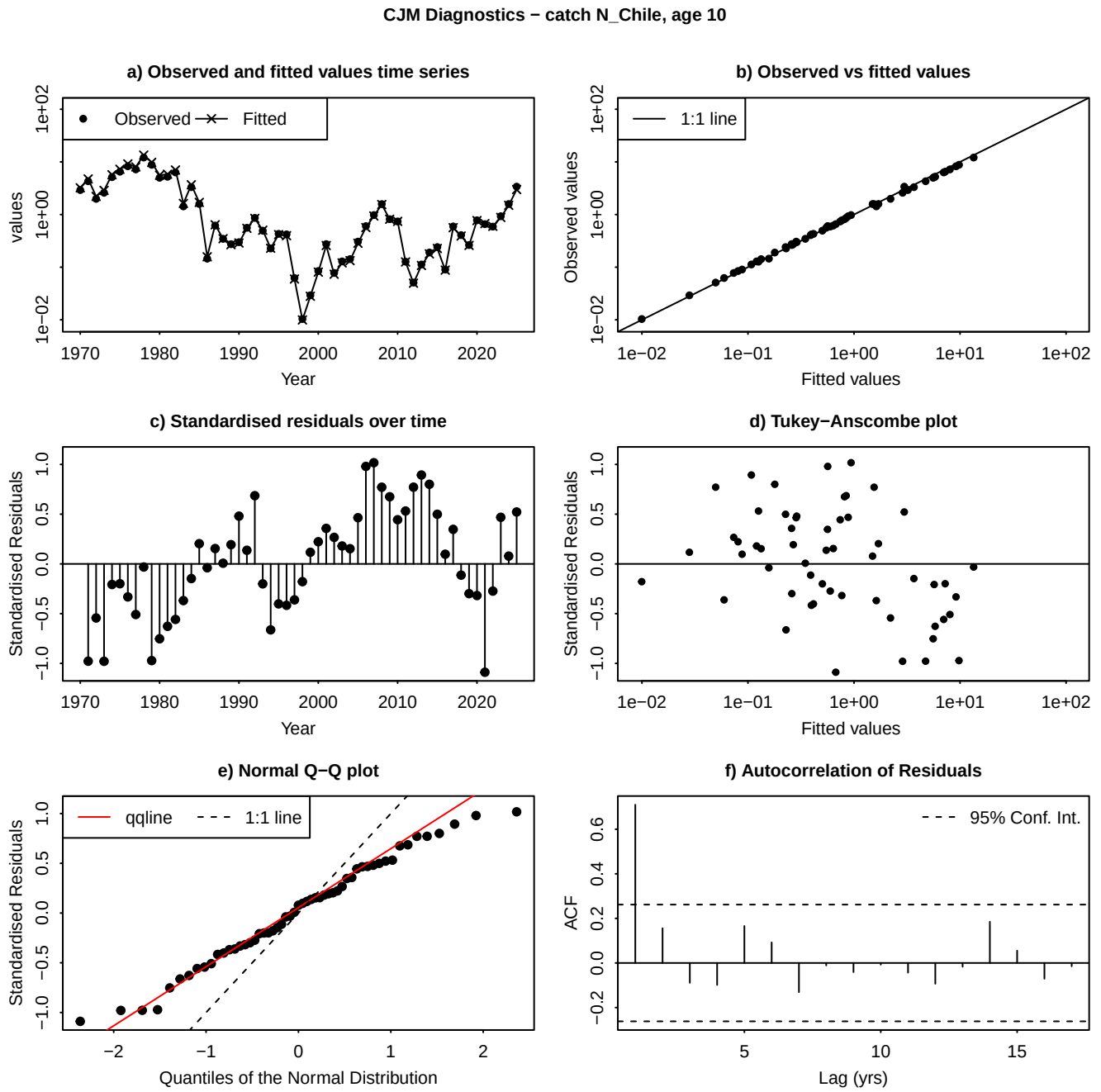

CJM Diagnostics − catch N_Chile, age 5
a) Observed and fitted values time series
Observed
Fitted
values
5e−01
1e+01
5e+02
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
5e−01
1e+01
5e+02
5e−01
5e+00
5e+01
5e+02
5e+03
Fitted values
c) Standardised residuals over time
Standardised Residuals
−6
−2
0
2
4
6
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
−6
−2
0
2
4
6
5e−01
5e+00
5e+01
5e+02
5e+03
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
−6
−2
0
2
4
6
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.0
0.2
5
10
15
Lag (yrs)

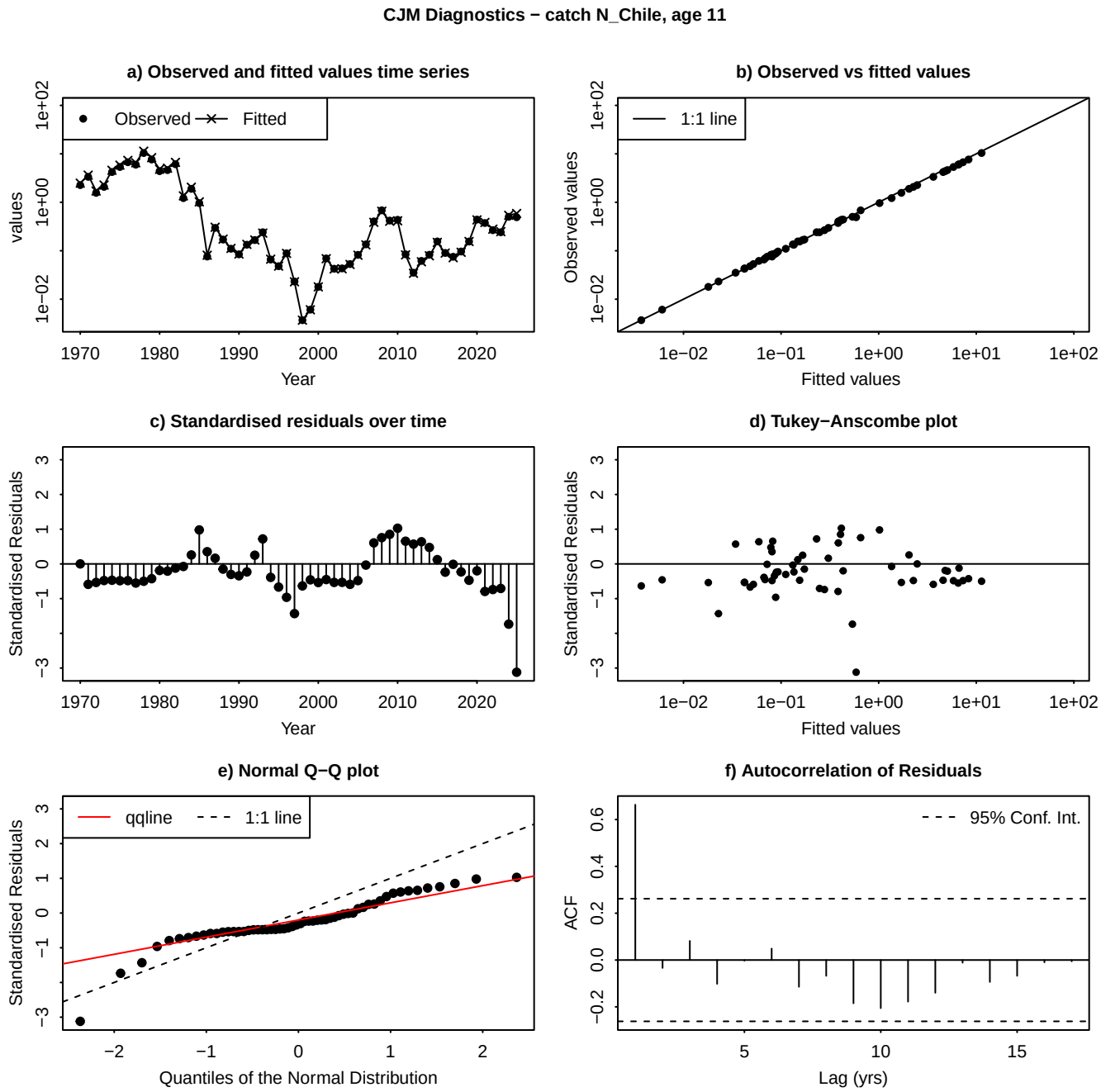

CJM Diagnostics − catch N_Chile, age 6
a) Observed and fitted values time series
Observed
Fitted
values
1e−01
1e+01
1e+03
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
1e−01
1e+01
1e+03
1e+00
1e+02
Fitted values
c) Standardised residuals over time
Standardised Residuals
−3
−2
−1
0
1
2
3
Year
d) Tukey−Anscombe plot
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Quantiles of the Normal Distribution
−2
−1
0
1
2
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.0
0.2
0.4
5
10
15
Lag (yrs)

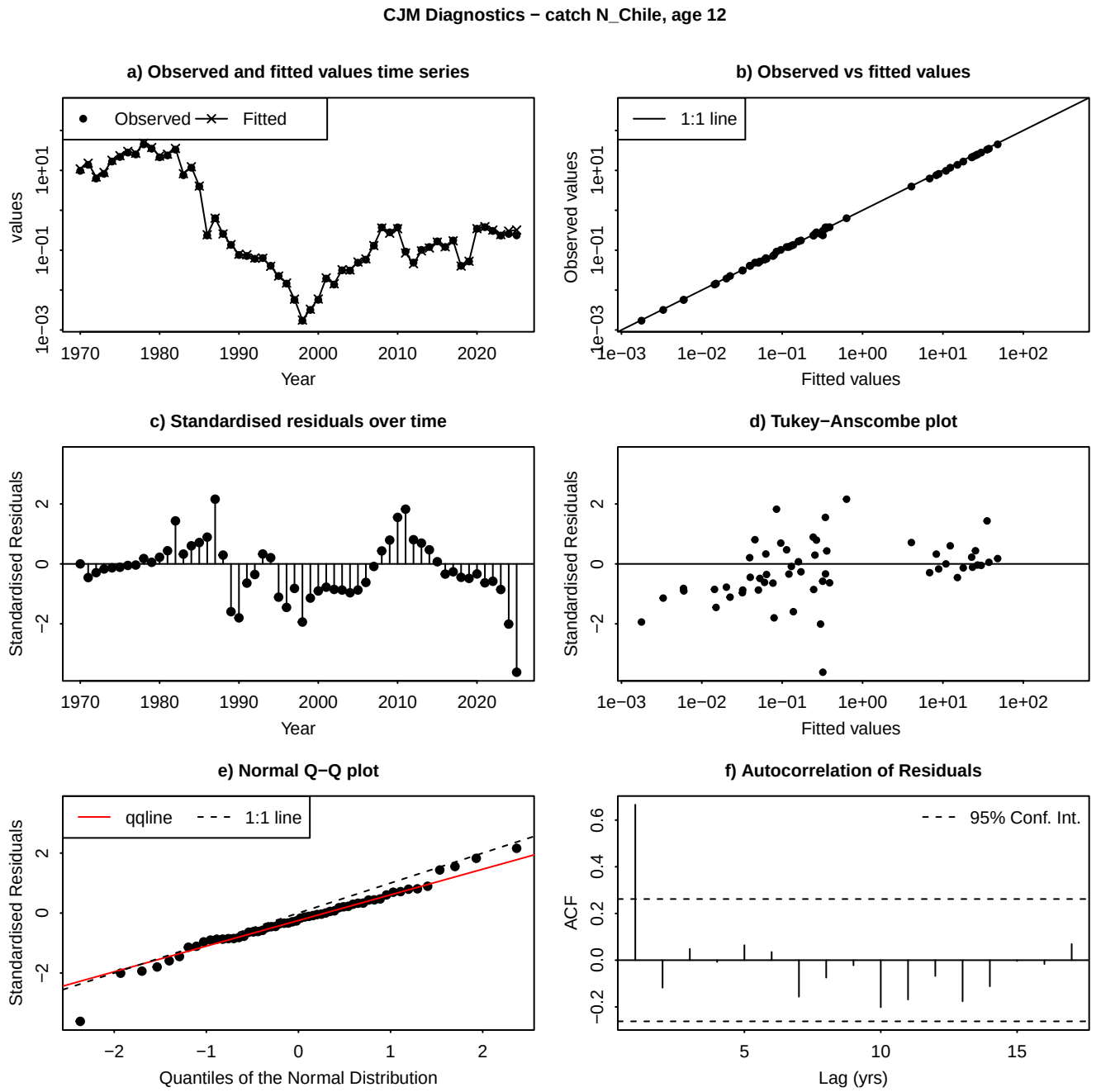

CJM Diagnostics − catch N_Chile, age 7
a) Observed and fitted values time series
Observed
Fitted
values
5e−01
1e+01
5e+02
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
5e−01
1e+01
5e+02
5e−01
5e+00
5e+01
5e+02
Fitted values
c) Standardised residuals over time
Standardised Residuals
−1.5
−0.5
0.5
1.5
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
−1.5
−0.5
0.5
1.5
5e−01
5e+00
5e+01
5e+02
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
−1.5
−0.5
0.5
1.5
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.0
0.2
0.4
5
10
15
Lag (yrs)

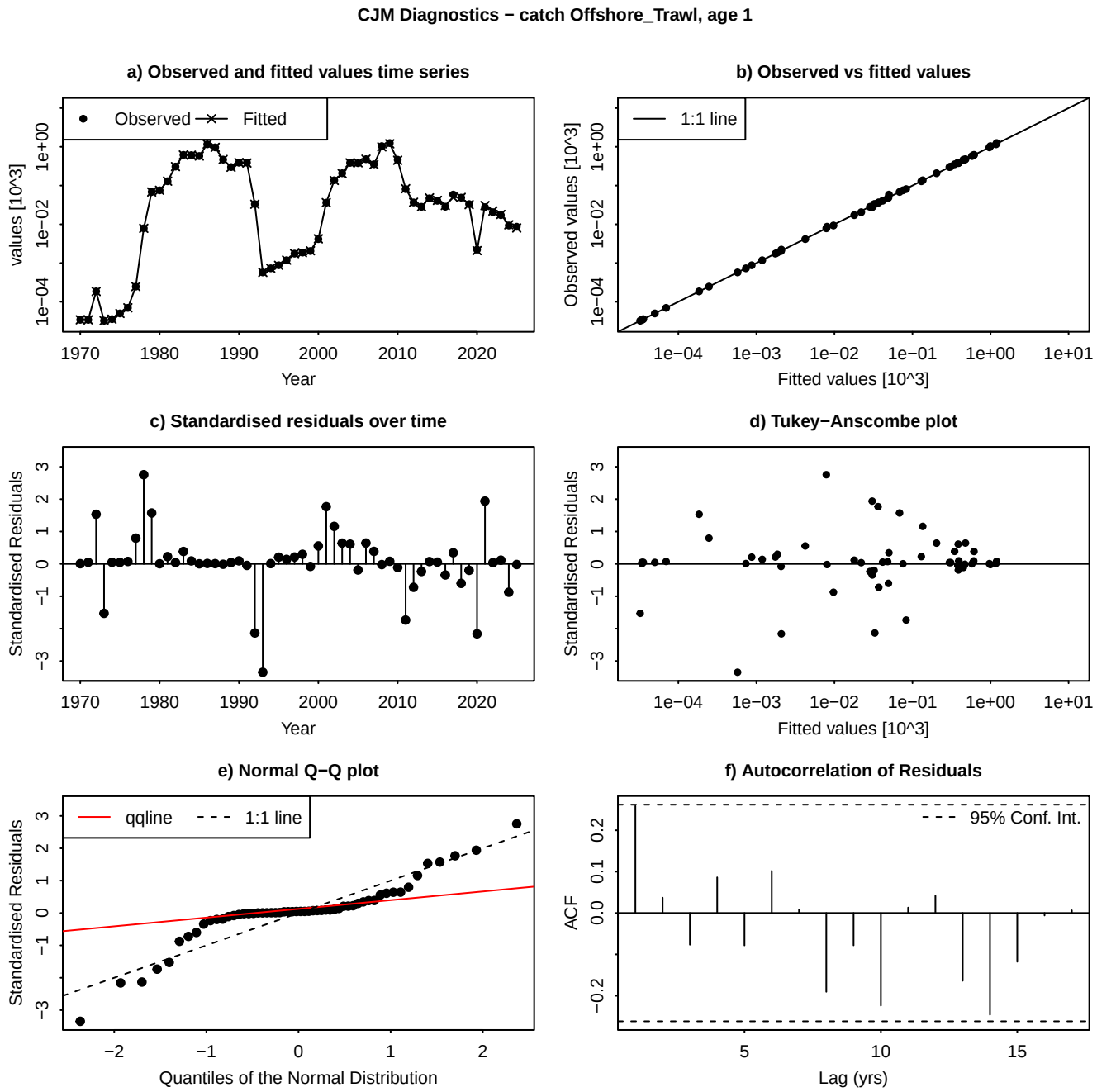

CJM Diagnostics − catch N_Chile, age 8
a) Observed and fitted values time series
Observed
Fitted
values
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

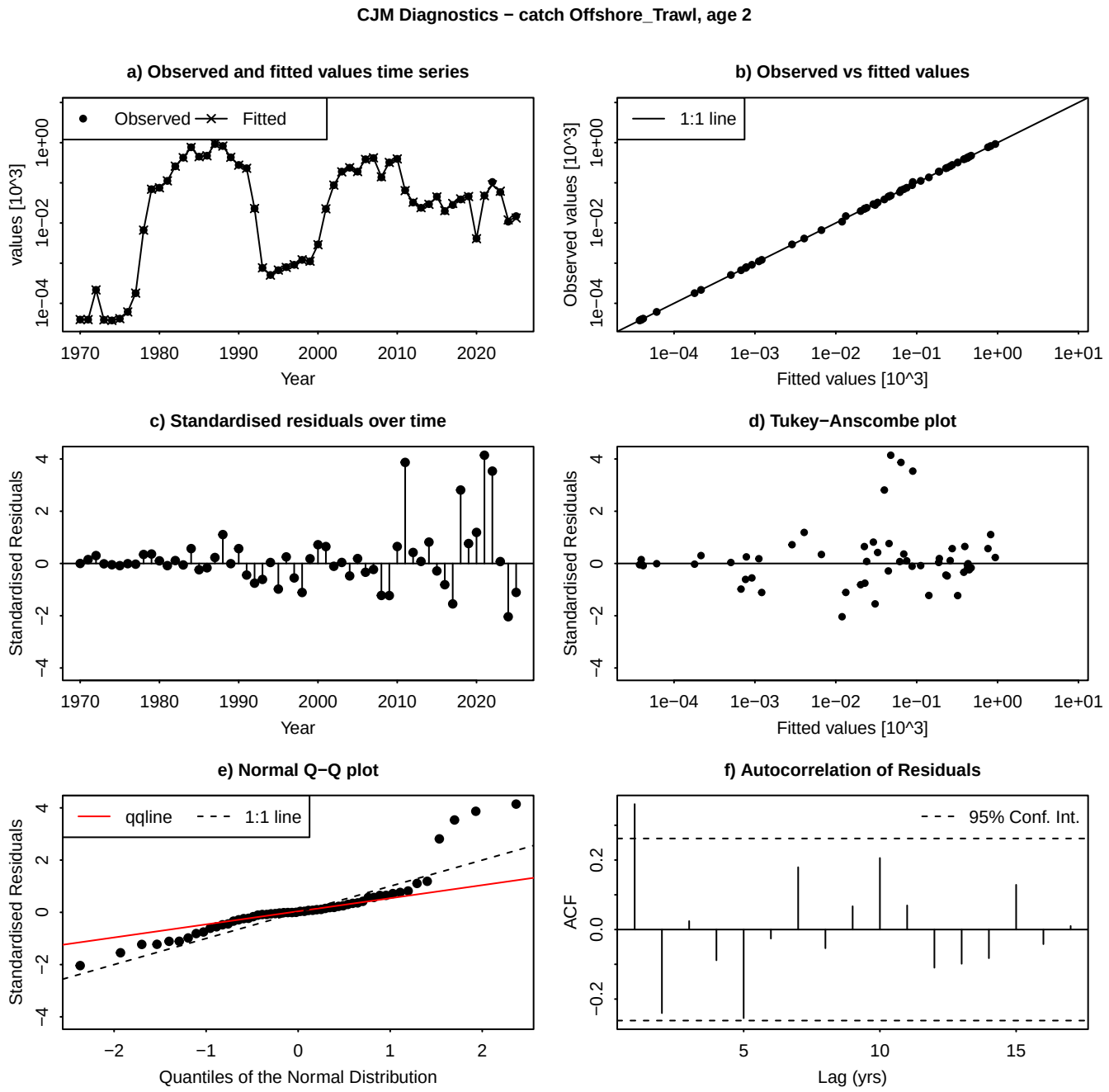

CJM Diagnostics − catch N_Chile, age 9
a) Observed and fitted values time series
Observed
Fitted
values
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

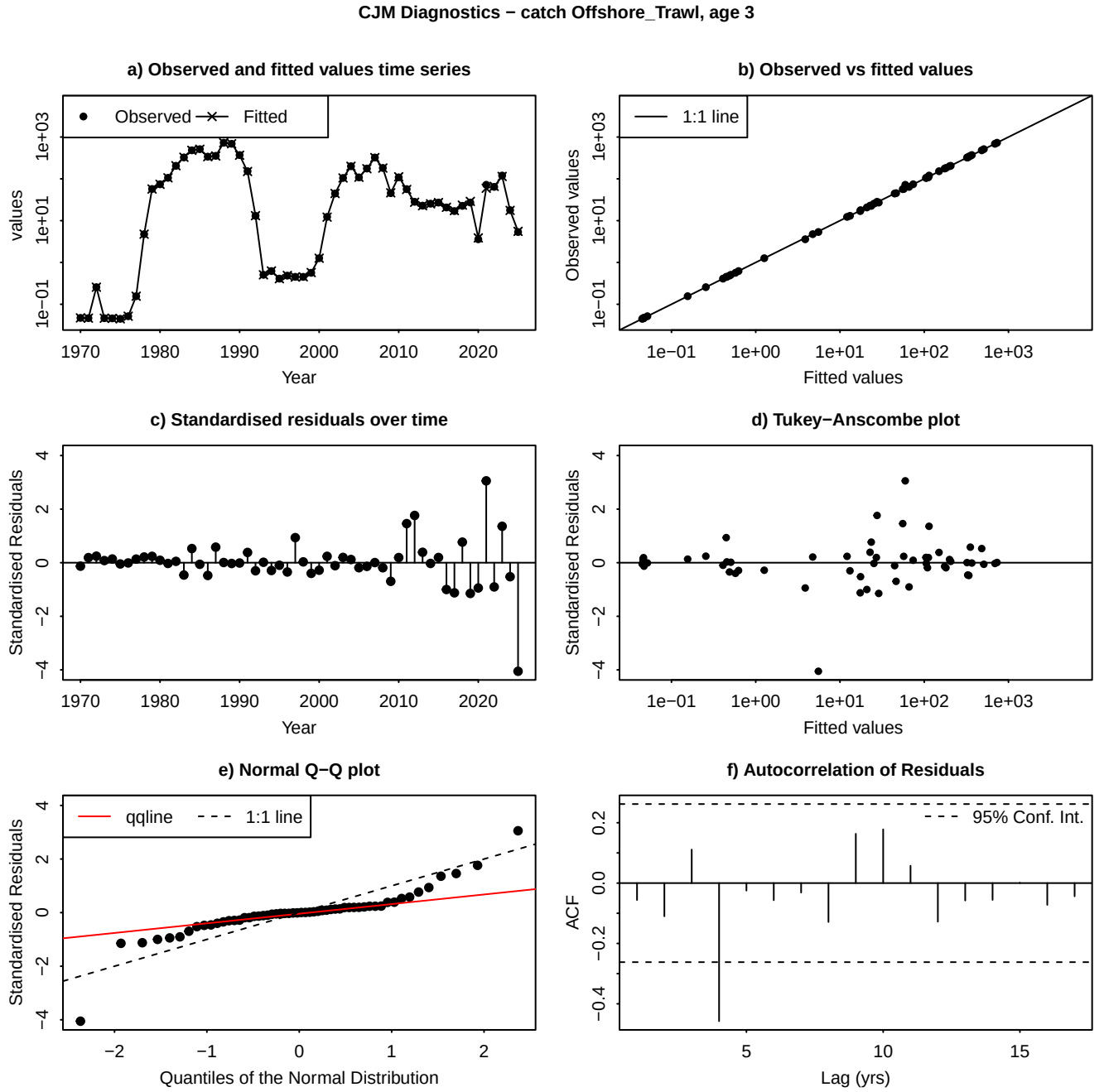

CJM Diagnostics − catch N_Chile, age 10
a) Observed and fitted values time series
Observed
Fitted
values
1e+02
1e+00
1e−02
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
1e−02
1e−01
1e+00
1e+01
1e+02
c) Standardised residuals over time
Standardised Residuals
1.0
0.5
0.0
−0.5
−1.0
Year
d) Tukey−Anscombe plot
e) Normal Q−Q plot
qqline
1:1 line
Quantiles of the Normal Distribution
−2
−1
0
1
2
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
0.6
0.4
0.2
0.0
−0.2
5
10
15
Lag (yrs)

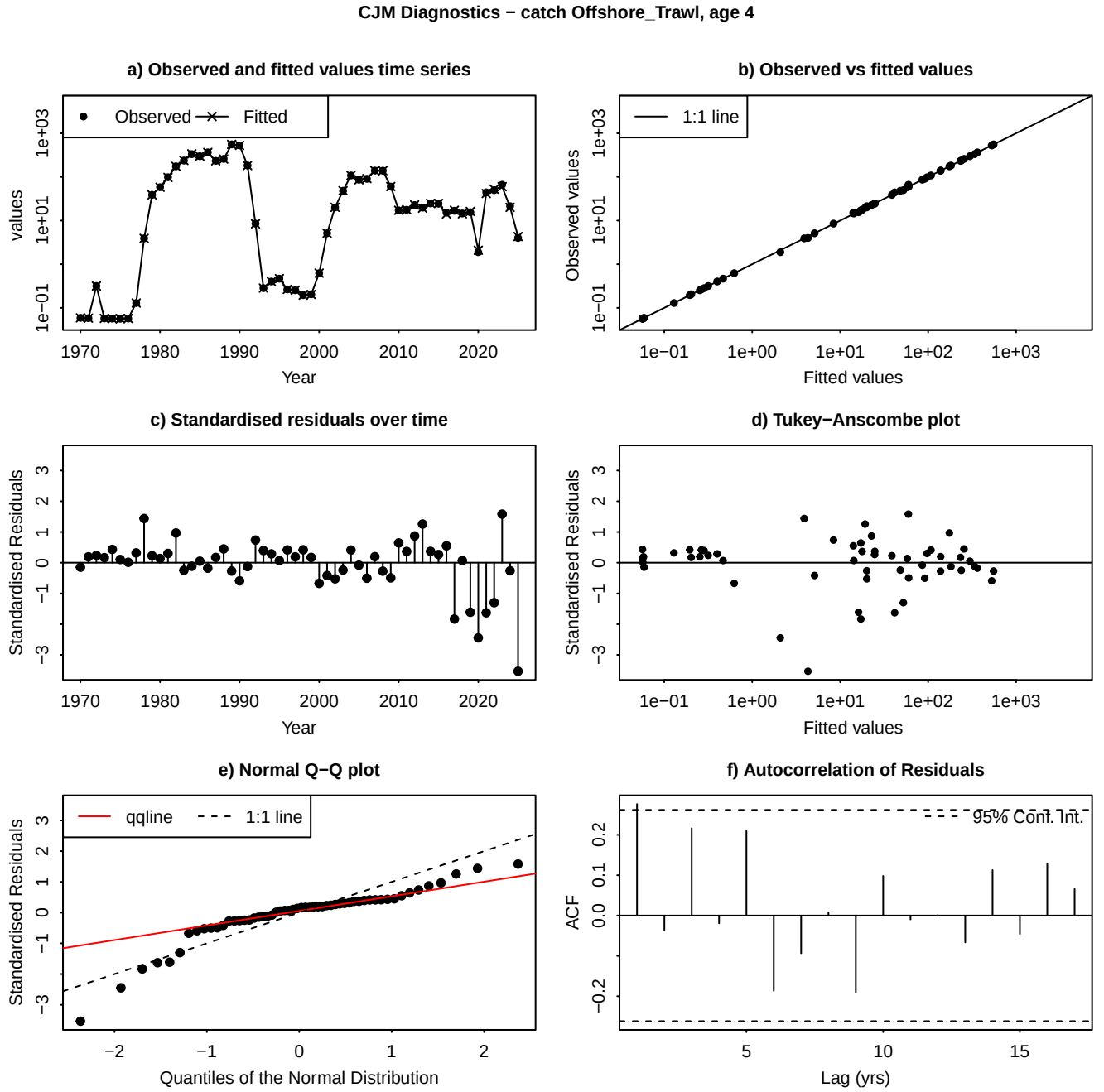

CJM Diagnostics − catch N_Chile, age 11
a) Observed and fitted values time series
Observed
Fitted
values
1e+02
1e+00
1e−02
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
1e−02
1e−01
1e+00
1e+01
1e+02
Fitted values
c) Standardised residuals over time
Standardised Residuals
3
2
1
0
−1
−3
Year
d) Tukey−Anscombe plot
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Quantiles of the Normal Distribution
−2
−1
0
1
2
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
0.6
0.4
0.2
0.0
−0.2
5
10
15
Lag (yrs)

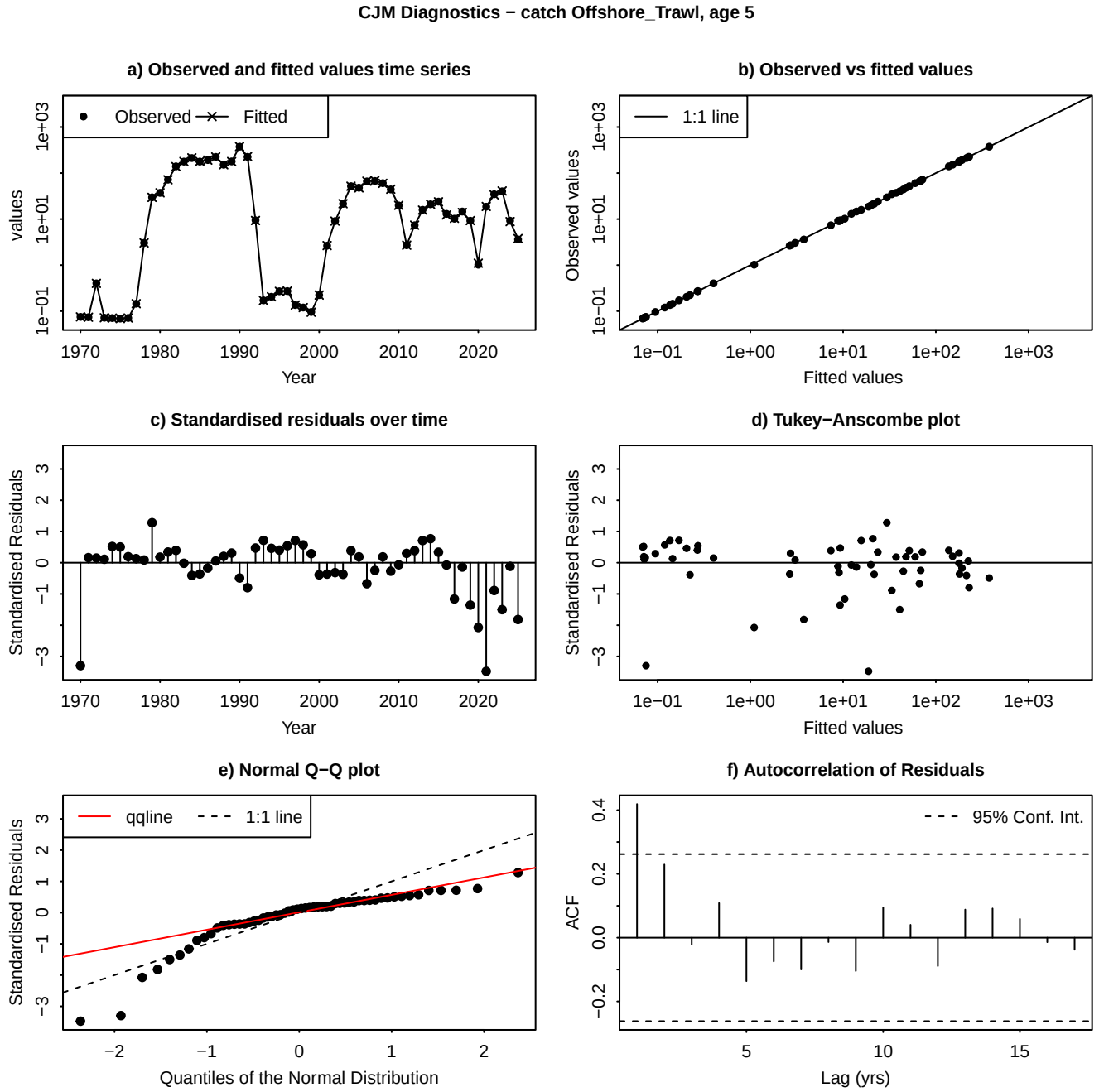

CJM Diagnostics − catch N_Chile, age 12
a) Observed and fitted values time series
Observed
Fitted
values
1e−03
1e−01
1e+01
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
1e−03
1e−01
1e+01
1e−03
1e−02
1e−01
1e+00
1e+01
1e+02
Fitted values
c) Standardised residuals over time
Standardised Residuals
−2
0
2
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
−2
0
2
1e−03
1e−02
1e−01
1e+00
1e+01
1e+02
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
−2
0
2
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.0
0.2
0.4
0.6
5
10
15
Lag (yrs)

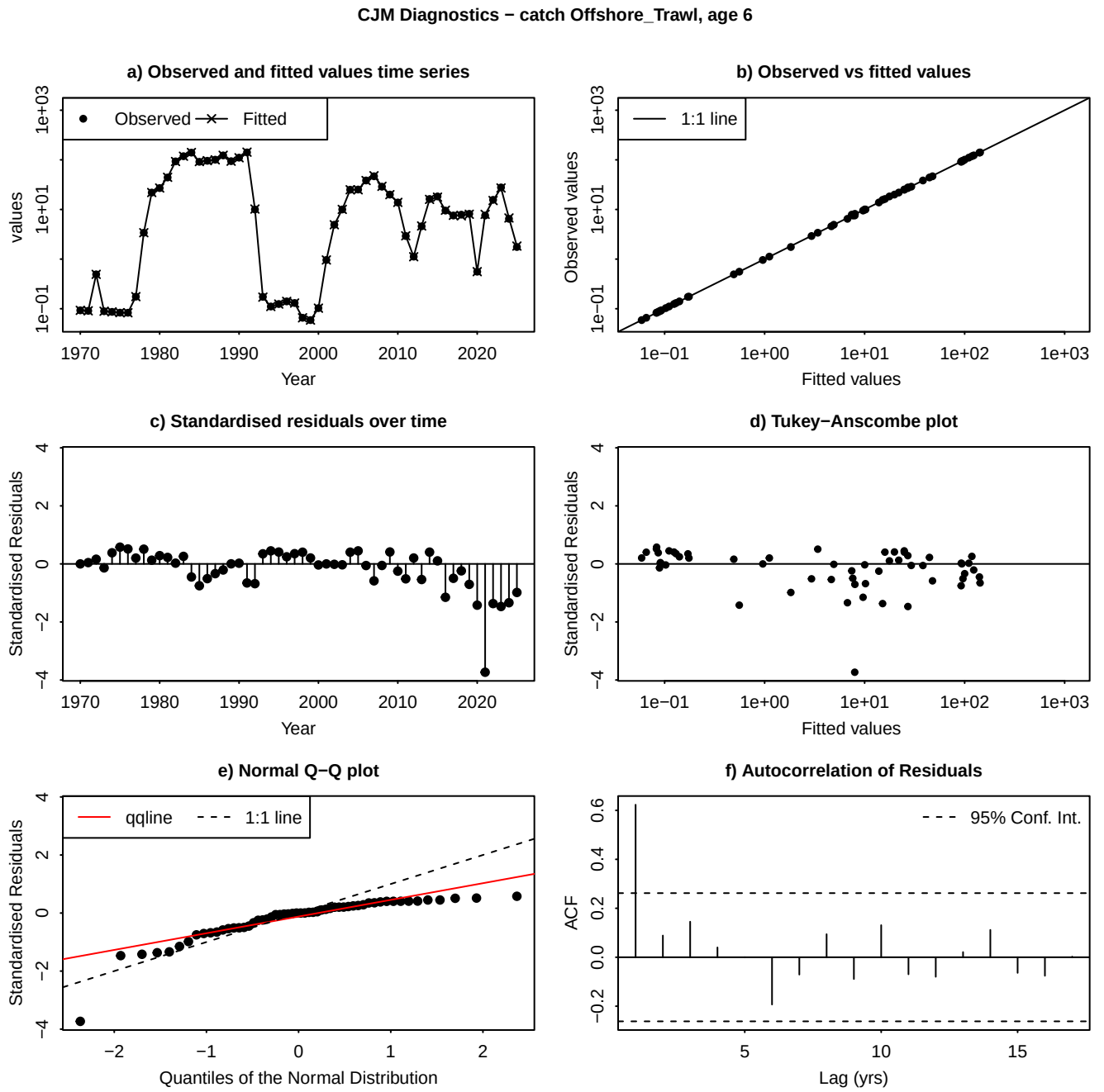

CJM Diagnostics − catch Offshore_Trawl, age 1
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
1e−04
1e−02
1e+00
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
1e−04
1e−03
1e−02
1e−01
1e+00
1e+01
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
−3
−1
0
1
2
3
Year
d) Tukey−Anscombe plot
e) Normal Q−Q plot
qqline
1:1 line
Quantiles of the Normal Distribution
−2
−1
0
1
2
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.0
0.1
0.2
5
10
15
Lag (yrs)

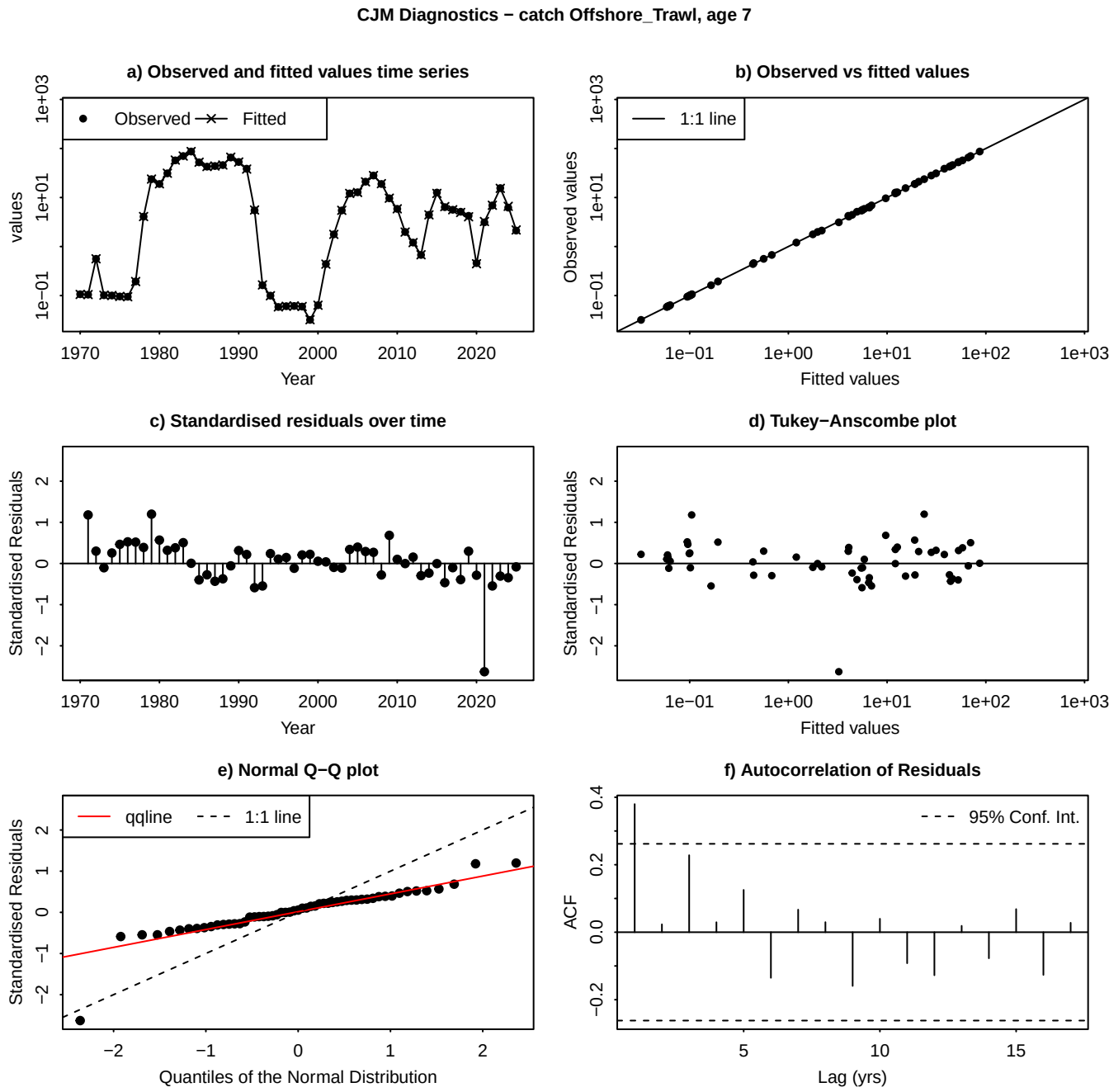

CJM Diagnostics − catch Offshore_Trawl, age 2
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
1e+00
1e−02
1e−04
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
1e+00
1e−02
1e−04
1e−04
1e−03
1e−02
1e−01
1e+00
1e+01
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
4
2
0
−2
−4
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
4
2
0
−2
−4
1e−04
1e−03
1e−02
1e−01
1e+00
1e+01
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
4
2
0
−2
−4
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
0.2
0.0
−0.2
5
10
15
Lag (yrs)

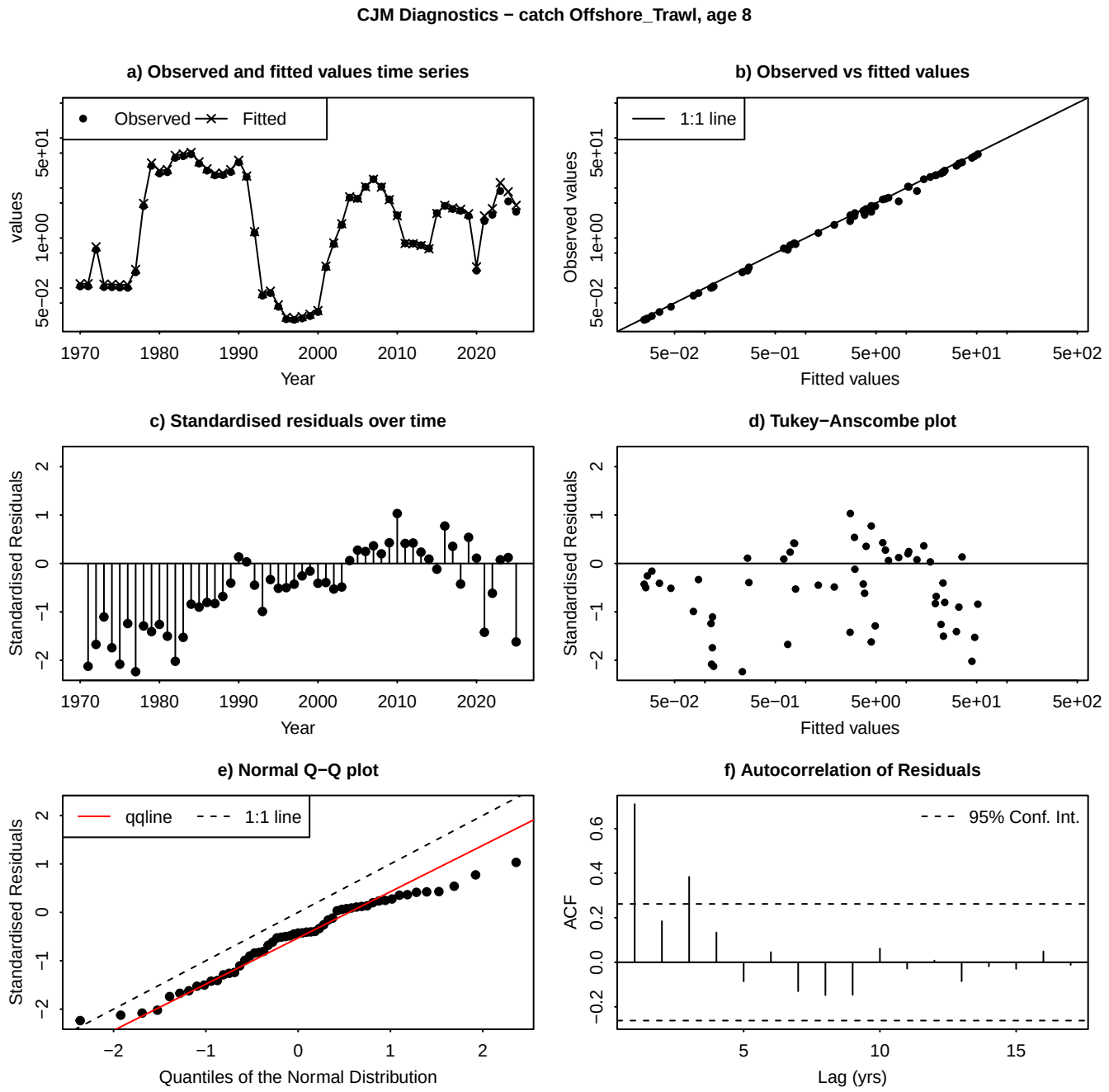

CJM Diagnostics − catch Offshore_Trawl, age 3
a) Observed and fitted values time series
Observed
Fitted
values
1e−01
1e+01
1e+03
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
1e−01
1e+01
1e+03
1e−01
1e+00
1e+01
1e+02
1e+03
Fitted values
c) Standardised residuals over time
Standardised Residuals
4
2
0
−2
−4
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
4
2
0
−2
−4
1e−01
1e+00
1e+01
1e+02
1e+03
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
4
2
0
−2
−4
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
0.2
0.0
−0.2
−0.4
5
10
15
Lag (yrs)

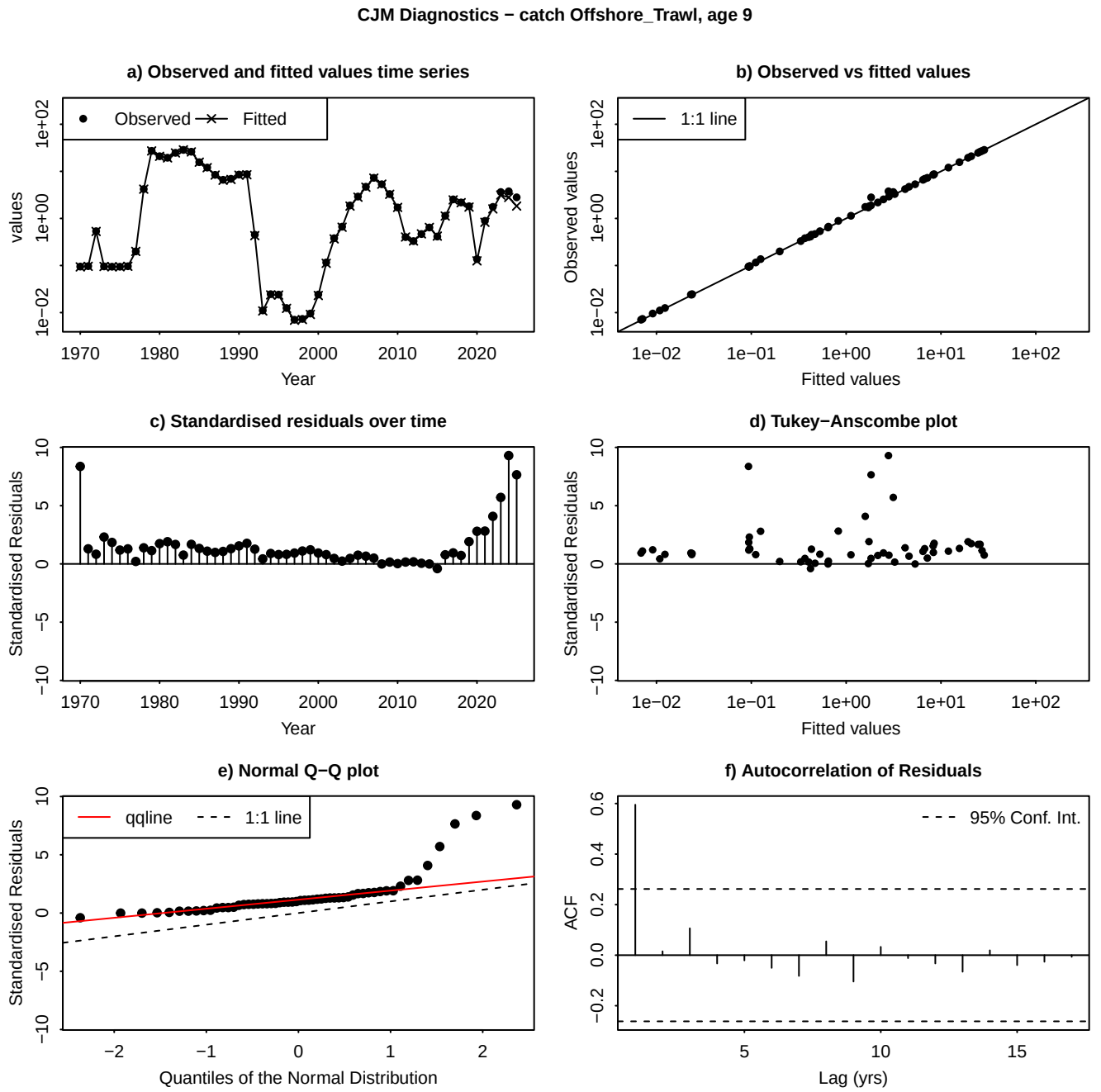

CJM Diagnostics − catch Offshore_Trawl, age 4
a) Observed and fitted values time series
Observed
Fitted
values
1e−01
1e+01
1e+03
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
1e−01
1e+01
1e+03
1e−01
1e+00
1e+01
1e+02
1e+03
Fitted values
c) Standardised residuals over time
Standardised Residuals
−3
−1
0
1
2
3
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
−3
−1
0
1
2
3
1e−01
1e+00
1e+01
1e+02
1e+03
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
−3
−1
0
1
2
3
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.0
0.1
0.2
5
10
15
Lag (yrs)

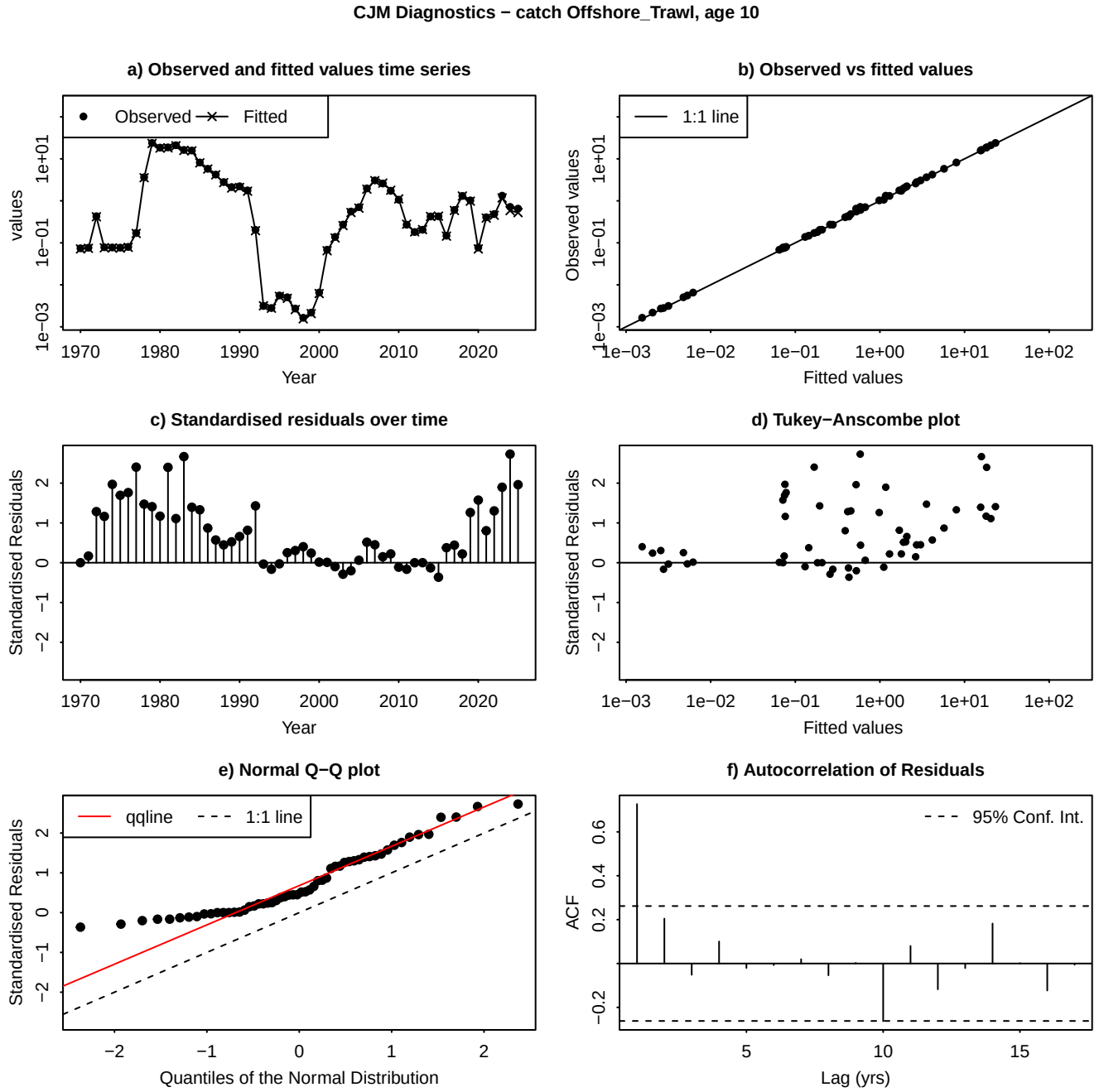

CJM Diagnostics − catch Offshore_Trawl, age 5
a) Observed and fitted values time series
Observed
Fitted
values
1e−01
1e+01
1e+03
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
1e−01
1e+01
1e+03
1e−01
1e+00
1e+01
1e+02
1e+03
Fitted values
c) Standardised residuals over time
Standardised Residuals
−3
−1
0
1
2
3
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
−3
−1
0
1
2
3
1e−01
1e+00
1e+01
1e+02
1e+03
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
−3
−1
0
1
2
3
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.0
0.2
0.4
5
10
15
Lag (yrs)

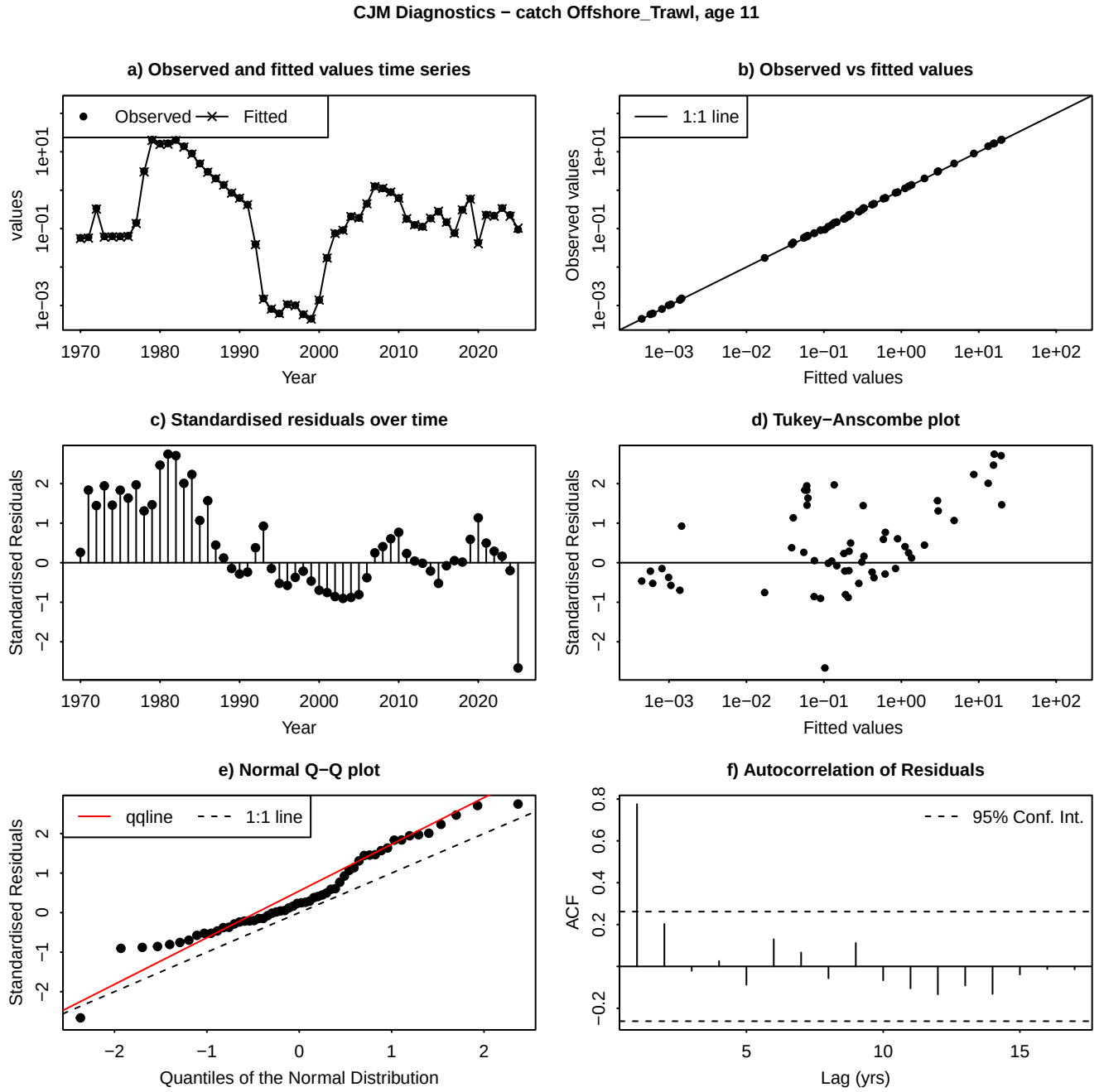

CJM Diagnostics − catch Offshore_Trawl, age 6
a) Observed and fitted values time series
Observed
Fitted
values
1e+03
1e+01
1e−01
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
1e+03
1e+01
1e−01
1e−01
1e+00
1e+01
1e+02
1e+03
Fitted values
c) Standardised residuals over time
Standardised Residuals
4
2
0
−2
−4
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
4
2
0
−2
−4
1e−01
1e+00
1e+01
1e+02
1e+03
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
4
2
0
−2
−4
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
0.6
0.4
0.2
0.0
−0.2
5
10
15
Lag (yrs)

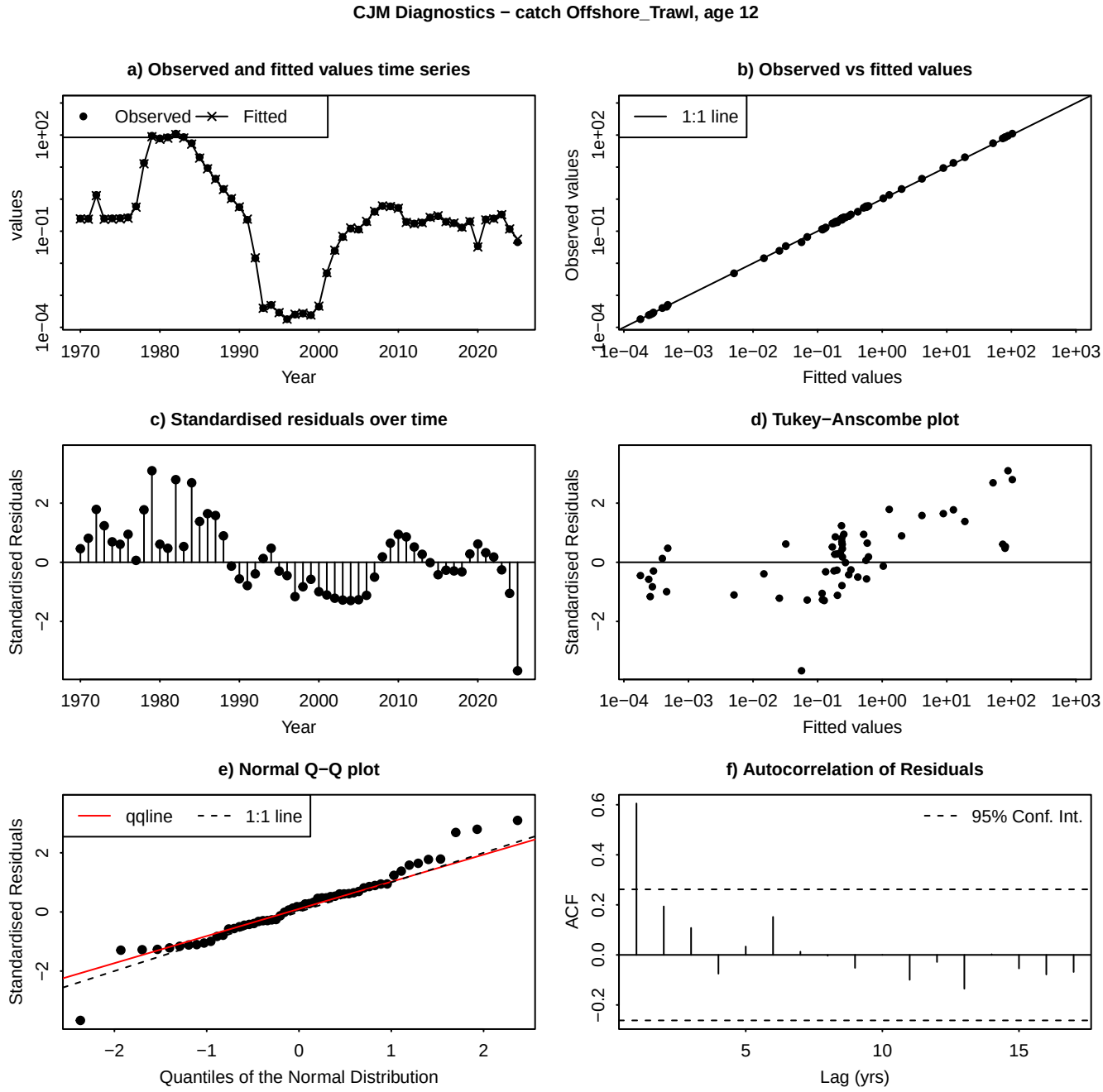

CJM Diagnostics − catch Offshore_Trawl, age 7
a) Observed and fitted values time series
Observed
Fitted
values
1e+03
1e+01
1e−01
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
1e+03
1e+01
1e−01
1e−01
1e+00
1e+01
1e+02
1e+03
Fitted values
c) Standardised residuals over time
Standardised Residuals
2
1
0
−1
−2
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
2
1
0
−1
−2
1e−01
1e+00
1e+01
1e+02
1e+03
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
2
1
0
−1
−2
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
0.4
0.2
0.0
−0.2
5
10
15
Lag (yrs)

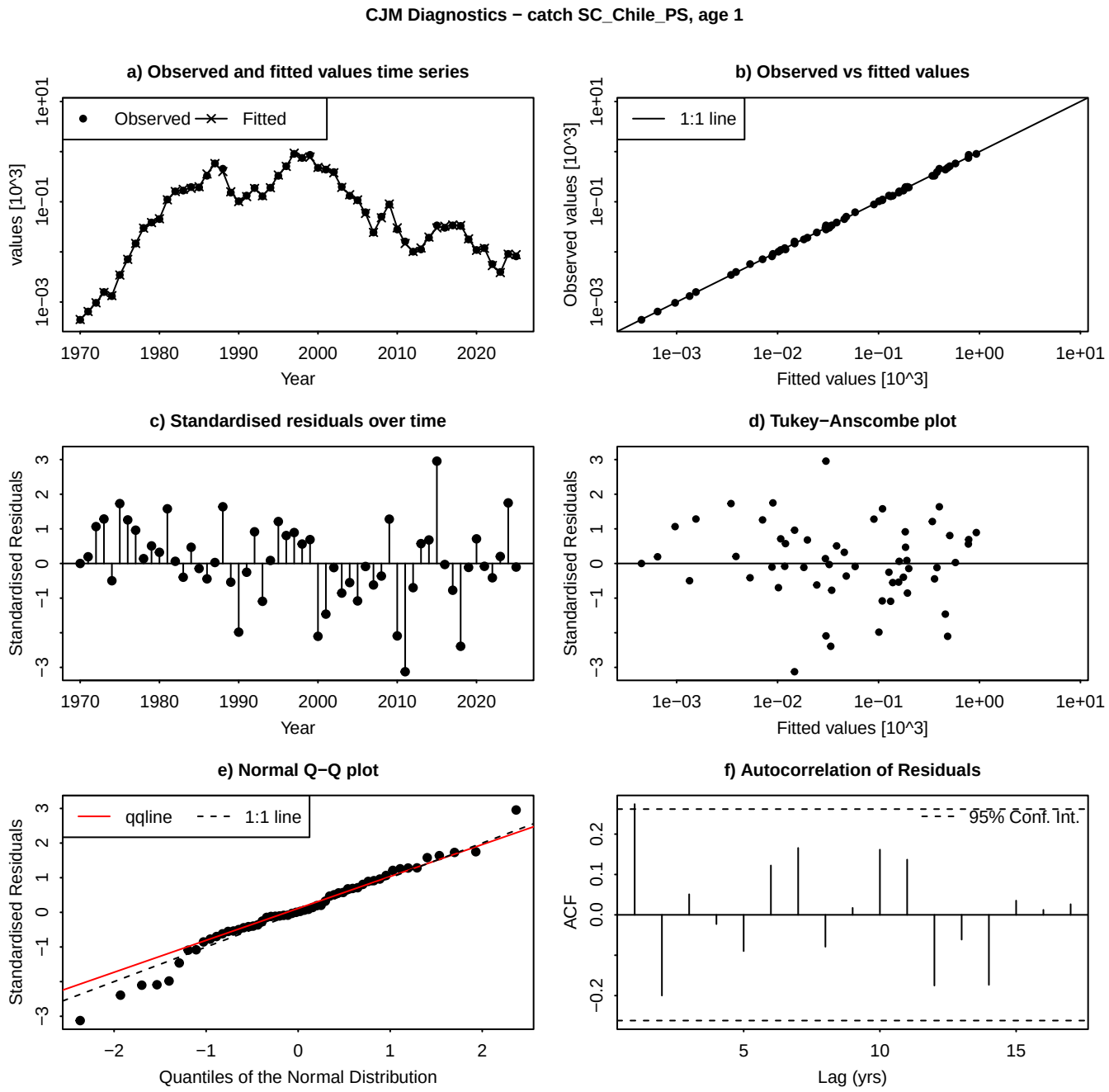

CJM Diagnostics − catch Offshore_Trawl, age 8
a) Observed and fitted values time series
Observed
Fitted
values
5e−02
1e+00
5e+01
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
5e−02
5e−01
5e+00
5e+01
5e+02
Fitted values
c) Standardised residuals over time
Standardised Residuals
−2
−1
0
1
2
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.0
0.2
0.4
0.6
5
10
15
Lag (yrs)

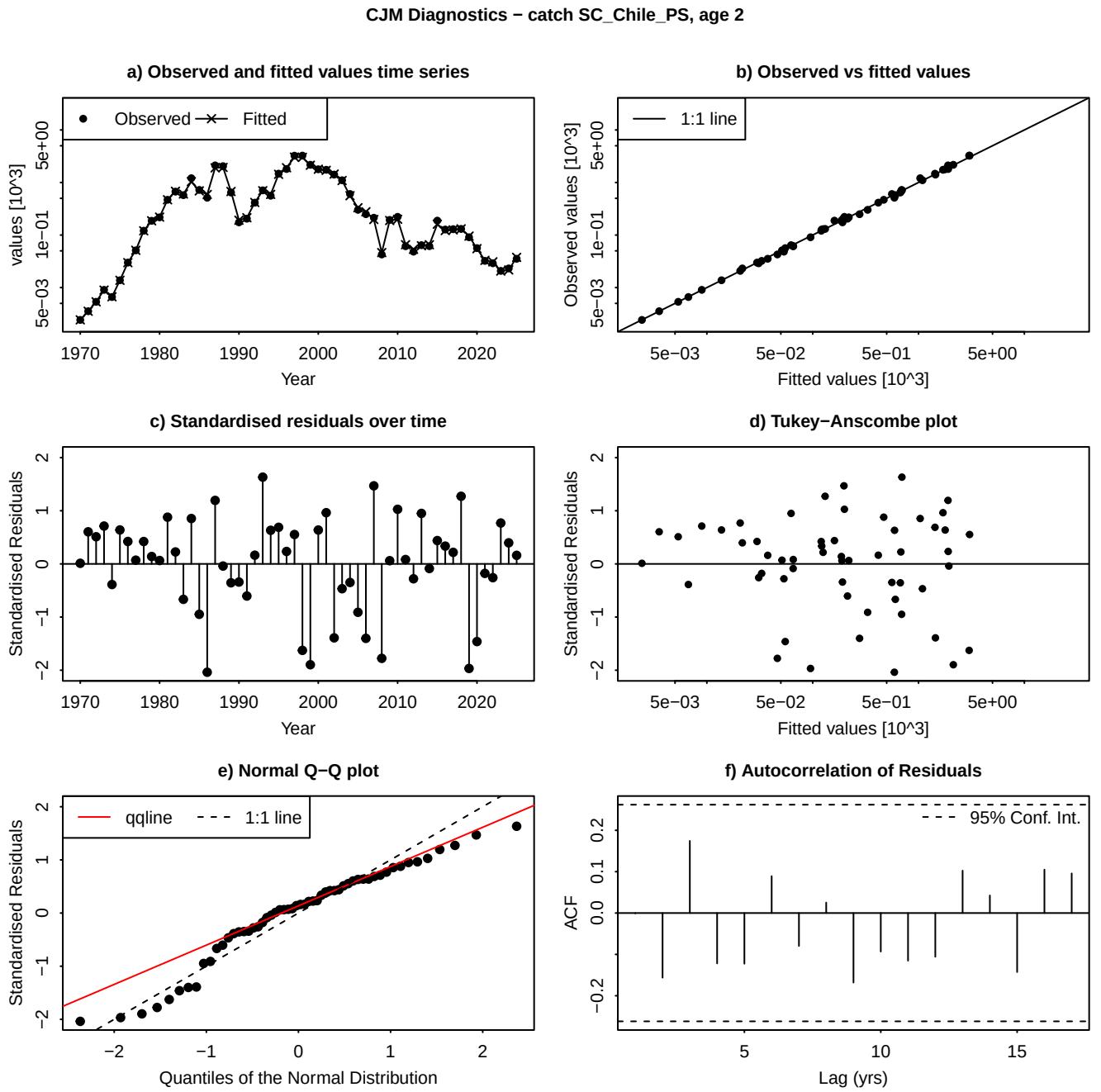

CJM Diagnostics − catch Offshore_Trawl, age 9
a) Observed and fitted values time series
Observed
Fitted
values
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

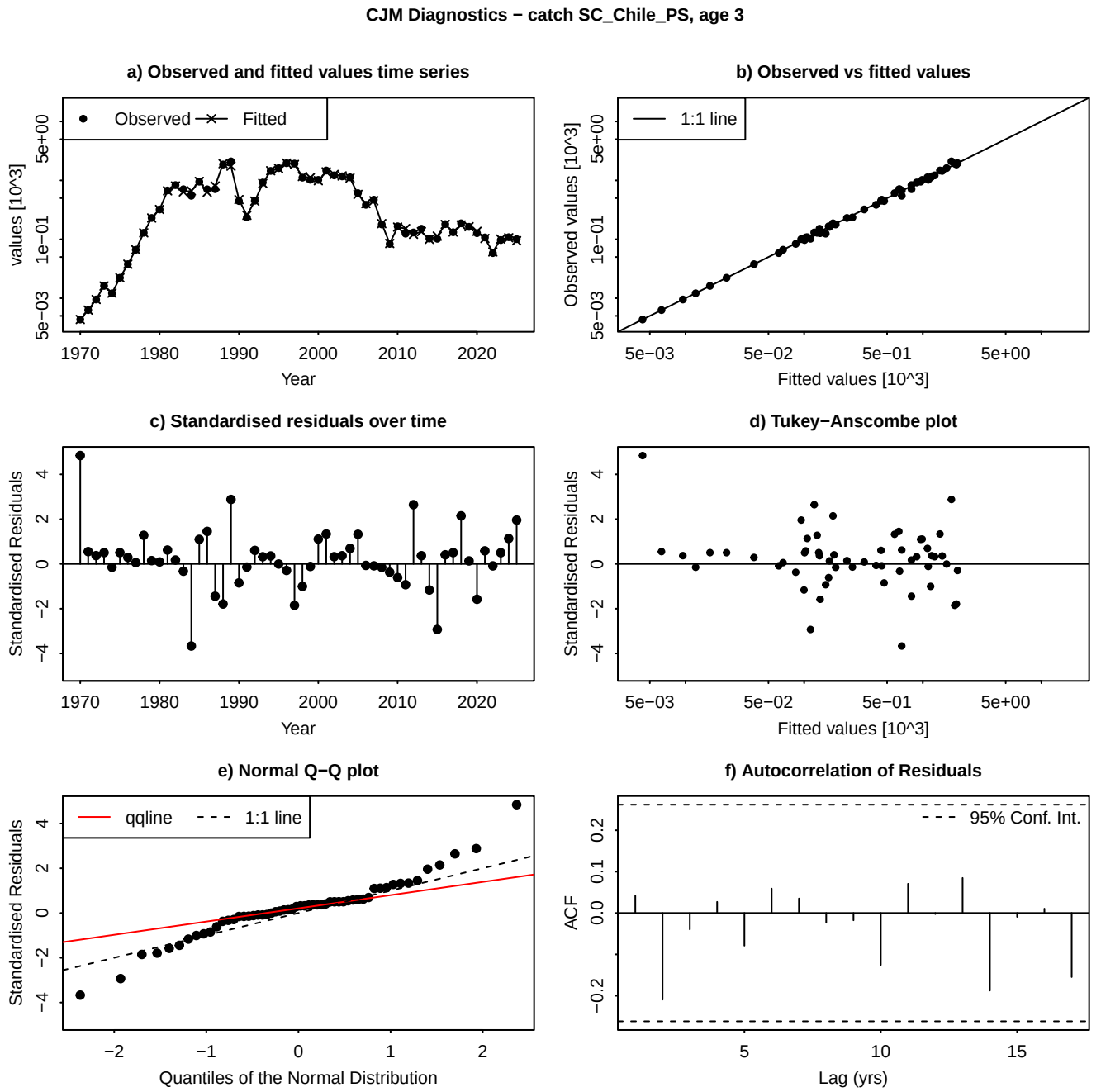

CJM Diagnostics − catch Offshore_Trawl, age 10
a) Observed and fitted values time series
Observed
Fitted
values
1e−03
1e−01
1e+01
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
1e−03
1e−01
1e+01
1e−03
1e−02
1e−01
1e+00
1e+01
1e+02
Fitted values
c) Standardised residuals over time
Standardised Residuals
−2
−1
0
1
2
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
−2
−1
0
1
2
1e−03
1e−02
1e−01
1e+00
1e+01
1e+02
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
−2
−1
0
1
2
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.2
0.4
0.6
5
10
15
Lag (yrs)

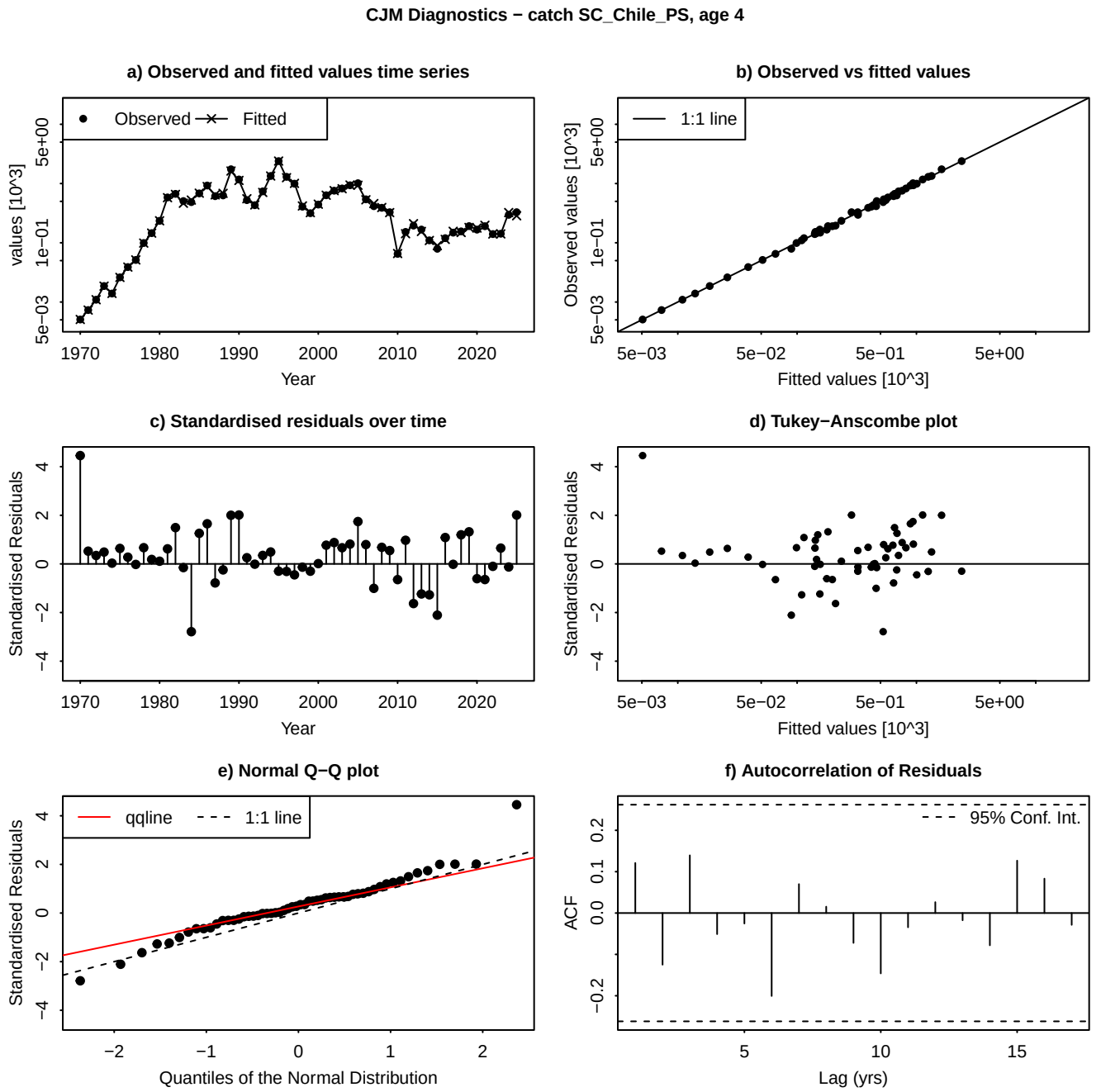

CJM Diagnostics − catch Offshore_Trawl, age 11
a) Observed and fitted values time series
Observed
Fitted
values
1e−03
1e−01
1e+01
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
1e−03
1e−01
1e+01
1e−03
1e−02
1e−01
1e+00
1e+01
1e+02
Fitted values
c) Standardised residuals over time
Standardised Residuals
−2
−1
0
1
2
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
−2
−1
0
1
2
1e−03
1e−02
1e−01
1e+00
1e+01
1e+02
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
−2
−1
0
1
2
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.2
0.4
0.6
0.8
5
10
15
Lag (yrs)

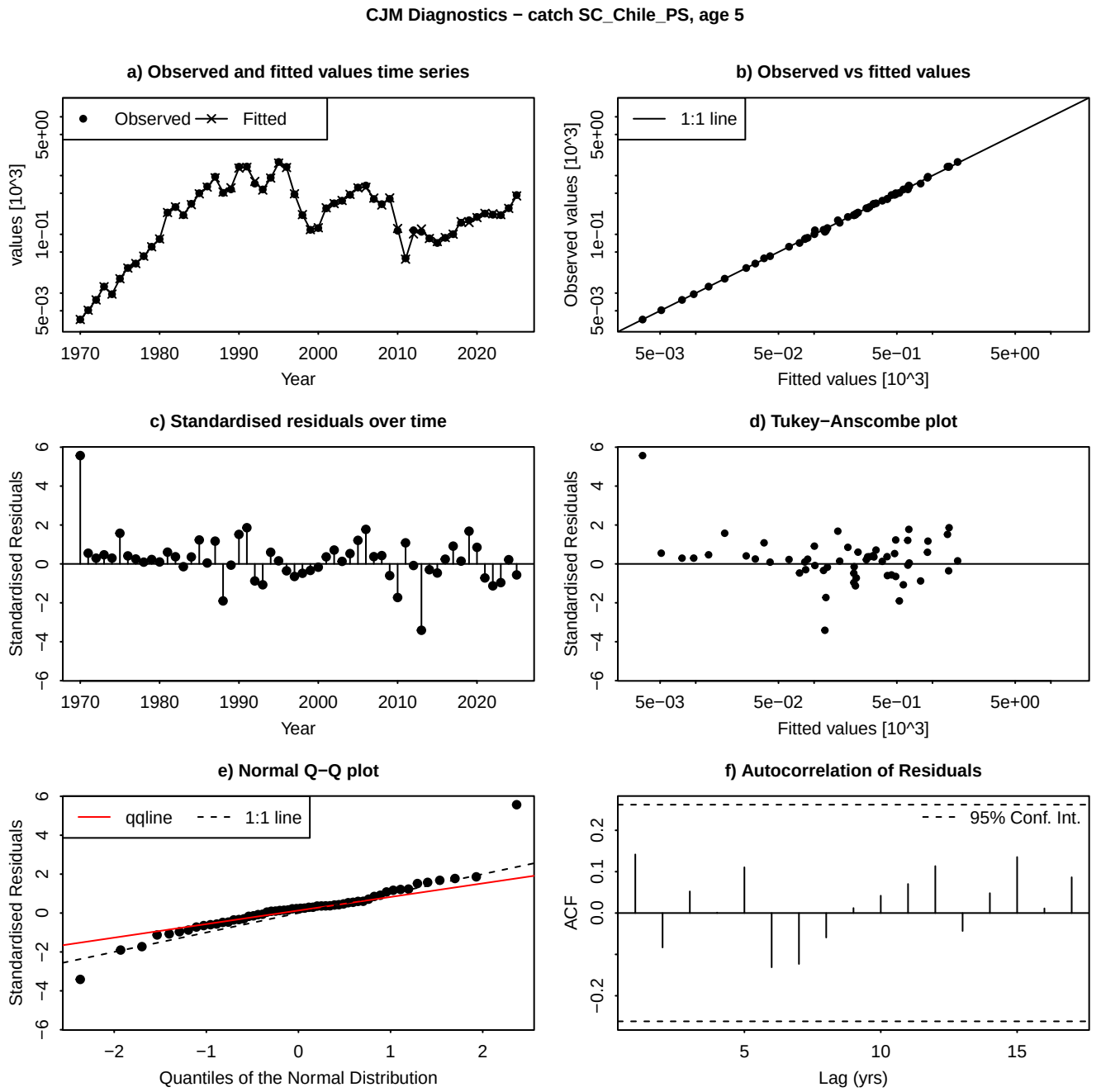

CJM Diagnostics − catch Offshore_Trawl, age 12
a) Observed and fitted values time series
Observed
Fitted
values
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

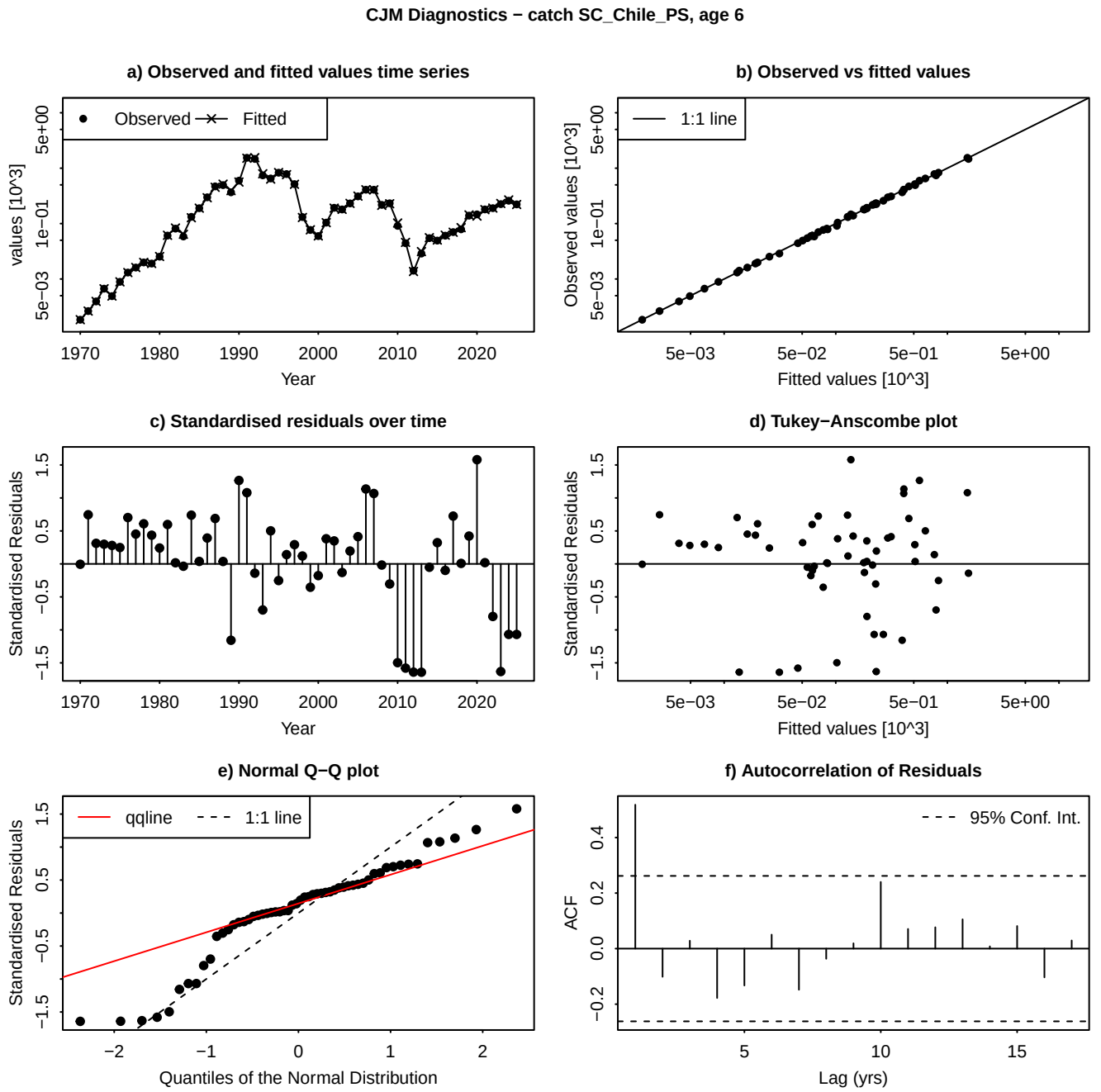

CJM Diagnostics − catch SC_Chile_PS, age 1
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
1e+01
1e−01
1e−03
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
1e+01
1e−01
1e−03
1e−03
1e−02
1e−01
1e+00
1e+01
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
3
2
1
0
−1
−3
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
3
2
1
0
−1
−3
1e−03
1e−02
1e−01
1e+00
1e+01
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
3
2
1
0
−1
−3
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
0.2
0.1
0.0
−0.2
5
10
15
Lag (yrs)

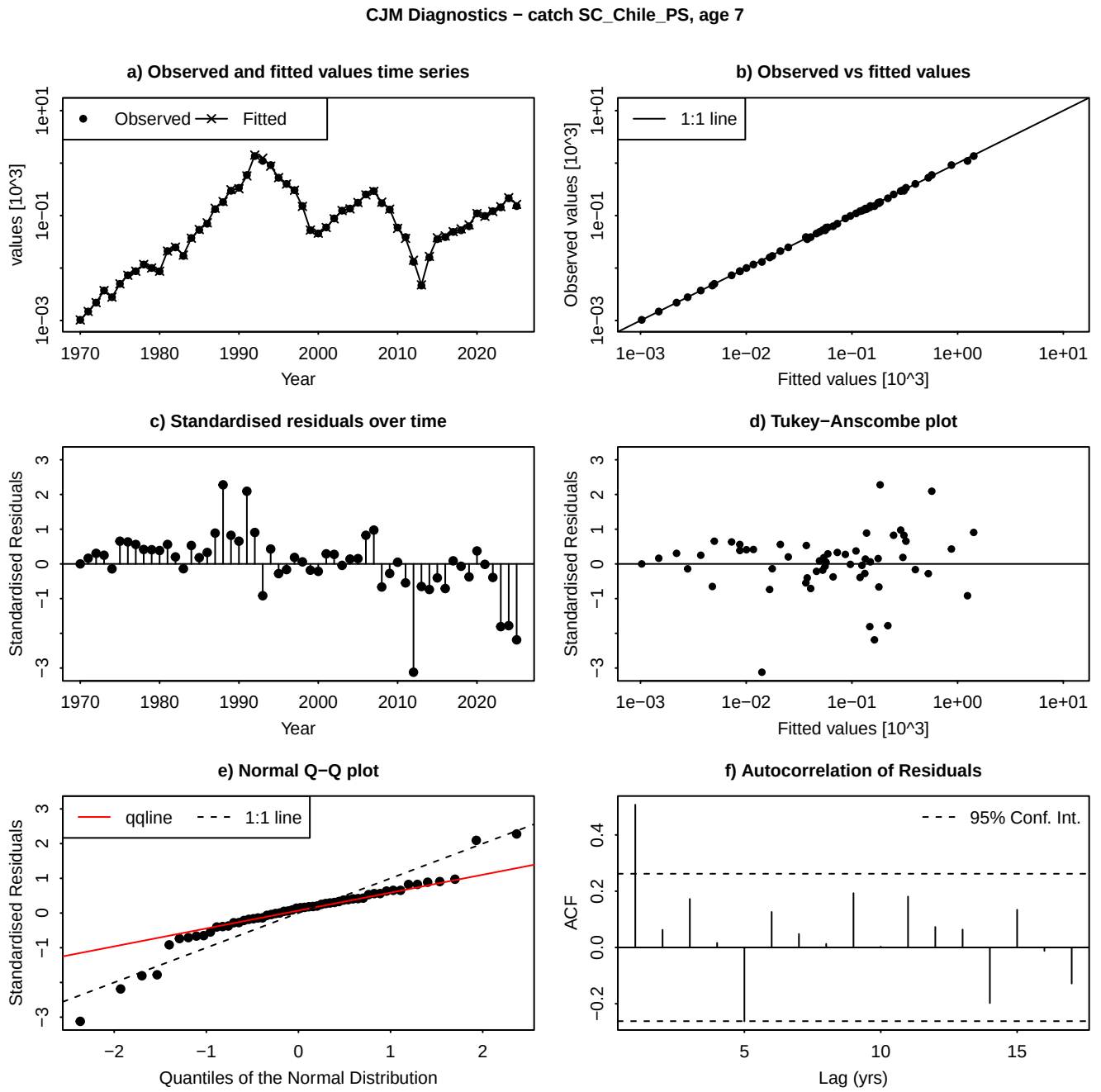

CJM Diagnostics − catch SC_Chile_PS, age 2
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
5e−03
1e−01
5e+00
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
5e−03
1e−01
5e+00
5e−03
5e−02
5e−01
5e+00
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
−2
−1
0
1
2
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
−2
−1
0
1
2
5e−03
5e−02
5e−01
5e+00
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
−2
−1
0
1
2
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.0
0.1
0.2
5
10
15
Lag (yrs)

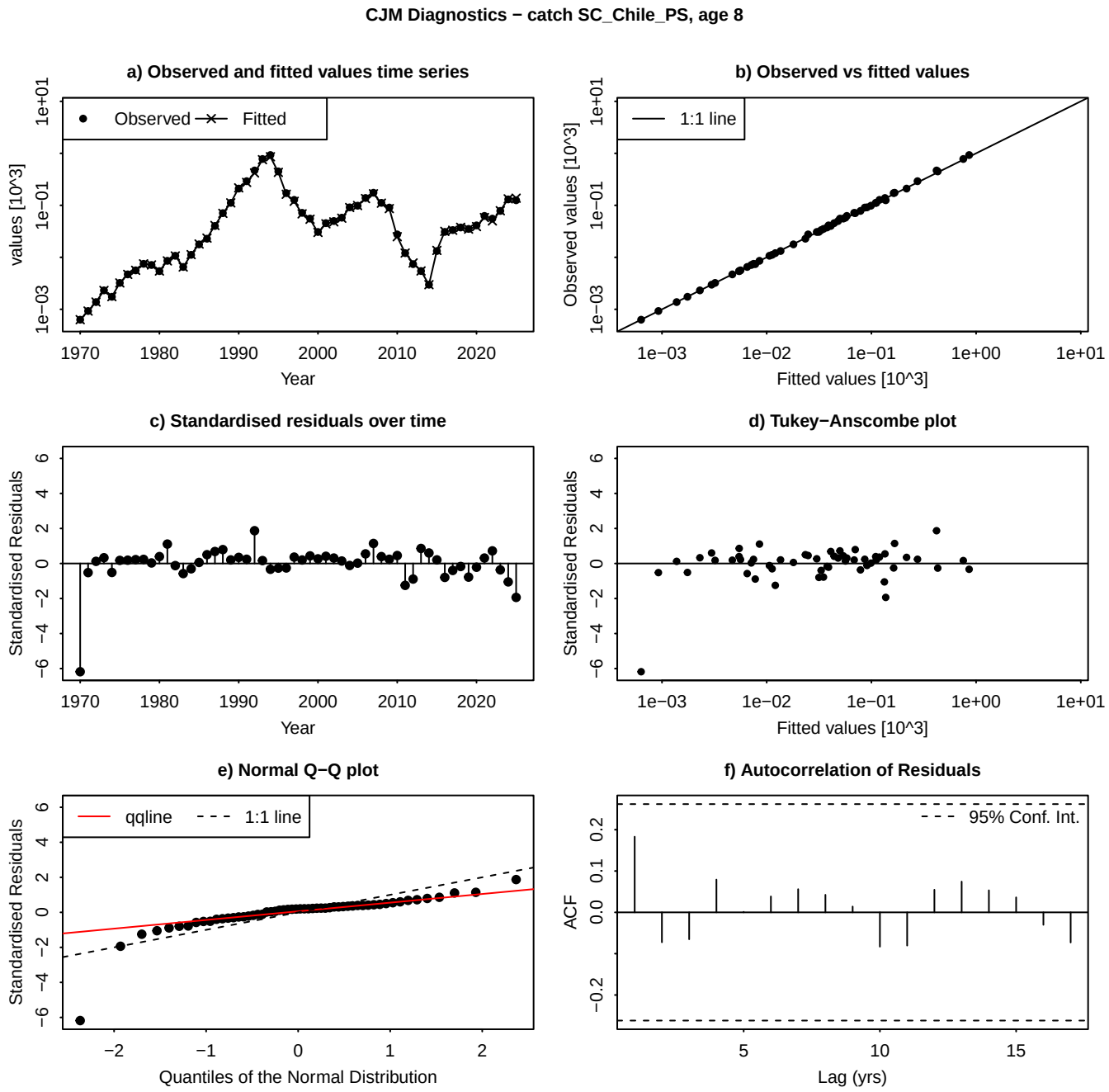

CJM Diagnostics − catch SC_Chile_PS, age 3
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
5e−03
1e−01
5e+00
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
5e−03
1e−01
5e+00
5e−03
5e−02
5e−01
5e+00
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
−4
−2
0
2
4
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
−4
−2
0
2
4
5e−03
5e−02
5e−01
5e+00
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
−4
−2
0
2
4
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.0
0.1
0.2
5
10
15
Lag (yrs)

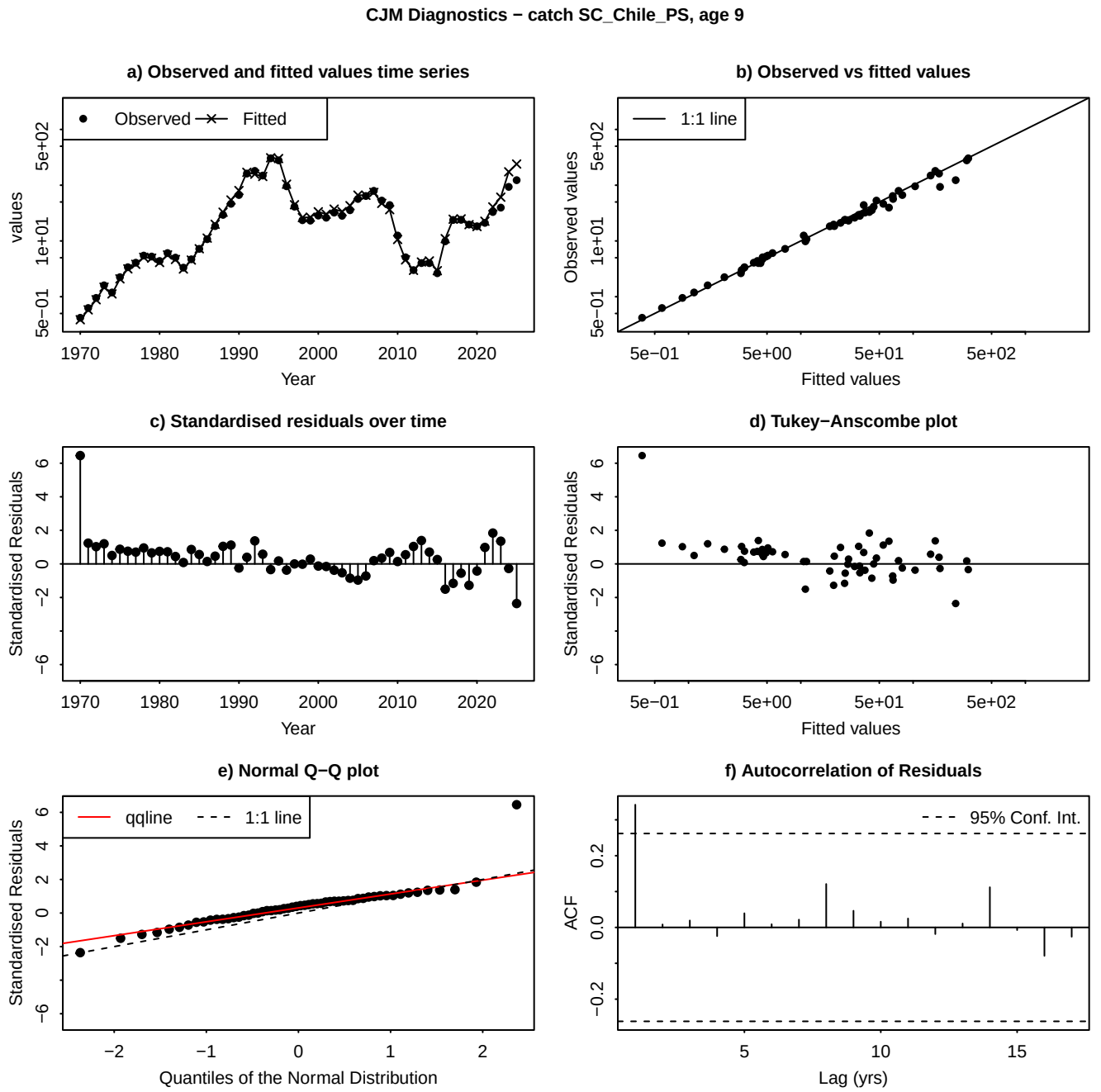

CJM Diagnostics − catch SC_Chile_PS, age 4
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
5e−03
1e−01
5e+00
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
d) Tukey−Anscombe plot
e) Normal Q−Q plot
qqline
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

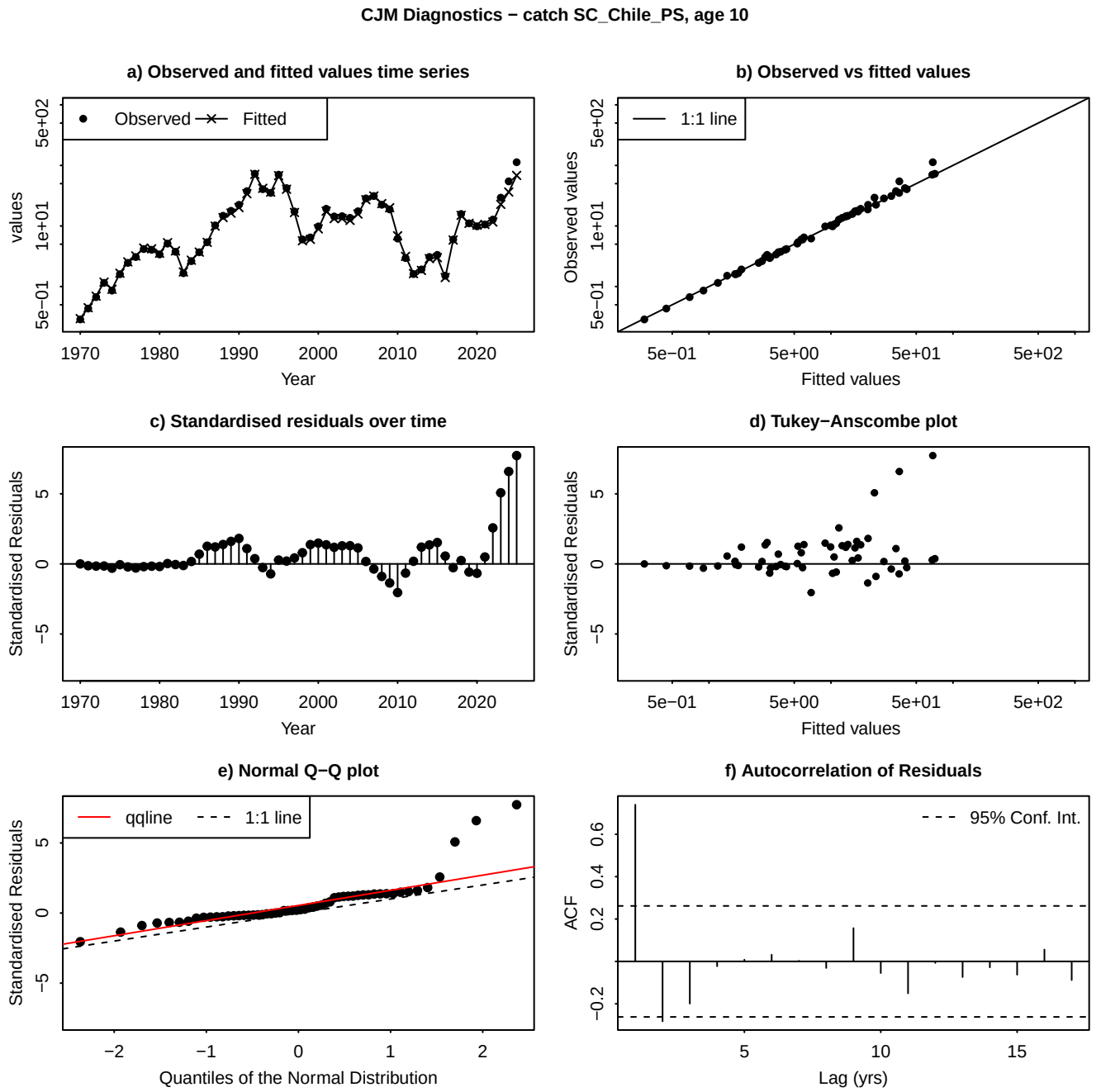

CJM Diagnostics − catch SC_Chile_PS, age 5
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
5e+00
1e−01
5e−03
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
5e+00
1e−01
5e−03
5e−03
5e−02
5e−01
5e+00
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
6
4
2
0
−2
−4
−6
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
6
4
2
0
−2
−4
−6
5e−03
5e−02
5e−01
5e+00
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
6
4
2
0
−2
−4
−6
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
0.2
0.1
0.0
−0.2
5
10
15
Lag (yrs)

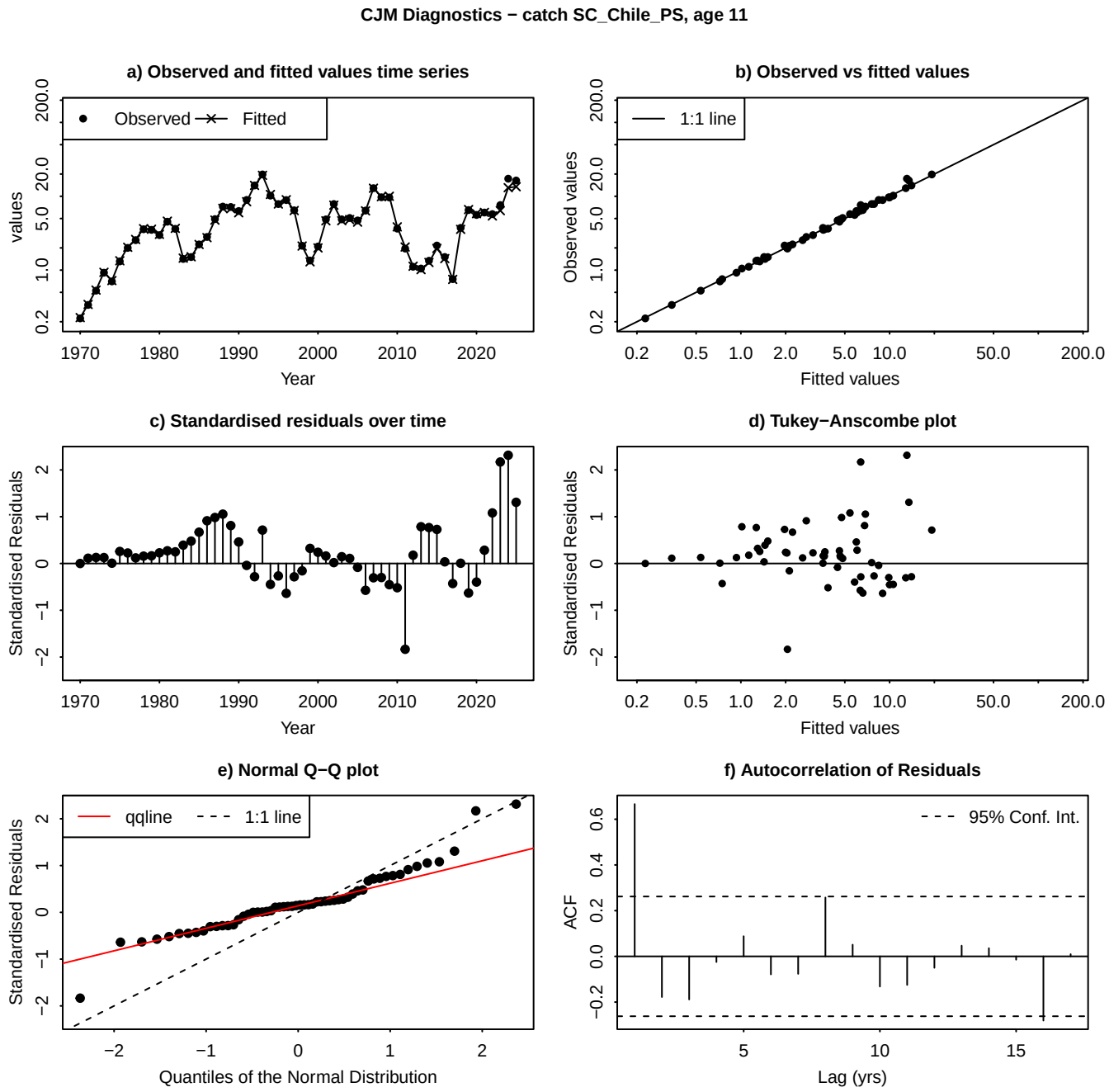

CJM Diagnostics − catch SC_Chile_PS, age 6
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
5e−03
1e−01
5e+00
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
5e−03
1e−01
5e+00
5e−03
5e−02
5e−01
5e+00
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
−1.5
−0.5
0.5
1.5
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
−1.5
−0.5
0.5
1.5
5e−03
5e−02
5e−01
5e+00
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
−1.5
−0.5
0.5
1.5
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.0
0.2
0.4
5
10
15
Lag (yrs)

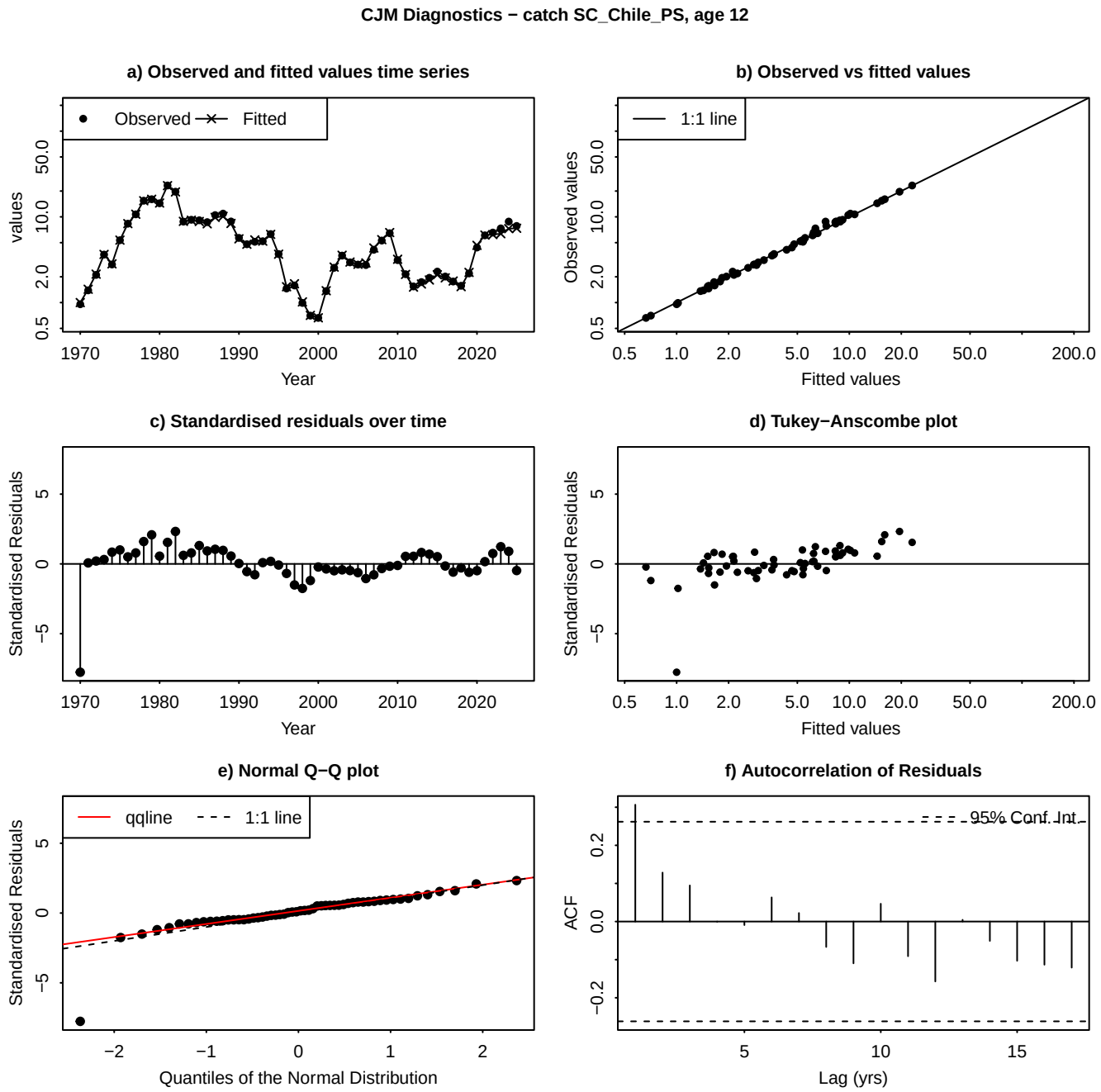

CJM Diagnostics − catch SC_Chile_PS, age 7
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
1e+01
1e−01
1e−03
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
Fitted values [10^3]
1e−03
1e−02
1e−01
1e+00
1e+01
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

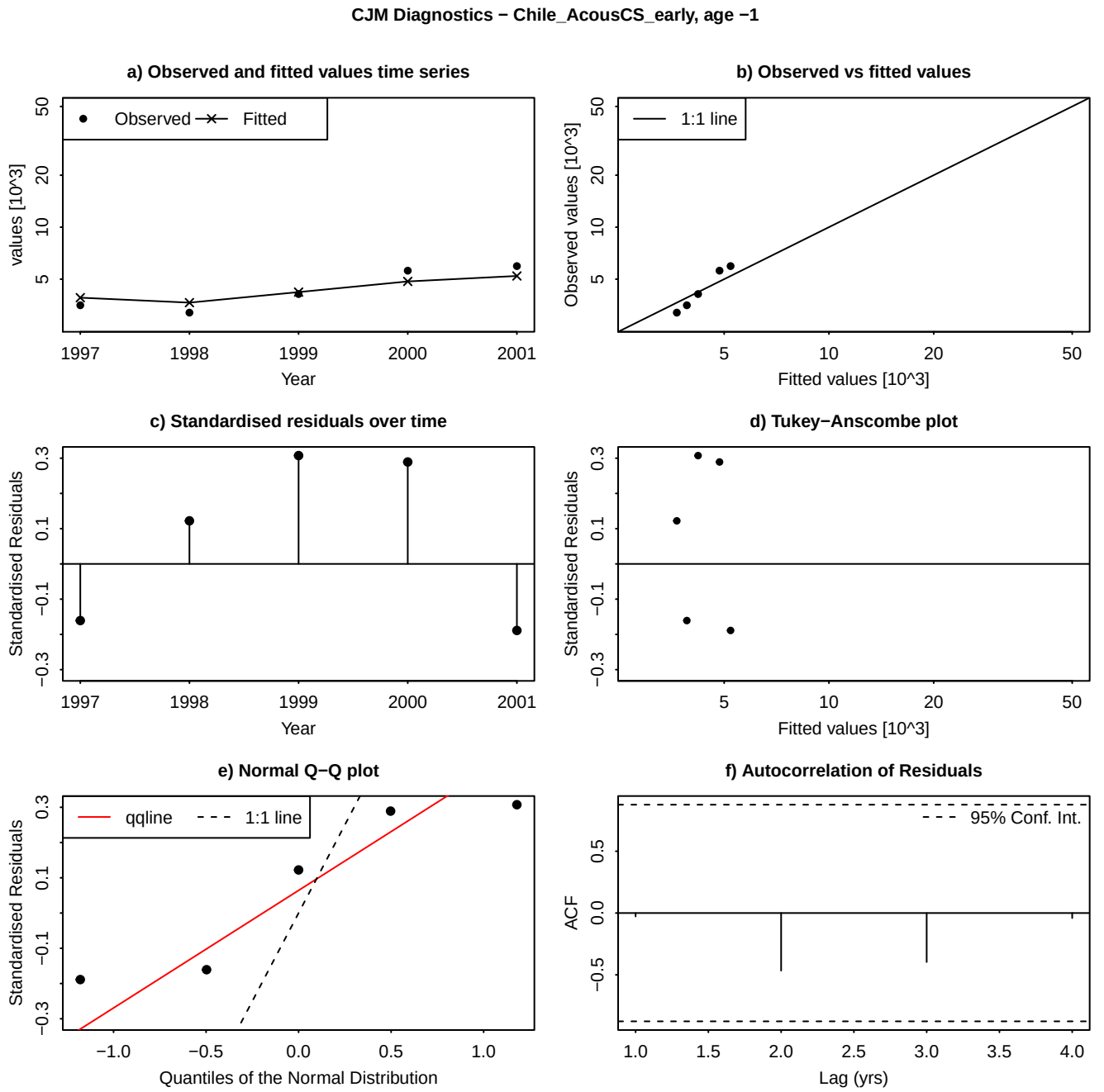

CJM Diagnostics − catch SC_Chile_PS, age 8
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
1e+01
1e−01
1e−03
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
1e+01
1e−01
1e−03
1e−03
1e−02
1e−01
1e+00
1e+01
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
6
4
2
0
−2
−4
−6
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
6
4
2
0
−2
−4
−6
1e−03
1e−02
1e−01
1e+00
1e+01
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
6
4
2
0
−2
−4
−6
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
0.2
0.1
0.0
−0.2
5
10
15
Lag (yrs)

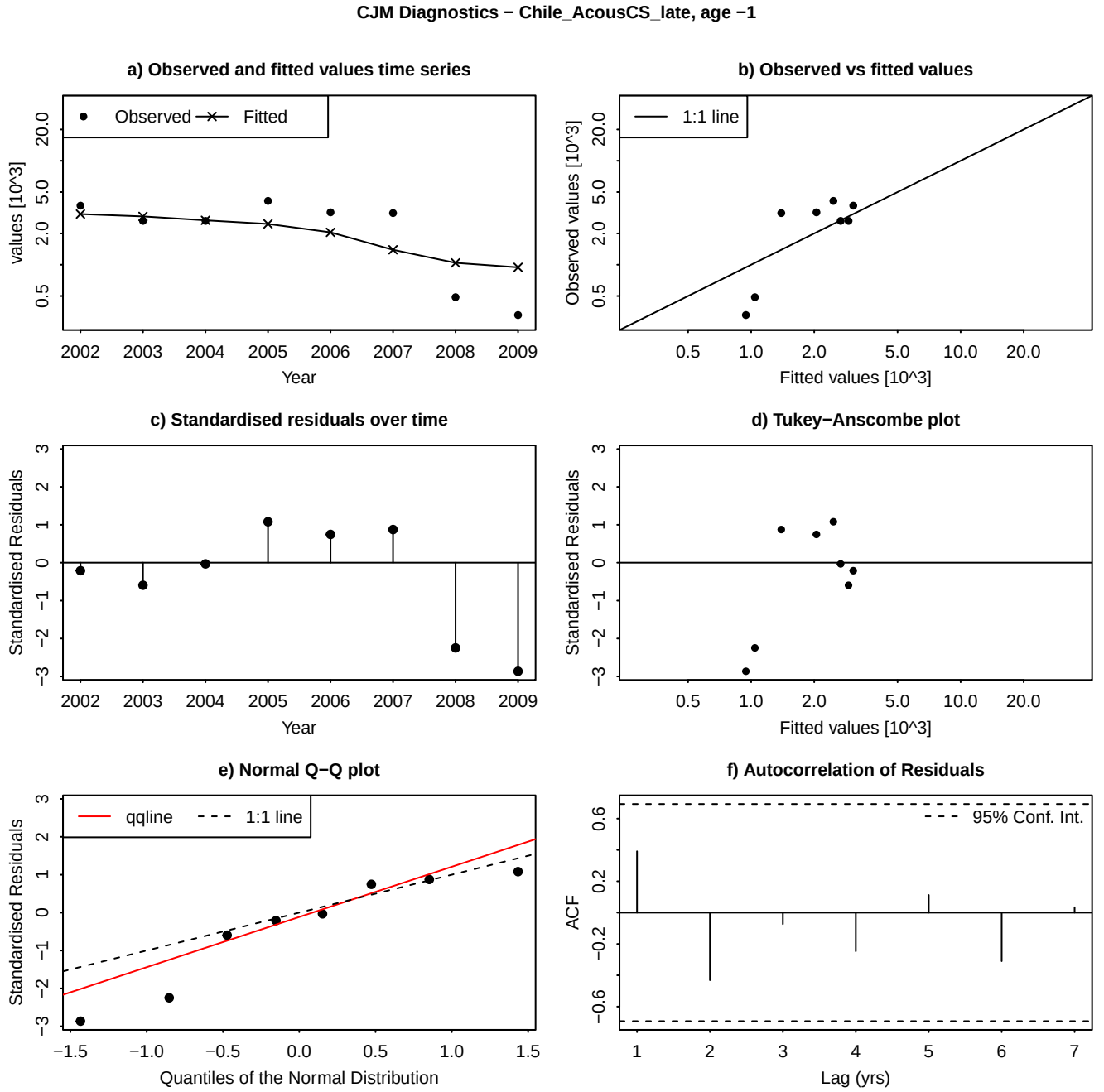

CJM Diagnostics − catch SC_Chile_PS, age 9
a) Observed and fitted values time series
Observed
Fitted
values
5e−01
1e+01
5e+02
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
5e−01
1e+01
5e+02
5e−01
5e+00
5e+01
5e+02
Fitted values
c) Standardised residuals over time
Standardised Residuals
−6
−2
0
2
4
6
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
−6
−2
0
2
4
6
5e−01
5e+00
5e+01
5e+02
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
−6
−2
0
2
4
6
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.0
0.2
5
10
15
Lag (yrs)

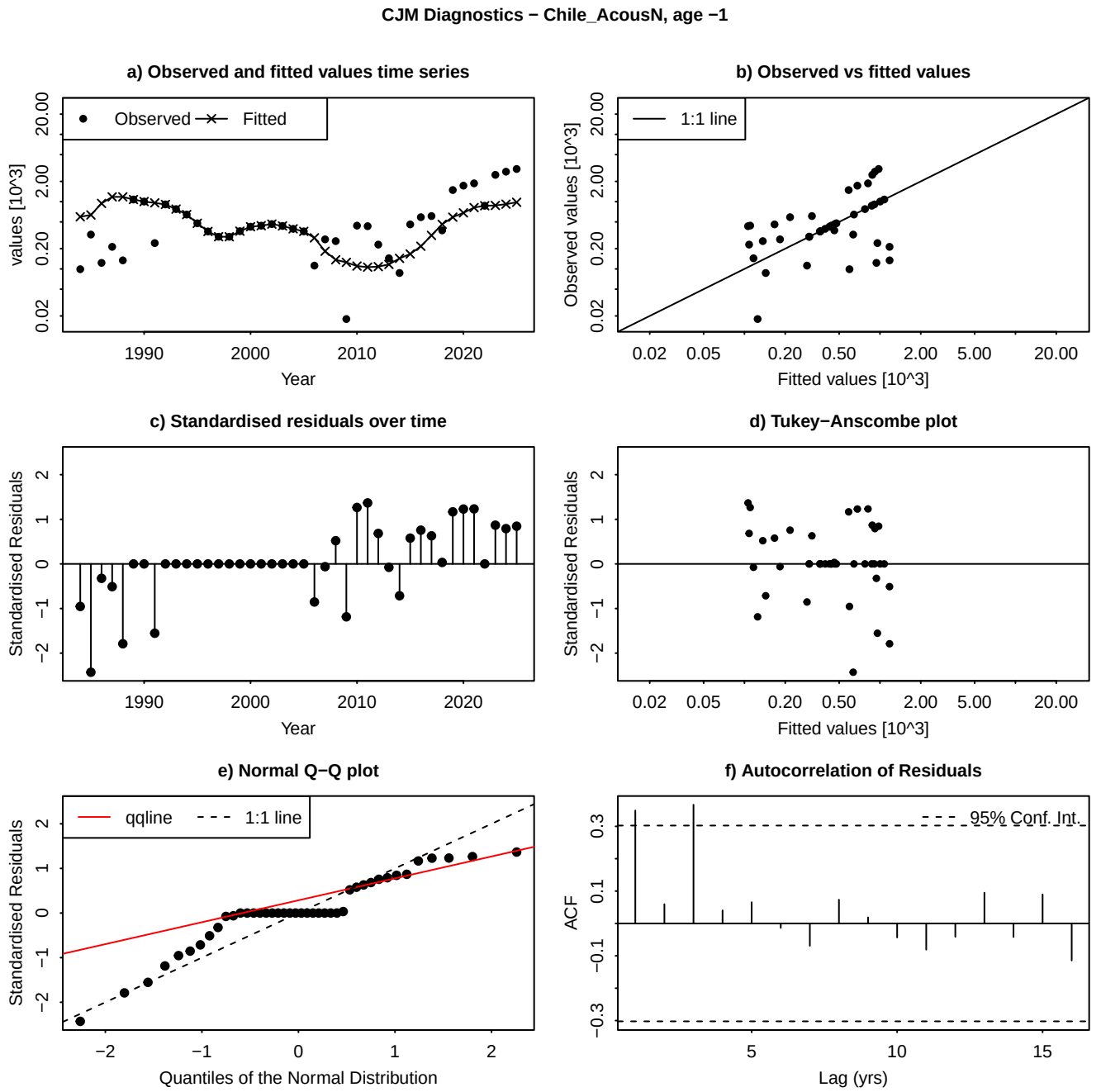

CJM Diagnostics − catch SC_Chile_PS, age 10
a) Observed and fitted values time series
Observed
Fitted
values
5e−01
1e+01
5e+02
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
5e−01
1e+01
5e+02
5e−01
5e+00
5e+01
5e+02
Fitted values
c) Standardised residuals over time
Standardised Residuals
−5
0
5
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
−5
0
5
5e−01
5e+00
5e+01
5e+02
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
−5
0
5
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.2
0.4
0.6
5
10
15
Lag (yrs)

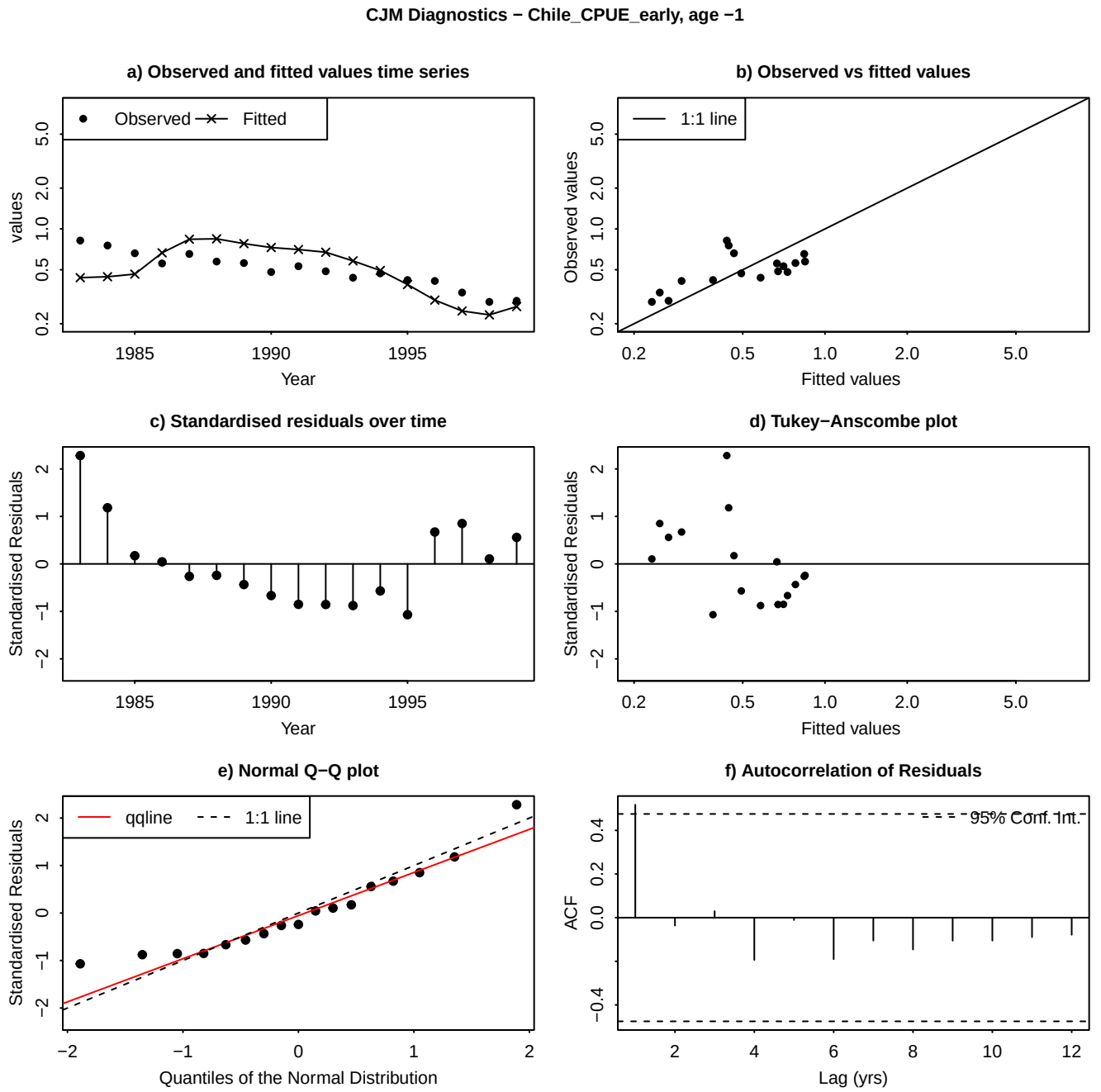

CJM Diagnostics − catch SC_Chile_PS, age 11
a) Observed and fitted values time series
Observed
Fitted
values
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

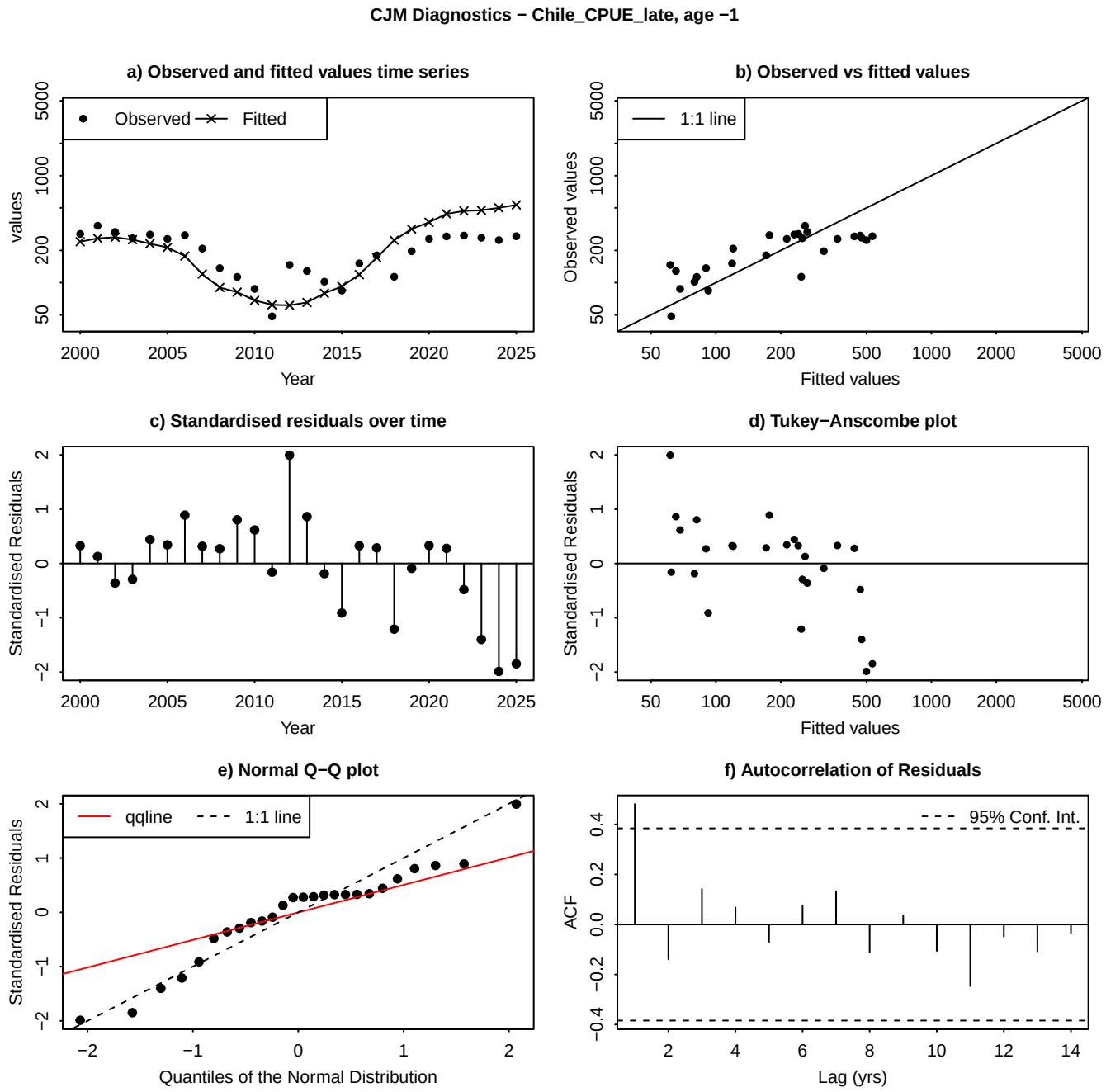

CJM Diagnostics − catch SC_Chile_PS, age 12
a) Observed and fitted values time series
Observed
Fitted
values
0.5
2.0
10.0
50.0
1970
1980
1990
2000
2010
2020
Year
b) Observed vs fitted values
1:1 line
Observed values
0.5
2.0
10.0
50.0
0.5
1.0
2.0
5.0
10.0
20.0
50.0
200.0
Fitted values
c) Standardised residuals over time
Standardised Residuals
−5
0
5
1970
1980
1990
2000
2010
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
−5
0
5
0.5
1.0
2.0
5.0
10.0
20.0
50.0
200.0
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
−5
0
5
−2
−1
0
1
2
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.2
0.0
0.2
5
10
15
Lag (yrs)

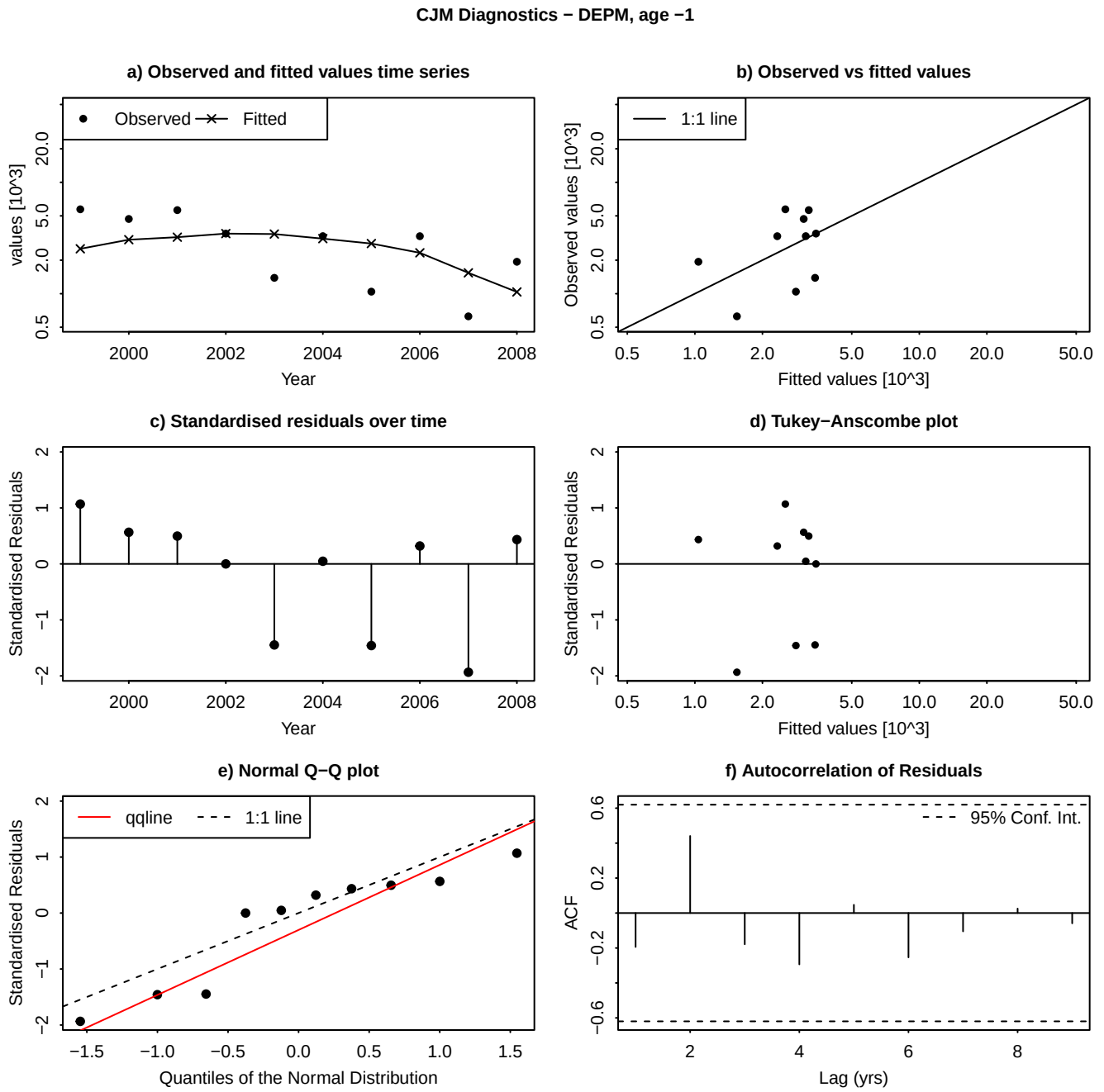

CJM Diagnostics − Chile_AcousCS_early, age −1
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

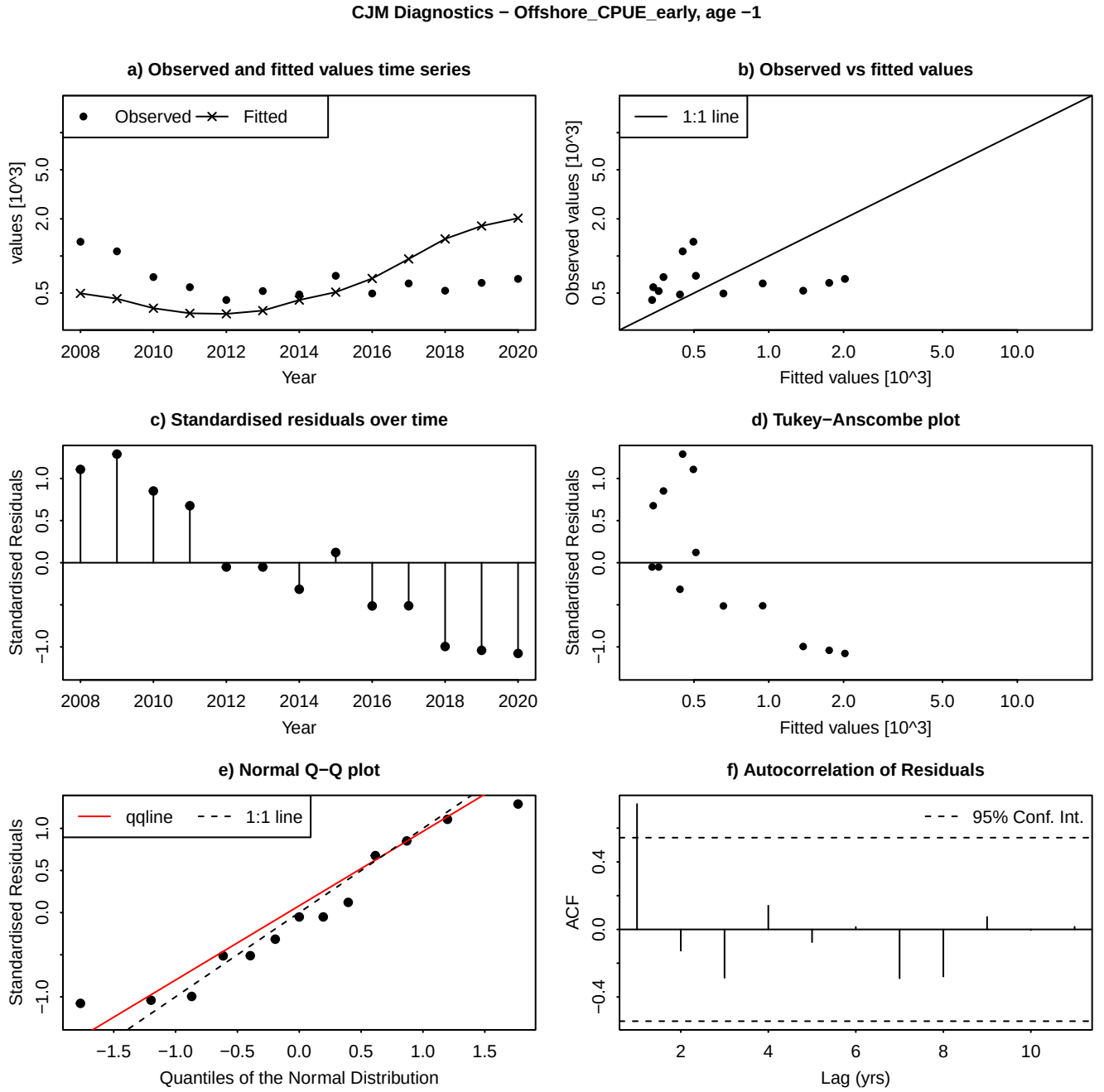

CJM Diagnostics − Chile_AcousCS_late, age −1
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

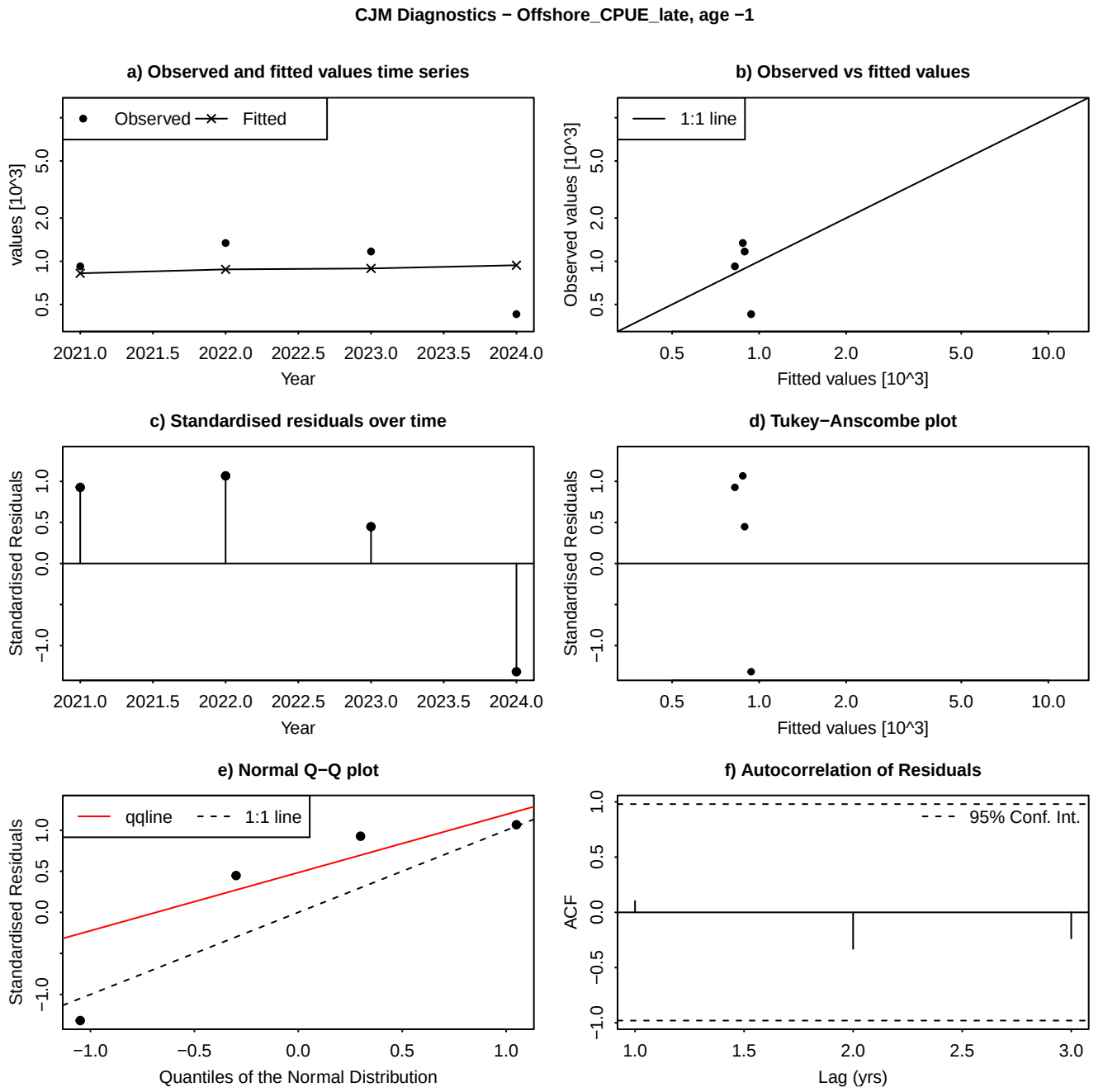

CJM Diagnostics − Chile_AcousN, age −1
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

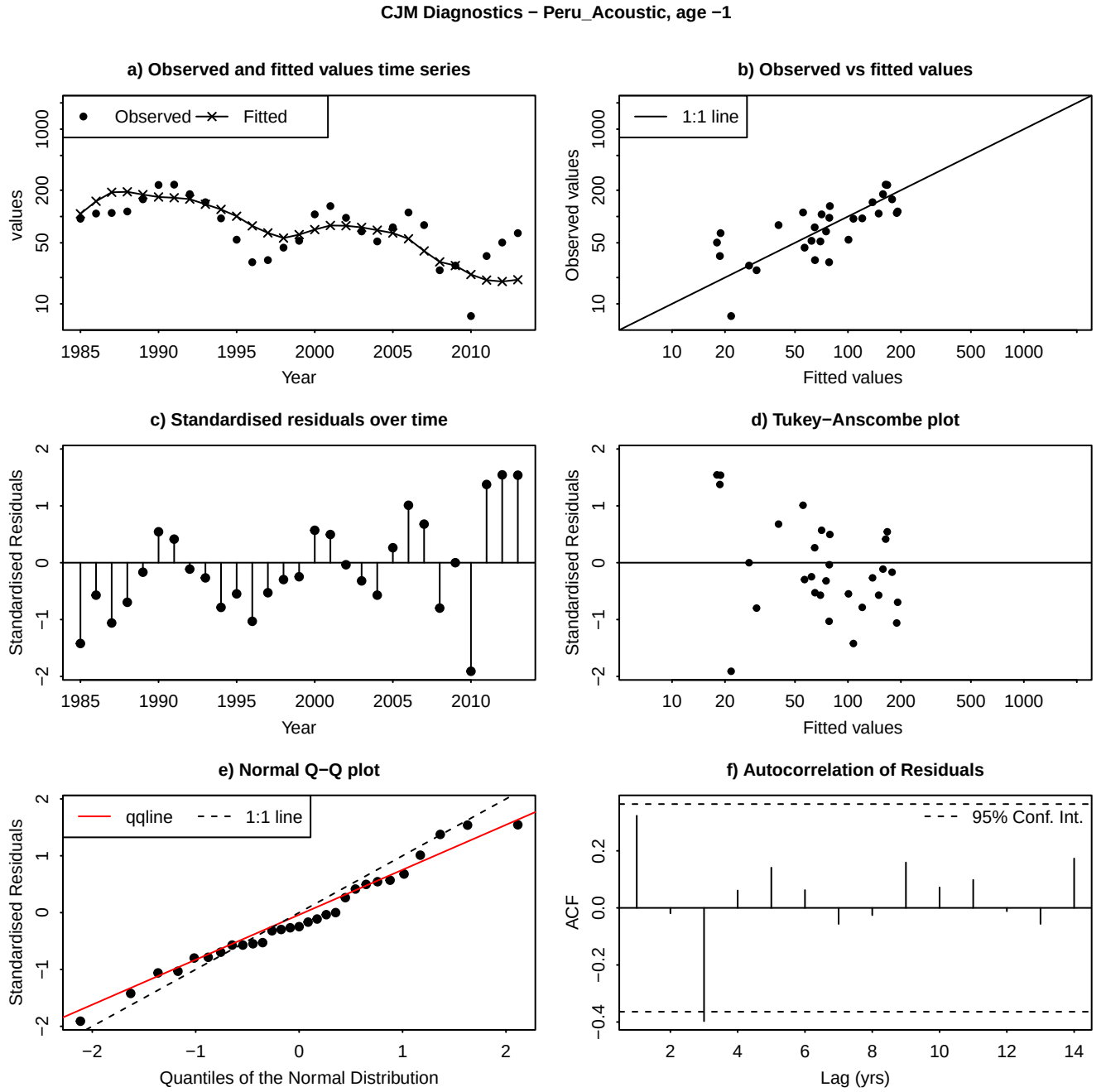

CJM Diagnostics − Chile_CPUE_early, age −1
a) Observed and fitted values time series
Observed
Fitted
values
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)

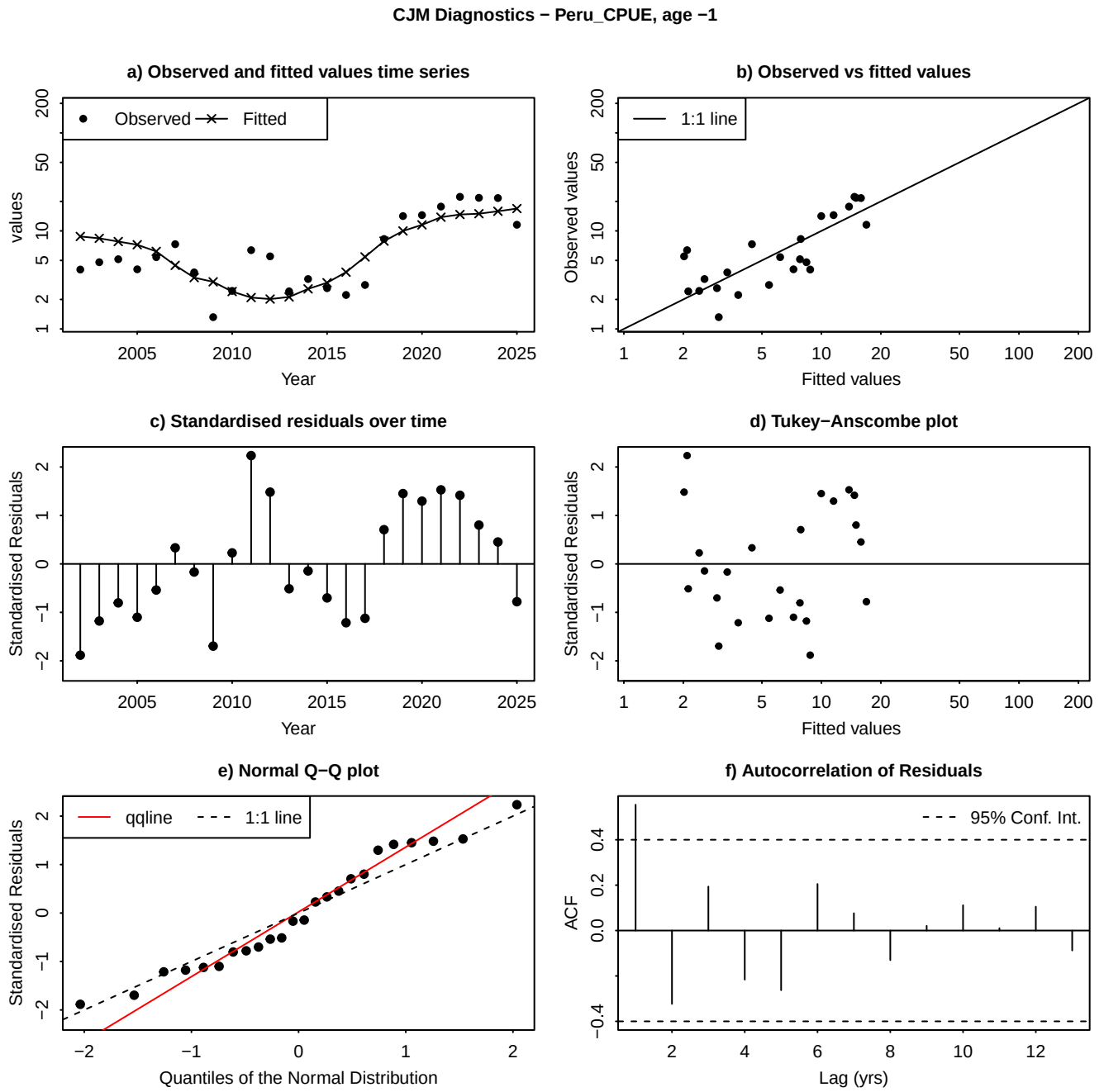

CJM Diagnostics − Chile_CPUE_late, age −1
a) Observed and fitted values time series
Observed
Fitted
values
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)


CJM Diagnostics − DEPM, age −1
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)


CJM Diagnostics − Offshore_CPUE_early, age −1
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
0.5
2.0
5.0
2008
2010
2012
2014
2016
2018
2020
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
0.5
2.0
5.0
0.5
1.0
2.0
5.0
10.0
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
−1.0
0.0
0.5
1.0
2008
2010
2012
2014
2016
2018
2020
Year
d) Tukey−Anscombe plot
Standardised Residuals
−1.0
0.0
0.5
1.0
0.5
1.0
2.0
5.0
10.0
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
−1.0
0.0
0.5
1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
−0.4
0.0
0.4
2
4
6
8
10
Lag (yrs)


CJM Diagnostics − Offshore_CPUE_late, age −1
a) Observed and fitted values time series
Observed
Fitted
values [10^3]
Year
b) Observed vs fitted values
1:1 line
Observed values [10^3]
Fitted values [10^3]
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values [10^3]
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)


CJM Diagnostics − Peru_Acoustic, age −1
a) Observed and fitted values time series
Observed
Fitted
values
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)


CJM Diagnostics − Peru_CPUE, age −1
a) Observed and fitted values time series
Observed
Fitted
values
Year
b) Observed vs fitted values
1:1 line
Observed values
Fitted values
c) Standardised residuals over time
Standardised Residuals
Year
d) Tukey−Anscombe plot
Standardised Residuals
Fitted values
e) Normal Q−Q plot
qqline
1:1 line
Standardised Residuals
Quantiles of the Normal Distribution
f) Autocorrelation of Residuals
95% Conf. Int.
ACF
Lag (yrs)